

# Ascent sequences with bounded multiplicity (OEIS A294220)

Factorial growth constants of every cap, fixed Taylor sectors of the large-cap expansion,  
and the ratio and endpoint laws

A merged research report built from three manuscripts  
(batch 77, manuscripts 47 and 52; batch 102, bundle Report 99)

Manuscripts dated 1 and 2 October 2026; merged 2 October 2026; Part III added 5 October  
2026

## Abstract

Let  $a_n^{(b)}$  count ascent sequences of length  $n$  in which each value occurs at most  $b$  times (the array OEIS A294220; cap three is A317784, cap two A202058). This report prints three manuscripts as three Parts. Part I proves, for every fixed  $b \geq 3$ , that  $(a_n^{(b)}/n!)^{1/n} \rightarrow \mu_b = 1/T_b$  with  $T_b = \int_0^\infty \log E_b(v)/(E_b(v) - 1) dv$  and  $E_b$  the truncated exponential, with a coarse error  $O_b(n/\log n)$  in  $\log a_n^{(b)}$ , a controlled Lambert- $W$  inverse in the length, the large-cap expansion  $T_b = \pi^2/6 + (b+1)(\zeta(b+2) - 1) + O(b^{5/2}(4/9)^b)$  with explicit constants, and a rounding-safe inverse in the cap with counterexamples to naive rounding; the cap-two input  $T_2 = 3\pi^2/8$  is taken from the sibling report on A202058. Part II writes  $T_b - \pi^2/6$  exactly as a convergent sum of positive Taylor sectors  $A_{r,b}$ , proves that the remainder after any fixed number  $K$  of sectors is asymptotic to sector  $K+1$ , and expands every fixed sector to all algebraic orders in  $1/b$ ; in particular the remainder of Part I is  $\sim \frac{8}{9\sqrt{\pi}}\sqrt{b}(4/9)^b$ . Part III, which writes the cap as  $r$ , proves for every fixed  $r \geq 2$  both the root limit and the consecutive-ratio limit  $a_{n+1}^{(r)}/((n+1)a_n^{(r)}) \rightarrow \mu_r$ , exponentially concentrated multiplicity, distinct-value and ascent proportions with a full large-deviation principle, and an  $O_r(n^{r+2})$  counting algorithm; it re-derives the large-cap analysis in the variable  $r+1$  and adds a continuous cap and exact-sector inverse envelopes. No prefactor or all-orders expansion in  $n$ , no joint regime  $b = b(n)$ , and no uniformity in a growing number of sectors is claimed. The report is unrefereed, and none of it is formalized.

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## Guide to this report

This guide was written when the two manuscripts were merged. Everything after it, from Part I on, is the manuscripts’ text, except for the passages marked **[write, 2 October 2026]** and the changes listed under “Conventions of the merge” below. Part III was added at the batch-102 write of 5 October 2026, with the passages marked **[write, 5 October 2026]** and the Guide’s statements about Part III.

**[write, 5 October 2026]** Until this write the subtitle read “Factorial growth constants of every cap  $b \geq 3$ , and fixed Taylor sectors of the large-cap expansion”, the abstract said that no ratio limit was claimed for  $b \geq 3$ , and the Guide described two Parts. Part III covers every cap  $r \geq 2$  ( $r$  is the  $b$  of Parts I and II) and proves the ratio limit. The title block, the abstract and this Guide were updated; the earlier statements remain true of Parts I and II.

## What the report proves

- **Part I** (source 47). Lemma 2.1 and Section 2: an exact compaction by a suffix involution, for every cap. Theorem 1.1: for every fixed  $b \geq 3$  the exponential generating function has radius  $T_b$ , and  $\log a_n^{(b)} = \log n! - n \log T_b + O_b(n/\log n)$ , by uniform positive-operator barriers (Sections 3–4) and a coefficient extraction at every length (Section 5). Section 6: the length inverse  $N_b(X) = n_0 + O_b(L/(\log L)^3)$ ,  $n_0 = L/W_0(L/(eT_b))$ . Theorem 1.2 and Section 7: monotone convergence  $T_b \downarrow \pi^2/6$ , the exact first variation  $(b+1)(\zeta(b+2)-1)$  and the explicit bound  $0 \leq T_b - \pi^2/6 - (b+1)(\zeta(b+2)-1) \leq 160b^{5/2}(4/9)^b$  for every  $b \geq 3$ . Section 8: the  $W_{-1}$  cap inverse with integer brackets (35), (36), and counterexamples to rounding by  $\lceil b_0 \rceil$ .
- **Part II** (source 52). Theorem 12.1:  $T_b - \pi^2/6 = \sum_{r \geq 1} A_{r,b}$  with strictly positive sectors, for every  $b \geq 2$ ; for fixed  $K$  the remainder  $\mathcal{R}_{K,b}$  is positive and  $\mathcal{R}_{K,b} \sim A_{K+1,b}$ . Theorem 13.1:  $A_{r,b} = C_r b^{(3-r)/2} \beta_r^b (\sum_j c_{r,j} b^{-j} + \dots)$  to every fixed order, with  $\beta_r = (r/(r+1))^r$ , rational  $c_{r,j}$ , explicit  $c_{r,1}$ , three corrections for  $r = 2, 3$ , and  $A_{1,b} = (b+1)(\zeta(b+2)-1)$  exactly. Corollary 14.1: the same for  $6/\pi^2 - \mu_b$ .
- **Part III** (bundle Report 99; cap written  $r$ ). Theorem 17.1: for every fixed  $r \geq 2$ ,  $(a_n^{(r)}/n!)^{1/n} \rightarrow \mu_r$  and  $a_{n+1}^{(r)}/((n+1)a_n^{(r)}) \rightarrow \mu_r$ ; the ratio limit is new for  $r \geq 3$ . Theorem 17.2: the proportions of values used  $k$  times, of distinct values and of ascents converge, with exponentially small deviations; Theorems 22.1 and 22.2: the weighted growth formula  $\lim N^{-1} \log(H_N(\mathbf{u})/N!) = -\log \chi(\mathbf{u})$  and a full large-deviation principle for the endpoint. Proposition 18.2 and Corollary 18.3: a histogram-preserving sorted recurrence and  $O_r(N^{r+2})$  enumeration. Corollary 23.2: a length inverse (weaker than Section 6). Section 24: the large-cap analysis again, in  $R = r+1$ , with a continuous cap, exact-sector envelopes  $R_K$ , their displacements and a two-candidate cap selection (Corollary 24.6).

## Sources and provenance

The report was built from two manuscripts, both dated 1 October 2026, both delivered in the arrival commit 096ee7b87 of 2 October 2026 as batch 77, manuscripts 47 and 52, and placed in this directory by commit aa7345800 (batch 77P2). Neither pins a ProveIt commit. Source 47 identifies the cap-two report (batch-77 manuscript 37) by the SHA-256 of its delivered  $\text{T}_{\text{E}}\text{X}$  and PDF; source 52 identifies source 47 the same way. All four hashes match the delivered files of 096ee7b87 (checked at the merge). Author lines: “A proof and reproducible research report” (47) and “A reproducible large cap addendum” (52); no tool or person is named.

| Part | Source | Archive and main file                                                                       | Pages | Labels | Prefix   |
|------|--------|---------------------------------------------------------------------------------------------|-------|--------|----------|
| I    | 47     | oeis-bounded-multiplicity-family-report.zip,<br>bounded-multiplicity-report.tex (612 lines) | 13    | 48     | amc:     |
| II   | 52     | oeis-fixed-sector-addendum.zip,<br>fixed-sector-addendum.tex (312 lines)                    | 7     | 29     | amc:sec: |

**Part III.** Bundle Report 99, archive `Ascent_Sequences_with_Bounded_Multiplicities_Source.zip`, arrived in commit 60f54ea06 and was placed by 6ab1f1979 (batch 102); it is dated 2 October 2026, its author line is “Research report prepared with OpenAI”, and it pins no commit and names neither this repository nor Parts I and II. Its `MANIFEST.sha256` (31 entries) and the payload hashes of its `PROVENANCE.json` (30) matched the delivery at placement. Its companion for the cap two, bundle Report 97, is Part V of the sibling report a202058-ascent-000-growth.

| Part | Source    | Archive and main file                                                                                        | Pages | Labels | Prefix  |
|------|-----------|--------------------------------------------------------------------------------------------------------------|-------|--------|---------|
| III  | Report 99 | Ascent_Sequences_with_<br>Bounded_Multiplicities_Source.zip, article.tex<br>(63 lines) with 11 section files | 28    | 119    | amc:bm: |

**Nesting.** Source 52 is an addendum of source 47: it refines the large-cap analysis of  $T_b$  and takes the identification  $\mu_b = 1/T_b$  and both inverses from source 47 (bibliography key **main**), without repeating their proofs. It does not bundle source 47; its archive carries four replay outputs byte-identical to its own saved results, which were not staged.

**Where the merge had to choose.** (i) The Parts follow the dependency order 47, 52. (ii) Both sources are printed in full; nothing of either is omitted except the title blocks and the bibliographies, which are merged below. (iii) Source 52 re-derives source 47’s first variation  $(b+1)(\zeta(b+2)-1)$  by a different computation; it is kept as a marked second proof. (iv) Source 47’s remainder bound  $O(b^{5/2}(4/9)^b)$  is not sharp in its power of  $b$ ; source 52’s sharp asymptotic is printed beside it in dated notes, and both are kept, because source 47’s bound has explicit constants and its integer brackets use them. (v) No symbol was renamed; the notation table below separates the letters the two sources use differently. At the batch-102 write: (vi) Report 99 is printed in full as Part III, after Part II, although it re-proves much of Parts I and II; its re-proofs are marked as second routes in dated notes, with the equations they reproduce. (vii) Report 99 and its special case Report 97 (cap two) stay in their two hosts, as the batch-77P2 split of the two reports requires; each Part names the other. (viii) Report 99’s large-cap coefficients are printed in its variable  $R = b+1$ , and a note converts them to Part II’s variable  $b$ . (ix) No symbol of Report 99 was renamed, although its cap is  $r$  and Part II’s sector index is  $r$ ; the notation table has a block for Part III.

## How the Parts depend on one another

Part II is an analysis of the integral  $T_b$  alone; its only combinatorial input is Part I’s identification  $\mu_b = 1/T_b$  (Theorem 1.1), used in Corollary 14.1. Part I’s Theorem 1.2 is the case  $K = 1$  of Theorem 12.1 with the first sector (50), except that Part I also proves explicit constants, valid for every  $b \geq 3$ , and Part II does not. Neither Part strengthens the other’s results in the length  $n$ , and Part II proves no new inverse. Part III is self-contained and uses neither earlier Part. It re-proves the root limit of Theorem 1.1 (weaker error), the compaction of Lemma 2.1, the monotonicity in Theorem 1.2, a weaker length inverse, and, in the variable  $R = b+1$ , Section 7 and Theorems 12.1 and 13.1; it strengthens Part I in the length  $n$  by the ratio limit and the endpoint laws.

## Relation to other reports

[**write, 2 October 2026**] *Sibling report on cap two.* The cap-two case  $b = 2$  (OEIS A202058, ascent sequences avoiding 000) is the subject of the report `SetTheory/Cardinals/docs/reports/generating-functions-and-asymptotics/oeis-sequence-asymptotics/a202058-ascent-000-growth`, opened by batch 77 from four manuscripts. Its Part I is the “separately audited cap-two report” that source 47 cites: its main theorem (label `a58:fd:thm:main` there) gives  $T_2 = 3\pi^2/8$ , the root limit and the error  $O(n^{9/13}(\log n)^3)$ , and its later Parts sharpen the cap-two asymptotics and prove the cap-two ratio limit. This report uses from it only  $T_2$  and the case  $b = 2$  of (2); source 47 proves nothing for  $b = 2$  and its barrier proof excludes  $b = 2$  (after (19)). By the coordinator’s decision at the merge the two are separate reports, not Parts of one: the sibling is about the fine asymptotics of a single cap, this report about the growth constants of all caps  $b \geq 3$  and their dependence on  $b$ .

No other repository report concerns ascent sequences with a multiplicity cap. The 201-avoiding ascent sequences (OEIS A202062) and the 120-avoiding ones (A202061) have their own batch-77 reports in the same directory, `a202062-ascent-201-enumeration` and `a202061-ascent-120-deficit`; both cite the paper of Conway, Conway, Elvey Price and Guttmann [1], but they share no theorem with this report.

[**write, 5 October 2026**] The sibling report now has a Part V, bundle Report 97, which is the case  $r = 2$  of Part III by the same method: its Theorems 42.1 (root limit) and 46.1 (ratio limit), Lemma 44.2 ( $J(1) = 3\pi^2/8$ ), Corollary 46.4 (endpoint laws, in closed form there), Lemma 47.1 and Corollary 47.2 are the cases  $r = 2$  of Theorems 17.1 and 17.2, Proposition 17.3, Lemma 23.1 and Corollary 23.2, and its Proposition 46.3 (an upper bound) is superseded by Theorem 22.1. So the two reports now share these theorems. The split stays: the cap-two material of Report 97 (a two-catalytic functional equation, closed forms, exact coefficients through 395) belongs with that report, and the material on all caps with this one. Two further batch-102 reports are neighbours; they are named at Question 4 in Section 10.

## Notation

Each Part keeps its source's letters; no symbol was renamed. The two sources agree on  $E_b$ ,  $T_b$ ,  $\mu_b$ ,  $f$ ,  $\tau = \pi^2/6$  and  $R_b(v) = e^v - E_b(v)$ . Symbols not in the table are local to the Part that uses them. The table prints the tempting false reading beside the true one.

| Symbol            | Meaning and where                                                                                                                                                                                                | Do not read it as                                                                                      |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| $L_b$ , $A_{1,b}$ | $(b+1)(\zeta(b+2)-1)$ : Part I's first variation $L_b$ (30) is Part II's first sector $A_{1,b}$ (50).                                                                                                            | $L$ is also the log-target $\log X$ in Section 6.                                                      |
| $H_b$ , $H_r$     | Part I: $H_b = T_b - \tau - L_b$ , the nonlinear remainder, which is Part II's $\mathcal{R}_{1,b}$ . Part II: $H_r = \sum_{j \leq r} 1/j$ , the harmonic number.                                                 | $H_3$ of Part II is $11/6$ , not a remainder. Part I also writes $H = h(i)$ and $H_q(v)$ in Section 5. |
| $\beta$           | Part I, Section 2: $\beta_0, \dots, \beta_{m-1}$ are the remaining label budgets of a state. Part II: $\beta_r = (r/(r+1))^r$ , the exponential base of sector $r$ ( $\beta_2 = 4/9$ , $\beta_3 = 27/64$ ).      | Not related.                                                                                           |
| $m$               | Part I: the number of active labels of a state (Sections 2–5); in Section 7, $m = b+1$ . Part II: $m = b+1$ throughout.                                                                                          |                                                                                                        |
| $q$ , $p$         | Part I: $q = \log E_b(v)$ and $p_j = \log E_{b-j}(v)$ , the characteristic curve; $p = p_1$ . Part II: $q = b^{-1}$ and $p = (3-r)/2$ in the coefficient algorithm.                                              |                                                                                                        |
| $c$ , $C$         | Part I: generic constants, $c_j$ , $c_b$ in (9), $C(q)$ in (17). Part II: $C_r$ and the rational coefficients $c_{r,j}$ of (46).                                                                                 |                                                                                                        |
| $D$ , $d$         | Part I: the weight sum $D$ of a state; $D = 9/\pi^4$ in (33); $d_r$ , $d_j$ , $d$ , $d_b = \mu_\infty - \mu_b$ . Part II: $D = 1 - t + tx$ , $D_r$ , $d_{r,j}$ , the integers $d_{k,n}$ , and a cutoff $d < 1$ . |                                                                                                        |
| $J$ , $K$         | Part I: a random index $J$ and its centre $K = NM(q)$ ; $K$ also a constant in (11). Part II: $J_{r,b}$ , the derivative-free sector integral (52), and $K$ , the number of sectors.                             |                                                                                                        |
| $a$               | Part I: the counts $a_n^{(b)}$ . Part II: the counts too, and $a_r(x)$ , the sector integrand (38), and a cutoff $a < s_r$ .                                                                                     |                                                                                                        |

| Symbol                     | Meaning and where                                                                                                                                                              | Do not read it as |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| $\mathcal{R}, \mathcal{E}$ | Part II: $\mathcal{R}_{K,b}$ , the remainder after $K$ sectors; $\tilde{\mathcal{R}}_{K,b}$ for $\mu_b$ . Part I: $\mathcal{E}(q)$ , the barrier error (18).                   |                   |
| $B$                        | Part I: a constant in Section 5. Part II: Bernoulli numbers $B_{2k}$ in the coefficient algorithm.                                                                             |                   |
| $\phi, h, \lambda$         | Part I: $\phi(y)$ and $\phi(x) = x^{3/2}(8/9)^x$ ; $h(y)$ and $h(x) = D(x+1)2^{-x}$ ; $\lambda = \dot{q}D$ . Part II: the phase $\phi_r(s)$ and $\lambda_r = -\phi_r''(s_r)$ . |                   |

**Part III against Parts I and II** (added 5 October 2026). Part III keeps Report 99's letters. The most dangerous swaps are the cap ( $r$  there,  $b$  here), the sector index ( $m$  there,  $r$  in Part II),  $T$  (a sector there, the growth integral in Part I),  $\tau$  and  $b_j$ .

| Symbol in Part III                | Meaning there, and its name in Parts I–II                                                                                                                                                                                            | Do not read it as                                                                               |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| $r$                               | The cap; Parts I–II: $b$ . $c_{N,r}$ is $a_N^{(b)}$ .                                                                                                                                                                                | Part II's sector index $r$ ; Part I's rank function $r(y)$ .                                    |
| $R$                               | $R = r + 1$ , the large-cap variable of Section 24, extended to real $R > 2$ ; Part II: $m = b + 1$ .                                                                                                                                | Part I's tail $R_b(v) = e^v - E_b(v)$ , which is $\delta_R(x)$ here.                            |
| $m$                               | Section 24: the sector index, Part II's $r$ ; Lemma 19.3: a permutation length.                                                                                                                                                      | Part I's number of active labels; Part II's $m = b + 1$ .                                       |
| $N, N_k, X_j$                     | $N$ is the word length ( $n$ in Parts I–II). $N_k$ : values used exactly $k$ times, as Part I's $N_k$ for $1 \leq k < r$ . $X_j$ : labels of residual capacity $j$ ; $X_j = N_{r-j}$ for $j < r$ , and $X_r$ is Part I's $N_0 = u$ . | $X_j$ is not Part I's $N_j$ .                                                                   |
| $J_r, \mathcal{J}(R), \mu_r$      | $J_r = \int_0^\infty g(E_r(w)) dw$ is Part I's $T_b$ , and $\mu_r = \mu_b$ ; $\mathcal{J}$ interpolates $J_r$ at $R = r + 1$ .                                                                                                       |                                                                                                 |
| $T_m(R)$                          | Sector $m$ ; at $R = b + 1$ it is Part II's $A_{m,b}$ .                                                                                                                                                                              | Part I's $T_b$ , the whole integral.                                                            |
| $g$                               | $g(v) = \log v/(v - 1)$ , Parts I–II's $f$ .                                                                                                                                                                                         | Part I's frozen integrand $g(y)$ and barriers $g_\pm$ .                                         |
| $A$                               | $A = \zeta(2) = \pi^2/6$ , Parts I–II's $\tau$ ; also the ascent count (Theorem 17.2) and the Eulerian numbers $A(m, k)$ .                                                                                                           | Part II's sectors $A_{r,b}$ ; Part I's generating function $A_b(t)$ .                           |
| $\tau$                            | A time variable $\tau(w) \in [0, 1]$ (Sections 20–22); the threshold $\tau(\epsilon)$ of (141).                                                                                                                                      | Parts I–II's $\tau = \pi^2/6$ .                                                                 |
| $b, b_j$                          | $b$ : a block length (Sections 19–21). $b_j$ : the endpoint limits (64).                                                                                                                                                             | The cap $b$ of Parts I–II.                                                                      |
| $a_m, C_m, c_{m,j}$               | Part II's $\beta_r, D_r$ and $d_{r,j}$ (normalized in $R$ ).                                                                                                                                                                         | Part II's $C_r = a_m C_m$ and its $c_{r,j}$ (in $b$ ); the counts $a_n^{(b)}$ .                 |
| $L, L(R), L_0$                    | $L$ : the threshold of the compact state (Section 18). $L(R) = R(\zeta(R+1) - 1)$ , Part I's $L_b$ and Part II's $A_{1,b}$ at $R = b + 1$ ; $L_0(R) = R2^{-R-1}$ .                                                                   | Part I's $L = \log X$ ; $\mathcal{L}$ is an entropy.                                            |
| $D, C_0$                          | $D = \sum_j u_{j-1} \partial_{u_j}$ (Section 20); $D$ also the number of distinct values. $C_0 = 18/\pi^4$ .                                                                                                                         | Part I's weight sum $D$ and its $D = 9/\pi^4 = C_0/2$ ; Part II's $D = 1 - t + tx$ .            |
| $\lambda$                         | $\lambda = \log 2$ (Section 24); an eigenvalue in Lemma 19.1.                                                                                                                                                                        | Part I's $\lambda = \dot{q}D$ ; Part II's $\lambda_r$ .                                         |
| $H_N, H_m$                        | $H_N(\mathbf{u})$ : the histogram polynomial; $H_m$ : the harmonic number, as Part II's $H_r$ ; $H_2(u)$ : a bracket width after Corollary 24.6.                                                                                     | Part I's remainder $H_b$ .                                                                      |
| $\beta_j, \alpha_j, \theta, \rho$ | Prefix and initial slopes, log-marks $\theta_j = \log u_j$ , and the block symbol (71).                                                                                                                                              | Part I's budgets $\beta_i, \theta = (b-1)/b$ and weights $\rho_j$ ; Part II's bases $\beta_r$ . |

| Symbol in Part III               | Meaning there, and its name in Parts I–II                                                                                                                                    | Do not read it as                                    |
|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------|
| $F, f_\ell$                      | $F(q)$ : the pressure; $F_R(x)$ : the tail factor;<br>$F(v) = (v - 1)/\log v$ (Lemma 19.1);<br>$F(x) = x \log(\mu_r x/e)$ (Section 23); $f_\ell$ : tail-factor coefficients. | Part I’s series $F(t, x)$ and Parts I–II’s $f = g$ . |
| $q, p_j, M$                      | Controls of the block construction and real marks;<br>$M = \sum_j X_j$ , the number of next labels, and $M(t)$ , a moment generating function.                               | Part I’s $q, p_j, M(q)$ ; Part II’s $q = 1/b$ .      |
| $x_\epsilon, r_{\min}(\epsilon)$ | $x_\epsilon = b_0 + 1$ for Part I’s $b_0$ (34); $r_{\min}(\epsilon) = b_\epsilon$ .                                                                                          |                                                      |

## Status, scope and non-claims

Neither manuscript states how it was produced; both reached the repository through its incoming channel for AI-assisted research deliveries and are treated as AI-assisted. Both are unrefereed: source 47 calls its audits “mathematical checks of this argument, not journal peer review”, and source 52’s verification records are research reviews, not formal verification. Their quadratures are high-precision numerical checks, not interval-certified enclosures. Nothing in this report is formalized in Lean or Rocq, and placing it in this repository confers no formal status.

The union of the sources’ non-claims, all kept in the Parts, is as follows.

- No fixed-cap prefactor, ratio limit or all-orders coefficient expansion in  $n$  for  $b \geq 3$ ; the error  $O_b(n/\log n)$  is deliberately coarse and is not the true subexponential scale;  $a_n^{(b)} \sim C_b n! \mu_b^n$  is not implied.
- The limits in length and cap are kept separate: the large-cap expansion concerns the growth constant after the length limit; no uniform asymptotics for  $b = b(n)$  and no exchange of limits.
- The length inverse is a controlled leading inverse, not an exact rounding rule. The cap inverse is an inverse for the cap of the limiting growth constant, not a cap-selection guarantee for finite-length counts; near integer transitions no unconditional rounding is justified.
- The sector results hold for a fixed number of sectors and a fixed algebraic order. No optimal truncation, summability, uniformity in  $r = r(b)$ , justified sum of all sector expansions, or theorem at the accumulation scale  $e^{-b}$ ; finite truncations of an earlier sector do not resolve later sectors; the reciprocal remainder is positive only eventually for each fixed  $K$ ; no new inverse. The word “transseries” in Part II means only the exact positive-sector decomposition.
- Priority is not established; the literature checks are scoped. The cap-two compacted recurrence and the constant  $8/(3\pi^2)$  (conjectured) are due to Conway, Conway, Elvey Price and Guttmann; source 47’s involution generalizes their compaction mechanism. The standalone OEIS page of A317784 could not be retrieved by source 47; its identification rests on A294220’s cross-reference.

[write, 5 October 2026] *Part III.* Report 99’s author line is “Research report prepared with OpenAI”; it is AI-assisted, unrefereed and not formalized, and its exact checks are finite. The first item above is narrowed: Parts I and II prove no ratio limit, but Part III proves it for every fixed cap (without a rate); a prefactor and an all-orders expansion in  $n$  remain unclaimed. Part III’s own non-claims, all printed there, are: no coefficient prefactor  $B_r(N)$ , amplitude, complete word-length equivalent or word-length transseries; no central limit theorem, covariance limit or local limit theorem; no uniform theorem for  $r = r(N)$  and no exchange of the limits



$N \rightarrow \infty$  and  $r \rightarrow \infty$ ; no uniform expansion with a growing number of sectors and nothing at the accumulation scale; no combinatorial meaning of a non-integer cap; no unconditional integer rounding of the Lambert approximations, for the length or for the cap; quadratures not interval certified; exact checks finite, corroborating and not proving; a bounded literature search and no priority claim. Its unproved claims are collected in Section 26.6.

## Instances of results elsewhere in the repository

Several steps are instances of results proved in the transseries volume `Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/`. No novelty is claimed for these mechanics.

- The length inverse  $n_0 = L/W_0(L/(eT_b))$  solves  $x \log(x/(eT_b)) = L$ ; with  $Y = \log(x/(eT_b))$  this is  $Y + \log Y = \log(L/(eT_b))$ , the linear–logarithmic equation  $aX + b \log X = L'$  with  $a = b = 1$ , solved on the principal branch by the volume’s Lambert core, theorem `p0:thm:lambert-core`.
- The cap model  $D(b+1)2^{-b} = \varepsilon$  is, in  $X = b+1$ , the equation  $(\log 2)X - \log X = \log(2D/\varepsilon)$ , the case  $a > 0 > b$  of the same theorem, whose branch rule selects  $W_{-1}$  for the large root, as (34) does.
- The integer brackets (35) and (36) are instances of the separation step of the volume’s theorem `p0:thm:staircase`. That step’s separation inequality is formalized in Lean as `staircase_separation`, in the file `StaircaseInversion.lean` of the directory `Analysis/FabiusFunction/Lean/FabiusFunction/`. Nothing specific to ascent sequences or to  $T_b$  is formalized.
- [5 October 2026] Part III’s length inverse (98) is the same linear–logarithmic equation, and its cap root  $x_\epsilon$  (146) is  $b_0 + 1$  for the  $W_{-1}$  root (34); its two-candidate bracket (Corollary 24.6) is again a separation step.

## Conventions of the merge

- Every label carries the report prefix: `amc:` in Part I and `amc:sec:` (“sectors”) in Part II. Source 47’s own section labels began with `sec:`, so its section labels read `amc:sec:compaction`, `amc:sec:checks` and so on; they belong to Part I. Part II’s labels all have the form `amc:sec:eq:...`, `amc:sec:thm:...`, `amc:sec:cor:...` or `amc:sec:sec:...`. Guide labels are `amc:guide:`, Part labels `amc:part:`.
- Sections are numbered through the report: source 47’s Section  $k$  is Section  $k$  here (Sections 1–10), source 52’s is Section  $k + 10$  (Sections 11–16). Theorems are numbered within sections, as in both sources, so source 52’s Theorem 2.1 is Theorem 12.1. Equations are numbered through the report.
- Source 52’s citation `\cite{main}` and its phrase “the separately supplied report” are now references to Part I; elsewhere “the main report” means Part I, with references added where it names a result. Its bibliography entry `main` is dropped. Source 52’s abstract is printed as delivered except for a bracketed pointer to Part I.
- Sentences about “the accompanying source archive”, “the package”, “the ZIP”, `verify.sh`, `build.sh`, `checks/`, `sources/` or `provenance.json` describe the delivered archives, which survive in commit 096ee7b87. Which of their files this report ships, and under which names,

is listed in the README; bracketed notes in the bibliography give the shipped names of the files it cites.

- **[5 October 2026] Part III.** Labels carry the prefix `amc:bm:` (“bounded”) before the delivered name, for example `amc:bm:thm:growth` and `amc:bm:cap:sharp`. Report 99’s Section  $k$  is Section  $k + 16$  and its equation  $(k)$  is  $(k + 61)$ ; statements are numbered within sections, so its Theorem 1.1 is Theorem 17.1. Its title block became the Part heading and its abstract is printed as delivered; its table of contents, its `\clearpage` commands and its preamble were dropped, and its macros were added to this preamble unchanged (its `\asc` is this report’s). Its citation key `OEISarray` is now `array`; its entries CCEG, DS and `OEISarray` were merged into the existing ones, keeping their details, and BMCDK and `OEIS000` were added. Sentences about “the accompanying package”, `run_checks.sh` or `checks/` describe the delivered archive, which survives in commit `60f54ea06`; a dated note in Section 25 gives the shipped names.

## Part I

# Factorial growth of ascent sequences with bounded multiplicity

*Source.* Batch-77 manuscript 47, archive `oeis-bounded-multiplicity-family-report.zip`, main file `bounded-multiplicity-report.tex` (612 lines, 13-page PDF), dated 1 October 2026. Author line: “A proof and reproducible research report”. No commit pin; it identifies the cap-two report by file hashes. Printed in full, with labels prefixed `\amc::`; one section label and the `\write` notes were added at the merge.

**Abstract (as delivered).** Let  $a_n^{(b)}$  count ascent sequences of length  $n$  in which each value occurs at most  $b$  times. For every fixed  $b \geq 3$  we prove the exact factorial root constant

$$\lim_{n \rightarrow \infty} \left( \frac{a_n^{(b)}}{n!} \right)^{1/n} = \left[ \int_0^\infty \frac{\log E_b(v)}{E_b(v) - 1} dv \right]^{-1}, \quad E_b(v) = \sum_{j=0}^b \frac{v^j}{j!}.$$

The separate cap-two proof gives the same formula. We establish exact compaction by a suffix involution, uniform positive-operator barriers, and a coefficient argument for every  $n$ . A non-optimal logarithmic error supports a controlled inverse for the required length. We also prove a large-cap expansion of the constant, including an exact zeta first variation and an exponentially smaller remainder, and obtain a rounding-safe cap inverse. This report does not determine fixed-cap prefactors, ratio limits, or all-orders coefficient expansions. The limits in length and cap are kept separate.

## 1 Results and interpretation

An ascent sequence is a word  $w_1 \cdots w_n$  of nonnegative integers with  $w_1 = 0$  and

$$0 \leq w_i \leq 1 + \text{asc}(w_1 \cdots w_{i-1}) \quad (i \geq 2), \quad \text{asc}(w) = \#\{i : w_i < w_{i+1}\}.$$

Write  $a_0^{(b)} = 1$  and let  $a_n^{(b)}$  count the words with every label used at most  $b$  times. Equivalently, these words avoid the constant pattern of length  $b+1$ . This is the array in OEIS A294220; cap two is A202058 and cap three is A317784 [4, 6].

**Theorem 1.1.** *For every fixed integer  $b \geq 3$ , set*

$$T_b = \int_0^\infty \frac{\log E_b(v)}{E_b(v) - 1} dv, \quad \mu_b = T_b^{-1}. \quad (1)$$

*The integrand is interpreted as 1 at zero. The exponential generating function  $A_b(t) = \sum_{n \geq 0} a_n^{(b)} t^n / n!$  has radius  $T_b$ , and*

$$\log a_n^{(b)} = \log(n!) - n \log T_b + O_b\left(\frac{n}{\log n}\right). \quad (2)$$

*Consequently  $(a_n^{(b)} / n!)^{1/n} \rightarrow \mu_b$  and*

$$a_n^{(b)} = n! \mu_b^n \exp\left(O_b\left(\frac{n}{\log n}\right)\right).$$

The subscript  $b$  permits constants and starting indices to depend on the fixed cap. The logarithmic error is deliberately coarse; it is not an estimate of the true subexponential scale. In particular, the theorem does not imply  $a_n^{(b)} \sim C_b n! \mu_b^n$  or  $a_{n+1}^{(b)} / ((n+1)a_n^{(b)}) \rightarrow \mu_b$ .

[write, 5 October 2026] The ratio limit  $a_{n+1}^{(b)} / ((n+1)a_n^{(b)}) \rightarrow \mu_b$  is proved for every fixed  $b \geq 2$  in Part III, Theorem 17.1, without a rate;  $a_n^{(b)} \sim C_b n! \mu_b^n$  remains open.

The separately audited cap-two report [7] proves  $T_2 = 3\pi^2/8$  and a stronger logarithmic error  $O(n^{9/13}(\log n)^3)$ , so (2) also holds for  $b = 2$ . Its analytic proof is not repeated here. For  $b = 1$ , the unique word at each length is strictly increasing,  $T_1 = \infty$ , and the normalized root is zero.

[write, 2 October 2026] *The cap-two report in this repository.* The report cited here is batch-77 manuscript 37, printed as Part I of the sibling report `a202058-ascent-000-growth` in this collection directory; its main theorem (label `a58:fd:thm:main` there) proves that the exponential generating function of A202058 has radius  $T = 3\pi^2/8$ , the root limit  $(a_n/n!)^{1/n} \rightarrow 8/(3\pi^2)$ , and the error  $O(n^{9/13}(\log n)^3)$ . The SHA-256 values that source 47 records for that report's delivered `a202058-report.tex` and `.pdf` (`data/47-caps-provenance.json`) match the files of the arrival commit `096ee7b87`; this was checked at the merge. Later Parts of the sibling report sharpen the cap-two error and prove the cap-two ratio limit (see its README). The two reports are kept separate, by the coordinator's decision at the merge: the sibling treats the fine asymptotics of the single cap  $b = 2$ , this one the growth constants of every cap  $b \geq 3$  and their large-cap behaviour, and this report takes from the sibling only  $T_2$  and the  $b = 2$  case of (2).

**Theorem 1.2.** *As  $b \rightarrow \infty$  through integers,*

$$T_b = \frac{\pi^2}{6} + (b+1)(\zeta(b+2) - 1) + O\left(b^{5/2}(4/9)^b\right), \quad (3)$$

$$T_b - \frac{\pi^2}{6} \sim \frac{b+1}{2^{b+2}}, \quad \frac{6}{\pi^2} - \mu_b \sim \frac{9(b+1)}{\pi^4 2^b}. \quad (4)$$

Moreover  $T_b$  strictly decreases to  $\pi^2/6$ , and  $\mu_b$  strictly increases to  $6/\pi^2$ .

This is an expansion of the fixed-cap growth constant after the length limit. It does not assert uniform coefficient asymptotics when  $b = b(n)$  grows. Sections 6 and 8 give different inverses for the length and the cap.

[write, 2 October 2026] Part II (source 52) sharpens the remainder in (3). The exact identity  $T_b - \pi^2/6 = \sum_{r \geq 1} A_{r,b}$  (Theorem 12.1) has first sector  $A_{1,b} = (b+1)(\zeta(b+2) - 1)$ , the first variation of (3), and its remainder after one sector is asymptotic to the second sector:

$$T_b - \frac{\pi^2}{6} - (b+1)(\zeta(b+2) - 1) \sim A_{2,b} \sim \frac{8}{9\sqrt{\pi}} \sqrt{b} \left(\frac{4}{9}\right)^b$$

(equations (43) and (48)). So the exponential rate  $(4/9)^b$  of (3) is sharp, while its power  $b^{5/2}$  is not: the true power is  $b^{1/2}$ . The two statements are consistent. The bound of this Part has explicit constants and holds for every  $b \geq 3$  (Section 8); Part II's is sharp but asymptotic, with constants that are not explicit.

## 2 Exact compaction and the counting operator

For any ascent-sequence prefix,  $\max(w) \leq \text{asc}(w)$ . Indeed a new maximum creates an ascent, whereas a nonascent cannot raise the maximum. Thus the top admissible label  $1 + \text{asc}(w)$  is unused and strictly larger than the last value.

Delete labels whose budgets have been exhausted. The remaining active labels, in numerical order, have positive budgets  $\beta_0, \dots, \beta_{m-1} \in \{1, \dots, b\}$ . Let  $k$  be the number of active labels at or below the actual last value. This can be a virtual boundary when that value has just been deleted. Choosing active rank  $i$  is an ascent exactly when  $i \geq k$ . Decrease  $\beta_i$  by one; remove it and set  $k' = i$  if it reaches zero, and otherwise set  $k' = i + 1$ . An ascent creates precisely one additional top label of budget  $b$ .

**Lemma 2.1** (Adjacent-budget interchange). *Suppose  $x < y$  are adjacent active labels with both at or below the last value. Exchanging their positive remaining budgets, while retaining the last boundary, preserves the number of legal suffixes of every length.*

*Proof.* In a suffix, reverse each maximal contiguous block consisting only of  $x, y$ , and then interchange  $x$  and  $y$  throughout that block. This is an involution that exchanges their total usages. Every intermediate usage is at most the total, so the exchanged quotas hold at every stage.

Inside a binary block, a comparison  $p < q$  becomes the same comparison between the complemented pair in reverse order. Its total number of ascents is preserved. No possible outside label lies strictly between  $x$  and  $y$ : the two labels were adjacent among the initially active labels, exhausted labels never return, and future labels are all above the initial alphabet. Thus boundary comparisons with outside letters are unchanged. Entry into an initial binary block is nonascending both before and after the map, since the initial last value is at least  $y$ .

Ascent totals therefore agree at each block end and before every unchanged outside letter. The timing inside a block can change, but both block labels were already admissible initially, and the admissible upper bound never decreases. This proves legality as well as invertibility.  $\square$

The lemma allows permutations below the boundary, not arbitrary sorting above it. Start with a canonical state in which budgets increase with rank. After choosing budget  $r > 1$ , move its decremented entry left to the beginning of the old budget- $r$  group. Every swap lies among the first  $i + 1$  entries, hence below the new boundary  $i + 1$ , which stays fixed. Exhaustion leaves an already sorted tuple and boundary  $i$ . A new top label is appended. This gives an exact reduction at every step. The last boundary does not follow the moved budget.

Write  $N_0 = u$  for the number of unused labels and  $N_j = s_j$  for those used  $j$  times,  $1 \leq j < b$ . Groups occur in increasing rank order as

$$N_{b-1}, N_{b-2}, \dots, N_1, N_0, \quad m = \sum_{j=0}^{b-1} N_j, \quad u \geq 1, \quad 0 \leq k < m.$$

If rank  $i$  belongs to group  $j$ , put  $\alpha = \mathbf{1}_{i \geq k}$ . Its child is

$$N' = N - e_j + \mathbf{1}_{j+1 < b} e_{j+1} + \alpha e_0, \quad k' = \begin{cases} i, & j = b - 1, \\ i + 1, & j < b - 1. \end{cases} \quad (5)$$

These transitions preserve  $u \geq 1$  and  $k < m$ : a choice of the final unused top label is always ascending and creates another. They satisfy  $m' \leq m + 1$ . Define the positive operator and its series by

$$(\mathcal{T}f)(x) = \sum_{i=0}^{m-1} f(x_i), \quad F(t, x) = \sum_{r \geq 0} \mathcal{T}^r \mathbf{1}(x) \frac{t^r}{r!}. \quad (6)$$

Let  $x_1$  have  $s_1 = u = k = 1$  and all other  $s_j = 0$ . Then

$$\mathcal{T}^r \mathbf{1}(x_1) = a_{r+1}^{(b)}, \quad F(t, x_1) = A'_b(t).$$

All series identities below allow  $+\infty$  and use nonnegative sums when necessary.

### 3 Characteristic curve and a uniform barrier

For this section fix  $b \geq 3$  and write  $\theta = (b-1)/b > 1/2$ . Parameterize time by  $v \geq 0$ :

$$q = \log E_b(v), \quad p_j = \log E_{b-j}(v) \quad (1 \leq j < b), \quad p_0 = q, \quad p_b = 0, \quad t(v) = \int_0^v \frac{\log E_b(z)}{E_b(z) - 1} dz. \quad (7)$$

Its endpoint is  $T_b < \infty$ : the integrand is  $O_b(\log v/v^b)$  at infinity. A dot denotes a time derivative. Direct differentiation gives

$$\dot{q} = e^{p_1} \frac{1 - e^{-q}}{q}, \quad \dot{p}_j = \frac{e^q - 1}{q} e^{p_{j+1} - p_j}. \quad (8)$$

At zero the removable values are all 1, so every  $p_j, q$  is strictly increasing. Polynomial asymptotics give

$$p_j = \frac{b-j}{b} q + c_j + O_b(e^{-q/b}), \quad (9)$$

$$e^{p_1} \sim c_b e^{\theta q}, \quad \dot{q} \sim c_b e^{\theta q}/q, \quad c_b = \frac{(b!)^\theta}{(b-1)!}. \quad (10)$$

Fixed-order differentiated estimates follow from the underlying rational functions of  $v$ , rather than differentiation of an arbitrary error term.

Define

$$\rho_0 = 1, \quad \rho_b = 0, \quad \rho_j = \frac{\dot{p}_j}{\dot{q}} = \frac{E_b(v)E_{b-j-1}(v)}{E_{b-1}(v)E_{b-j}(v)} \quad (1 \leq j < b).$$

They satisfy  $1 = \rho_0 \geq \rho_1 \geq \dots \geq \rho_{b-1} > 0$ . To see this, put  $d_r = v^r/r!$  and note

$$E_r^2 - E_{r-1}E_{r+1} = d_r \left( E_r - \frac{v}{r+1} E_{r-1} \right) \geq 0;$$

the bracket has positive coefficients. Also  $\rho_j \rightarrow (b-j)/b$ . Positivity, continuity, and rational asymptotics yield fixed-cap constants  $c > 0, K < \infty$  such that

$$c \leq \rho_j \leq 1, \quad |(\rho_j)_q| \leq K \quad (0 \leq j < b, q \geq 0). \quad (11)$$

For a state  $x$ , let

$$D = \sum_{j=0}^{b-1} \rho_j N_j, \quad h(y) = \int_0^y \rho_{\text{group}(z)} dz, \quad r(y) = h(y)/D, \quad r = r(k),$$

where group weights are constant on unit rank intervals. Put

$$\psi(t, x) = \exp \left( \sum_{j=1}^{b-1} p_j s_j + qu - qr \right). \quad (12)$$

Here  $D \geq u \geq 1$ ,  $cm \leq D \leq m$ ,  $0 \leq r \leq 1$ , and  $\psi(0, x) = 1$ .

### 3.1 Exact child rank and its bounds

If rank  $i$  belongs to group  $j$ , write  $H = h(i)$  and  $d_j = \rho_j - \rho_{j+1} \in [0, 1]$ . Every compaction swap is below the new boundary, so its weight sum is unchanged. Consequently

$$D_i = D - d_j + \alpha, \quad h_{x_i}(k_i) = H + \rho_{j+1}, \quad r_i = \frac{H + \rho_{j+1}}{D - d_j + \alpha}. \quad (13)$$

This includes saturation through  $\rho_b = 0$ . Since  $D_i \geq 1$  and  $D_i \geq D - 1$ , we have  $D_i \geq D/2$  and

$$|r_i - r(i)| \leq \frac{4}{D}. \quad (14)$$

For an ascent,  $i \geq k$  and  $D_i \leq D + 1$ , whence the crucial one-sided bound

$$r(k) - r_i \leq \frac{H}{D} - \frac{H}{D+1} \leq \frac{1}{D+1} \leq \frac{1}{2}. \quad (15)$$

### 3.2 Frozen integral and residual

For real  $y$  in group  $j$  set

$$g(y) = \exp(p_{j+1} - p_j + q\mathbf{1}_{y \geq k} - q(r(y) - r(k))).$$

The exact transition ratio  $\psi(t, x_i)/\psi(t, x)$  replaces  $r(i)$  by  $r_i$  in  $g(i)$ . The characteristic equations imply

$$\int_0^m g(y) dy = \dot{q}D = \sum_{j=1}^{b-1} \dot{p}_j s_j + \dot{q}u =: \lambda. \quad (16)$$

Indeed, for  $\phi(y) = e^{-qh(y)/D}$  one has  $\phi_y = -(q\rho_j/D)\phi$  and  $e^{p_{j+1}-p_j} = q\dot{q}\rho_j/(e^q - 1)$ . Using  $\phi(0) = 1, \phi(m) = e^{-q}$ , the integrals before division by  $\phi(k)$  are

$$\frac{\dot{q}D}{e^q - 1}(1 - \phi(k)), \quad \frac{\dot{q}De^q}{e^q - 1}(\phi(k) - e^{-q}),$$

whose sum is  $\dot{q}D\phi(k)$ . Continuity handles  $q = 0$ .

By (9), every frozen ratio is at most  $C_b e^{\theta q}$ . On each of at most  $b + 1$  intervals cut at group boundaries and  $k, g$  decreases. All endpoints are integers; hence the left-sum error is bounded by the sum of left endpoint values:

$$\left| \sum_i g(i) - \lambda \right| \leq C_b e^{\theta q}.$$

Exact descending ratios obey the same bound; exact ascending ratios are at most  $C_b e^{(\theta+1/2)q}$  by (15). Apply  $|e^z - e^w| \leq |z - w| \max(e^z, e^w)$ , (14), and  $m/D \leq 1/c$  to obtain

$$\left| \frac{\mathcal{T}\psi}{\psi} - \sum_i g(i) \right| \leq C_b q e^{(\theta+1/2)q}.$$

Finally  $|h_q|, |D_q| \leq Km$  imply  $|r_q| \leq 2K/c$ . Thus

$$\frac{\partial_t \psi}{\psi} = \lambda - \dot{q}(r + qr_q), \quad \left| \frac{\partial_t \psi}{\psi} - \lambda \right| \leq C_b(1 + q)e^{\theta q}.$$

Together these give the state-uniform estimate

$$\left| \frac{\partial_t \psi}{\psi} - \frac{\mathcal{T} \psi}{\psi} \right| \leq C_b(1+q)e^{(\theta+1/2)q} =: C(q). \quad (17)$$

Let

$$\mathcal{E}(q) = \int_0^q \frac{C(z)}{\dot{q}(z)} dz = O_b((1+q)^3 e^{q/2}), \quad g_{\pm} = e^{\pm \mathcal{E}} \psi. \quad (18)$$

Then  $\partial_t g_- \leq \mathcal{T} g_-$  and  $\partial_t g_+ \geq \mathcal{T} g_+$ , and both start at 1. Since  $\theta > 1/2$ ,

$$\mathcal{E}(q) = o(e^{p_1(q)}). \quad (19)$$

This strict inequality explains why this particular proof excludes  $b = 2$ .

## 4 Comparison on the infinite state space

Let  $X$  be the continuous-time chain with each rank-choice edge at rate 1. Its generator is  $L = \mathcal{T} - mI$  and its total rate is  $m$ . Because  $m$  grows by at most one per jump, comparison with a linear pure-birth process proves nonexplosion. Expanding by finite jump paths cancels survival factors against the potential factor and gives the extended-real identity

$$F(t, x) = \mathbb{E}_x \exp \left( \int_0^t m(X_s) ds \right). \quad (20)$$

Let  $\tau_M$  be the first exit from  $m \leq M$  and  $Y_s = \exp(\int_0^s m(X_r) dr)$ . The stopped region is finite. Applying stopped Dynkin comparison to  $Y_s g_+(t-s, X_s)$  and discarding its nonnegative exit term gives  $F(t, x) \leq g_+(t, x)$  upon letting  $M \rightarrow \infty$ .

For the lower bound fix  $\delta > 0$  with  $t + \delta < T_b$ . Strict increase of all  $p_j, q$  gives an  $\eta > 0$ , uniform for  $0 \leq v \leq t$ , such that

$$g_-(v, y) \leq C e^{-\eta m(y)} g_+(v + \delta, y). \quad (21)$$

Indeed the population terms each decrease by at least  $\eta$  in the log ratio; its rank term is bounded above by  $q(t + \delta)$ , and the  $\mathcal{E}$  terms have favorable signs. The stopped subsolution inequality is

$$g_-(t, x) \leq \mathbb{E}_x[Y_t; \tau_M > t] + \mathbb{E}_x[Y_{\tau_M} g_-(t - \tau_M, X_{\tau_M}); \tau_M \leq t].$$

On exit  $m(X_{\tau_M}) = M + 1$ . Equation (21) and the stopped supersolution inequality at horizon  $t + \delta$  bound the second term by  $C e^{-\eta(M+1)} g_+(t + \delta, x) \rightarrow 0$ . Nonexplosion handles the first term. We have proved

$$e^{-\mathcal{E}(q)} \psi(t, x) \leq F(t, x) \leq e^{\mathcal{E}(q)} \psi(t, x), \quad 0 \leq t < T_b. \quad (22)$$

No uniqueness assertion for an infinite differential system is required.

## 5 The radius and the coefficient limit

Let  $x_N$  have  $s_1 = N, u = 1, k = N$  and all other populations zero; the increasing prefix of length  $N$  reaches it. Its rank formula yields

$$\log \psi(t, x_N) = N p_1(q) + \frac{q}{\rho_1 N + 1}, \quad \log F(t, x_N) = N p_1(q) + O_b(\mathcal{E}(q) + q), \quad (23)$$



with error independent of  $N$ . In particular  $F(t, x_N) \geq e^{Np_1 - \mathcal{E}}$ . The positive semigroup identity and one increasing-prefix path at every length give, for  $h > 0$ ,

$$\begin{aligned} F(t+h, x_1) &= \sum_{r \geq 0} \frac{h^r}{r!} (\mathcal{T}^r F(t, \cdot))(x_1) \\ &\geq \sum_{r \geq 0} \frac{h^r}{r!} F(t, x_{r+1}) \geq \exp(p_1 - \mathcal{E} + he^{p_1}). \end{aligned}$$

As  $t \uparrow T_b$ , (19) makes the last expression diverge. Monotonicity forces  $F(T_b + h, x_1) = \infty$ . Equation (22) gives finiteness below  $T_b$ . Thus  $F(t, x_1)$  and  $A_b$  have exact radius  $T_b$ , proving the normalized-root limsup. The next argument is needed for the full limit.

### 5.1 A concentration estimate from barrier values

Put  $p = p_1$ ,  $d = (\log t)_q$  and  $M(q) = p_q/d = tp$ . Direct characteristic asymptotics give

$$p_q \rightarrow \theta, \quad p_{qq} \rightarrow 0, \quad d \sim \frac{qe^{-\theta q}}{c_b T_b}, \quad d'/d \rightarrow -\theta, \quad M(q) \sim \frac{\theta c_b T_b e^{\theta q}}{q}. \quad (24)$$

These limits are uniform after bounded shifts in  $q$ . For  $c_j = \mathcal{T}^j \mathbf{1}(x_N)$  define

$$\mathbb{P}(J = j) = \frac{c_j t(q)^j}{j! F(t(q), x_N)}, \quad K = NM(q).$$

The deterministic center  $K$  need not be the exact mean. We use values of (23), never derivatives of its error. Set

$$H_q(v) = p(q+v) - p(q) - M(q)(\log t(q+v) - \log t(q)).$$

For  $|v| \leq 1$ ,  $H_q(0) = H'_q(0) = 0$  and

$$H''_q(v) \rightarrow \theta^2 e^{-\theta v}, \quad M(q)d(q+v) \rightarrow \theta e^{-\theta v}$$

uniformly. Consequently there are fixed-cap constants  $B, c > 0$  with

$$|H_q(v)| \leq Bv^2, \quad M(q)|\log t(q+v) - \log t(q)| \geq c|v|.$$

The exponential Markov bounds at  $q \pm \gamma\varepsilon$ , with fixed small  $\gamma > 0$ , now give

$$\mathbb{P}(|J - K| > \varepsilon K) \leq 2 \exp\{-c'_b N \varepsilon^2 + 2\mathcal{E}(q+1) + O_b(q+1)\} \quad (25)$$

for sufficiently large  $q$  and  $0 < \varepsilon < 1/2$ . Explicitly the main exponent is at most  $N(B\gamma^2\varepsilon^2 - c\gamma\varepsilon^2)$ ; choosing  $\gamma$  small makes it negative. Uniformity permits shrinking  $\varepsilon$ .

### 5.2 Extraction for each sufficiently large length

For every sufficiently large integer  $n$ , choose

$$q = \frac{1}{2} \log n, \quad \varepsilon = (\log n)^{-2}, \quad N = \left\lfloor \frac{(1 - 4\varepsilon)n}{M(q)} \right\rfloor.$$

Then

$$\begin{aligned} N &\asymp_b n^{1-\theta/2} \log n, & N\varepsilon^2 &\asymp_b \frac{n^{1-\theta/2}}{(\log n)^3}, \\ K &= (1-4\varepsilon)n + O_b(n^{\theta/2}/\log n), & \mathcal{E}(q+1) &= O_b(n^{1/4}(\log n)^3). \end{aligned}$$

Since  $1/2 < \theta < 1$ ,  $N\varepsilon^2$  dominates the error and both  $N$  and the floor loss in  $K$  are  $o(\varepsilon n)$ . Equation (25) puts probability at least  $1/2$  in  $|J - K| \leq \varepsilon K$ . Some integer  $j$  there satisfies

$$\frac{c_j t(q)^j}{j!} \geq \frac{F(t(q), x_N)}{4\varepsilon K + 6}.$$

For large  $n$  it obeys  $(1-6\varepsilon)n \leq j$  and  $N + j \leq n$ , giving

$$\log c_j \geq \log(j!) - j \log t(q) + Np(q) - \mathcal{E}(q) - O_b(\log n). \quad (26)$$

The increasing prefix reaches  $x_N$  in  $N-1$  transitions, so  $a_{N+j}^{(b)} \geq c_j$ . Appending the new maximum  $1 + \text{asc}(w)$  repeatedly gives an injective extension to length  $n$ , so  $a_n^{(b)} \geq c_j$ .

### 5.3 A quantitative logarithmic error

From (8),

$$T_b - t(q) = O_b((1+q)e^{-\theta q}). \quad (27)$$

In (26) the term  $Np(q)$  is nonnegative. Also  $\log(n!/j!) \leq 6\varepsilon n \log n = 6n/\log n$  and

$$|-j \log t(q) + n \log T_b| = O_b(\varepsilon n + n(1+q)e^{-\theta q}).$$

All latter errors, and  $\mathcal{E}(q)$ , are  $o_b(n/\log n)$  for  $q = \frac{1}{2} \log n$ . Hence

$$\log a_n^{(b)} \geq \log(n!) - n \log T_b - O_b(n/\log n).$$

For the matching upper estimate use a single coefficient of  $F(t, x_1)$ :

$$a_n^{(b)} \leq (n-1)! F(t(q), x_1) t(q)^{-(n-1)}.$$

Equations (23) and (27) bound its excess logarithm above  $\log(n!) - n \log T_b$  by

$$O_b(\mathcal{E}(q) + q + \log n + n(1+q)e^{-\theta q}) = o_b(n/\log n).$$

This proves (2), and therefore the full root limit. The proof uses no unproved ratio asymptotic.

## 6 Inverse in the sequence length

For fixed  $b \geq 2$  and  $X > 1$ , define the integer threshold

$$N_b(X) = \min\{n \geq 0 : a_n^{(b)} \geq X\}, \quad L = \log X.$$

The injective extension above makes the counts nondecreasing, and Theorem 1.1 makes them unbounded. Let  $W_0$  denote the principal real branch of Lambert  $W$ , characterized by  $W_0(z)e^{W_0(z)} = z$  for  $z > 0$ . As  $X \rightarrow \infty$  with  $b$  fixed,

$$n_0 = \frac{L}{W_0(L/(eT_b))}, \quad N_b(X) = n_0 + O_b\left(\frac{L}{(\log L)^3}\right) = n_0 \left(1 + O_b((\log L)^{-2})\right). \quad (28)$$

To prove it, Stirling and (2) give  $\log a_n^{(b)} = G(n) + O_b(n/\log n)$  for  $G(x) = x \log(x/(eT_b))$ . The threshold inequalities at  $N = N_b(X)$  and  $N - 1$ , together with  $G(N) - G(N - 1) = O_b(\log N)$ , imply  $L = G(N) + O_b(N/\log N)$ . Thus  $N \sim L/\log L$ , while  $G(n_0) = L$  exactly. Since  $G'(x) = \log(x/T_b) \asymp \log L$  between comparable  $N$  and  $n_0$ , the mean value theorem gives (28).

This is a controlled leading inverse, not an exact rounding rule: its error tends to infinity. One may expand  $W_0$  itself in logarithms, but such a formal expansion cannot improve the error inherited from the unknown coefficient correction. For  $b = 2$ , the separate report gives a sharper inverse using its stronger logarithmic bound.

## 7 Large caps and an exact first variation

Put  $f(x) = \log x/(x - 1)$ , continuously extended at 1, and  $\tau = \pi^2/6$ . The representation

$$f(x) = \int_0^1 \frac{dt}{1 + t(x - 1)} \quad (29)$$

shows  $f' < 0$  and  $f'' > 0$  on  $[1, \infty)$ . Thus  $E_b$  increasing strictly in  $b$  implies  $T_b$  decreasing strictly. The cap-two integrand dominates all later integrands and is integrable. Dominated convergence and the nonnegative geometric series give

$$T_b \downarrow \int_0^\infty \frac{v}{e^v - 1} dv = \sum_{k \geq 1} \frac{1}{k^2} = \tau.$$

Write

$$R_b(v) = e^v - E_b(v), \quad \Delta_b(v) = f(E_b(v)) - f(e^v), \quad L_b = \int_0^\infty [-f'(e^v)] R_b(v) dv.$$

Convexity gives  $\Delta_b(v) \geq [-f'(e^v)] R_b(v) \geq 0$ . We will prove

$$L_b = (b + 1)(\zeta(b + 2) - 1), \quad 0 \leq T_b - \tau - L_b = O(b^{5/2}(4/9)^b). \quad (30)$$

### 7.1 Evaluation of the linear term

For  $v > 0$ ,

$$-f'(e^v) = \sum_{h \geq 2} ((h - 1)v - 1)e^{-hv}, \quad R_b(v) = \sum_{k \geq b+1} \frac{v^k}{k!}.$$

Although individual terms can have either sign near zero, their integrated absolute values are bounded by

$$\sum_{k \geq b+1} \sum_{h \geq 2} \left( \frac{(h - 1)(k + 1)}{h^{k+2}} + \frac{1}{h^{k+1}} \right) < \infty \quad (b \geq 2).$$

Fubini is therefore legitimate. Integration and then summation in  $k$  yield

$$L_b = \sum_{h \geq 2} \sum_{k \geq b+1} \frac{(h - 1)k - 1}{h^{k+2}} = \sum_{h \geq 2} \frac{b + 1}{h^{b+2}} = (b + 1)(\zeta(b + 2) - 1).$$

## 7.2 A uniform bound for the nonlinear remainder

There is an absolute constant  $C$  with

$$0 \leq f''(z) \leq C \frac{1 + \log z}{z^3} \quad (z \geq 1). \quad (31)$$

This follows from smoothness at 1 and the derivative asymptotics at infinity, or directly from (29). For  $0 \leq v \leq b$ , the function  $e^{-v}E_b(v)$  decreases, since its derivative is  $-e^{-v}v^b/b!$ . Stirling therefore gives

$$e^{-v}E_b(v) \geq e^{-b}E_b(b) \geq e^{-b}b^b/b! \geq cb^{-1/2}$$

with absolute  $c > 0$ . Taylor's theorem on  $[E_b(v), e^v]$ , using  $\log z \leq v$ , gives

$$0 \leq \Delta_b(v) - [-f'(e^v)]R_b(v) \leq Cb^{3/2}(1+v)e^{-3v}R_b(v)^2. \quad (32)$$

Let  $m = b + 1$ . Expanding the square, grouping  $S = k + \ell$ , and using  $\sum_{k=m}^{S-m} \binom{S}{k} \leq 2^S$ , we obtain

$$\begin{aligned} \int_0^\infty (1+v)e^{-3v}R_b(v)^2 dv &\leq \sum_{S \geq 2m} 2^S \left( \frac{1}{3^{S+1}} + \frac{S+1}{3^{S+2}} \right) \\ &= O\left(b(4/9)^b\right). \end{aligned}$$

All these integrands and terms are nonnegative, so Tonelli applies. This bounds the integrated remainder on  $[0, b]$  by  $O(b^{5/2}(4/9)^b)$ .

For  $v \geq b \geq 3$ ,  $E_b(v) \geq v^b/b! \geq 2$  and  $\log E_b(v) \leq v$ . Convexity implies the remainder is at most  $\Delta_b(v) \leq f(E_b(v))$ , and hence its tail is at most

$$2b! \int_b^\infty v^{1-b} dv = \frac{2b!b^{2-b}}{b-2} = O(b^{3/2}e^{-b}) = O\left(b^{5/2}(4/9)^b\right).$$

This proves (30) and (3) with constants independent of  $b$ .

Since

$$L_b = \frac{b+1}{2^{b+2}} + O(b3^{-b}),$$

the remainder is smaller than the leading term by an exponentially decaying factor. The reciprocal identity

$$\mu_\infty - \mu_b = \frac{T_b - \tau}{\tau T_b}, \quad \mu_\infty = \frac{1}{\tau} = \frac{6}{\pi^2},$$

then gives the useful quantified form

$$\mu_\infty - \mu_b = D(b+1)2^{-b} \left[ 1 + O\left(b^{3/2}(8/9)^b\right) \right], \quad D = \frac{9}{\pi^4}. \quad (33)$$

Indeed the additive contributions  $O(b3^{-b})$  and  $O(b^24^{-b})$  are absorbed by  $O(b^{5/2}(4/9)^b)$ . This also proves (4). Equation (3) is a rigorously controlled first-variation expansion; it is not an all-orders transseries.

## 8 Inverse in the multiplicity cap

For small  $\varepsilon > 0$  let

$$b_\varepsilon = \min\{b \geq 2 : \mu_b \geq \mu_\infty - \varepsilon\}.$$

Monotonicity proves existence. The large real solution of the leading model  $D(b+1)2^{-b} = \varepsilon$  is

$$b_0 = -\frac{1}{\log 2} W_{-1}\left(-\frac{\varepsilon \log 2}{2D}\right) - 1, \quad (34)$$

where  $W_{-1}$  is the real branch with  $W_{-1}(z) \leq -1$  on  $[-1/e, 0)$ . The branch is essential;  $W_0$  gives the irrelevant small root.

There are constants  $C$  and  $\varepsilon_0 > 0$  such that, for  $0 < \varepsilon < \varepsilon_0$ , with

$$\eta = Cb_0^{3/2}(8/9)^{b_0},$$

one has the integer-safe bracket

$$[b_0 - \eta] \leq b_\varepsilon \leq [b_0 + \eta]. \quad (35)$$

No interpolation of the exact sequence  $T_b$  to real  $b$  is assumed.

Here is the discrete justification. For the real leading model  $h(x) = D(x+1)2^{-x}$ ,

$$(\log h)'(x) = \frac{1}{x+1} - \log 2$$

is negative and bounded away from zero for large  $x$ . Equation (33) first places  $b_\varepsilon$  within a fixed bounded distance of  $b_0$ : at integers several units below or above  $b_0$ , the fixed change of  $h$  dominates the vanishing relative error. In that bounded window the relative error is uniformly  $O(b_0^{3/2}(8/9)^{b_0})$ . Increasing  $C$  if necessary, every integer  $b < b_0 - \eta$  fails the threshold and every integer  $b \geq b_0 + \eta$  meets it. Monotonicity handles all integers outside the window, giving (35).

A useful explicit strengthening, proved by the same estimates, is

$$[b_0] + 1 \leq b_\varepsilon \leq [b_0 + 2000b_0^{3/2}(8/9)^{b_0}] \quad (b_0 \geq 13). \quad (36)$$

Here are details to make its constants and strict lower bound reproducible. The bound  $f''(z) \leq 16(1 + \log z)/z^3$  and  $b! \leq 3\sqrt{b}(b/e)^b$  make (32) valid with 216. Its integrated central bound is  $64b^{3/2}(b+4)(4/9)^b$ , and its tail is at most  $6b^{5/2}(4/9)^b$ . Thus  $0 \leq H_b := T_b - \tau - L_b \leq 160b^{5/2}(4/9)^b$  for every  $b \geq 3$ .

**[write, 2 October 2026]** In Part II's notation,  $L_b = A_{1,b}$  and  $H_b = \mathcal{R}_{1,b}$ , and Theorem 12.1 with (48) gives  $H_b \sim A_{2,b} \sim \frac{8}{9\sqrt{\pi}}\sqrt{b}(4/9)^b$ . The explicit bound  $160b^{5/2}(4/9)^b$  above is therefore valid but not sharp in its power of  $b$ ; it is kept because it is explicit, and the integer brackets below use it. ( $H_b$  is not Part II's harmonic number  $H_\tau$ .)

Put  $M_b = (b+1)2^{-b-2}$  and  $V_b = L_b - M_b$ . Integral comparison gives

$$L_b \leq (b/2)2^{-b}, \quad (b+1)3^{-b-2} \leq V_b \leq (b+4)3^{-b-2}.$$

Hence  $M_b L_b / V_b \leq (9b/8)(3/4)^b \leq 729/512 < \tau$ , so

$$d_b := \mu_\infty - \mu_b \geq \frac{L_b}{\tau(\tau + L_b)} > \frac{M_b}{\tau^2} = h(b).$$

The upper bounds above also give  $0 < d_b/h(b) - 1 \leq 644\phi(b)$ , where  $\phi(x) = x^{3/2}(8/9)^x$ . For  $x \geq 13$ ,  $\phi$  decreases, and for  $x \geq 2$ ,  $(\log h)'(x) < -1/3$ . Every integer  $b \leq b_0$  fails the threshold, proving its strict lower bracket. For  $m = \lceil b_0 + 2000\phi(b_0) \rceil$ ,

$$\frac{d_m}{\varepsilon} \leq e^{-2000\phi(b_0)/3}(1 + 644\phi(b_0)) < 1,$$

proving the upper bracket. The constants are conservative for moderate caps. In particular, at  $\varepsilon = h(n)$  for an integer  $n \geq 3$ , one has  $b_0 = n$  but  $b_\varepsilon \geq n + 1$ . This supplies actual counterexamples to unconditional rounding by  $\lceil b_0 \rceil$  at arbitrarily small tolerances.

Thus  $b_\varepsilon = b_0 + O(1)$ , and away from vanishingly small neighborhoods of integer transitions the threshold is the expected ceiling. Exactly near those transitions, an unconditional rounding prescription is not justified by an asymptotic equivalence. In particular this is an inverse for the cap of the limiting growth constant, not a cap selection guarantee for finite-length counts.

## 9 Numbers and reproducible checks

The following values are numerical quadratures of (1); the displayed digits are checks, not interval-certified enclosures.

| $b$      | $T_b$                | $\mu_b$              |
|----------|----------------------|----------------------|
| 2        | 3.701101650408509482 | 0.270189823046234057 |
| 3        | 2.212067760911192191 | 0.452065717728321870 |
| 4        | 1.871791135820212368 | 0.534247641664251966 |
| 5        | 1.748278343303498299 | 0.571991298656956259 |
| 6        | 1.695001120651513698 | 0.589970111415399178 |
| 8        | 1.657736605427835731 | 0.603232139970701640 |
| 10       | 1.648371562426905637 | 0.606659337490447259 |
| $\infty$ | 1.644934066848226436 | 0.607927101854026629 |

In particular, for cap three (A317784),

$$a_n^{(3)} = n! (0.452065717728321870 \dots)^n \exp(O(n/\log n)),$$

where the constant is exactly the reciprocal integral, not the rounded decimal. The first ten counts, indexed from  $n = 0$ , are

$$\begin{aligned} b = 3 : & \quad 1, 1, 2, 5, 14, 47, 180, 773, 3701, 19488, \\ b = 4 : & \quad 1, 1, 2, 5, 15, 52, 210, 964, 4960, 28272. \end{aligned}$$

The accompanying source archive contains independent reproducibility routes:

- Direct generation of numerical words, unsorted raw-budget recursion, and grouped recursion agree for caps 3, 4 through  $n = 9$ .
- All 8,460 adjacent-budget state comparisons and 5,364 canonical raw/grouped state comparisons pass in the stated finite ranges.
- The actual suffix involution, its inverse, quotas, and ascent total pass 69,500 individual checks.
- Seeded tests of exact child-rank identities and the implementation of the barrier cover 82,801 transitions for  $b = 3, \dots, 8$ . Finite residual tests do not prove a uniform bound; Section 3 does.

- High-precision quadrature reproduces the constants, large-cap ratios, exact first variation, and numerical instances of both inverse models. It is not substituted for the analytic error estimates.

The commands, dependency list, test results, proof-source provenance, independent audits, and checksums are included. All mathematics needed for  $b \geq 3$  is in this report; the separate cap-two document is identified by its file hash rather than duplicated. The PDF and T<sub>E</sub>X source can be rebuilt with the supplied script. No external OEIS edit or publication is implied.

[**write, 2 October 2026**] In this repository the five checks and their saved results are shipped as `code/47-caps-*.py` and `data/47-caps-*.json`, the proof records and audits as `47-caps-*.md`, and the provenance record as `data/47-caps-provenance.json`. The delivered PDF and checksum manifest are not shipped, and the delivered T<sub>E</sub>X source is the text of this Part. The README lists every file and gives a rerun recipe that leaves the shipped results untouched.

## 10 Literature and the remaining questions

Duncan and Steingrímsson [3] study pattern avoidance and growth rates for ascent sequences. Conway, Conway, Elvey Price and Guttmann [1] give the cap-two compacted recurrence and conjecture its factorial growth constant  $8/(3\pi^2)$ . The present arbitrary-budget involution generalizes that compaction mechanism. Their recurrence and conjectured constant are prior work, not a discovery claimed here.

The OEIS family entry A294220, retrieved on 1 October 2026, describes the maximum-letter-multiplicity array, identifies caps two and three as A202058 and A317784, and lists the finite values matching the checks above. Its retrieved formula section gives array relations and its program section gives exact enumeration. The standalone A317784 page could not be retrieved in this check; its identification here is verified through A294220's explicit cross-reference. No claim about A317784's current formula-section wording is made. Search-engine cache dates and the OEIS site footer do not certify an individual entry's revision date.

The report proves the statements presented here; it makes no exhaustive priority claim. The literature check is scoped, and conventional expert review and further literature screening remain appropriate before publication. The independent audits included with the package are mathematical checks of this argument, not journal peer review.

Several concrete questions remain:

1. Determine the subexponential factor of  $a_n^{(b)}$  for fixed  $b$ : a prefactor, any power or stretched-exponential correction, and eventually a full coefficient expansion. The present error permits many possibilities.
2. Establish or refute the normalized ratio limit. A root limit by itself does not answer it.
3. Sharpen the rank barrier enough to quantify the true singular behavior of  $A_b$  near  $T_b$ , rather than only its radius and coefficient exponential scale.
4. Determine a joint regime  $b = b(n)$  and the transition to unrestricted ascent sequences. The convergence of growth constants alone does not allow exchange of limits.
5. Extend the controlled large-cap first variation to further exponentially small sectors and their internal  $1/b$  expansions, with uniform remainder estimates. Such a cap expansion is logically distinct from an all-orders expansion in  $n$ .

6. Find a direct combinatorial explanation for the truncated-exponential integral. A standard ascent-sequence/Fishburn bijection must not be assumed to preserve label multiplicity without proof.

[**write, 2 October 2026**] Question 5 is answered for every *fixed* number of sectors in Part II: the exact positive sectors  $A_{r,b}$ , the positive remainder after  $K$  sectors with its sharp size (Theorem 12.1), and an expansion of each fixed sector to every algebraic order in  $1/b$  with rational coefficients (Theorem 13.1). Still open, as Part II says itself: estimates uniform in a growing sector index  $r = r(b)$ , the accumulation scale  $e^{-b}$ , optimal truncation and summability. For cap two, question 2 (the ratio limit) is answered in the sibling report **a202058-ascent-000-growth**; for every fixed  $b \geq 3$  questions 1–4 and 6 remain open.

[**write, 5 October 2026**] Question 2 is now answered for every fixed  $b \geq 3$ : Part III, Theorem 17.1, proves  $a_{n+1}^{(b)}/((n+1)a_n^{(b)}) \rightarrow \mu_b$  (no rate), and proves it again for  $b = 2$ . Questions 1, 3, 4 and 6 remain open for every fixed  $b \geq 3$ ; Section 26 of Part III restates 1 and 4 and adds questions on fluctuations and on the endpoint region (list in Section 26.6).

*Neighbouring reports on Question 4.* Two reports of batch 102 in this collection directory treat related families. **a336070-weak-ascents** (bundle Reports 105, 107 and 108) treats unrestricted weak ascent sequences and, for every fixed  $d$ , the difference- $d$  ascent sequences ( $d = 0$  is the ordinary unrestricted ascent sequences,  $d = 1$  the weak ones). It proves the factorial constant  $6/\pi^2$ , the limit of  $\mu_b$  as  $b \rightarrow \infty$ , for every fixed  $d$  (label **wasc:gr:thm:main** there), with  $\log(a_n^{(d)}/(n!(6/\pi^2)^n)) = \frac{d}{2}(\log n)^2 + O_d(\log n)$  for  $d \geq 1$  (**wasc:sh:thm:main**) and no such correction at  $d = 0$  (**wasc:sh:rem:dzero**). **a202059-ascent-100-110-growth** (bundle Reports 241 and 243) shows that the ascent sequences avoiding 100, or 110, have  $(u_n/n!)^{1/n} = \Theta((\log n)^{-2}) \rightarrow 0$ ; its classes avoiding 000 and 100, and 000, 100 and 110, lie inside the cap-two class, and Report 241 there notes that the cap-two lower bounds do not transfer to them. Report 241 cites this report, but not as an input. Neither report studies a multiplicity cap, and neither answers Question 4.

## Part II

# Fixed Taylor sectors for multiplicity cap growth constants

*Source.* Batch-77 manuscript 52, archive **oeis-fixed-sector-addendum.zip**, main file **fixed-sector-addendum.tex** (312 lines, 7-page PDF), dated 1 October 2026. Author line: “A reproducible large cap addendum”. No commit pin; it identifies source 47 by file hashes. An addendum to source 47 (Part I), printed in full, with labels prefixed **amc:sec:**; “the main report” and its citation **main** now refer to Part I. No symbol was renamed.

**Abstract (as delivered).** The cap integral has exact positive Taylor sectors and an exponentially sharp remainder after any fixed number of sectors. Each fixed sector has an algebraic expansion to arbitrary fixed order; its coefficients are rational and algorithmically computable. We give the leading constants, the first three corrections in sectors two and three, and a precise reciprocal corollary for the growth constant. This supplements the separately delivered bounded-multiplicity report [Part I of this report]: its combinatorial identification and inversion results are prerequisites, not reproved here. The statements are fixed-sector results. Finite algebraic truncations cannot generally resolve subsequent exponential sectors, and no uniform growing-sector or accumulation-scale theorem is asserted.



## 11 Objects and relationship to the main report

For integer  $b \geq 2$ , define

$$E_b(v) = \sum_{j=0}^b \frac{v^j}{j!}, \quad f(x) = \frac{\log x}{x-1}, \quad T_b = \int_0^\infty f(E_b(v)) dv, \quad (37)$$

with  $f(1) = 1$ . Set  $\tau = \pi^2/6$ ,  $m = b+1$ ,  $R_b(v) = e^v - E_b(v)$ , and  $H_r = \sum_{j=1}^r 1/j$ . The integral is finite for  $b \geq 2$  because its integrand is  $O_b(\log v/v^b)$  at infinity. Also  $\int_0^\infty f(e^v) dv = \int_0^\infty v/(e^v - 1) dv = \tau$ .

The main report, Part I of this report (Theorem 1.1), proves, for each fixed cap  $b \geq 3$ ,

$$\lim_{n \rightarrow \infty} \left( \frac{a_n^{(b)}}{n!} \right)^{1/n} = \mu_b = T_b^{-1},$$

where  $a_n^{(b)}$  counts ascent sequences in which every value occurs at most  $b$  times. Its cited cap-two theorem gives the same identification for  $b = 2$ . The present addendum refines the large- $b$  analysis of  $T_b$  after that fixed-cap length limit. It does not strengthen the main report's error in  $n$  or give joint  $b, n$  asymptotics.

The main report established the exact first variation and a nonlinear bound  $O(b^{5/2}(4/9)^b)$  (Part I, (30)). The theorem below sharpens that remainder to a full fixed-sector hierarchy, starting with  $A_{2,b} \asymp b^{1/2}(4/9)^b$ . The main report's controlled length inverse and rounding-safe cap inverse (Sections 6 and 8) remain the supplied inversion results; no sharper inverse is claimed in this addendum.

## 12 Exact positive sectors and the sharp remainder

Define, for  $r \geq 1$ ,

$$a_r(x) = \frac{(-1)^r}{r!} f^{(r)}(x) = \int_0^1 \frac{t^r dt}{(1-t+tx)^{r+1}}, \quad A_{r,b} = \int_0^\infty a_r(e^v) R_b(v)^r dv. \quad (38)$$

These sectors are strictly positive. Define their primary exponential bases by

$$\beta_r = \left( \frac{r}{r+1} \right)^r. \quad (39)$$

They strictly decrease to  $e^{-1}$ ; in particular every fixed  $\beta_r$  is greater than  $e^{-1}$ .

**Theorem 12.1.** *For each integer  $b \geq 2$ , the following is an exact convergent identity:*

$$T_b - \tau = \sum_{r \geq 1} A_{r,b}. \quad (40)$$

*For every fixed integer  $K \geq 1$ , as  $b \rightarrow \infty$  through integers,*

$$T_b - \tau = \sum_{r=1}^K A_{r,b} + \mathcal{R}_{K,b}, \quad (41)$$

$$0 < \mathcal{R}_{K,b} = O_K \left( b^{(2-K)/2} \beta_{K+1}^b \right), \quad (42)$$

$$\mathcal{R}_{K,b} = A_{K+1,b} \left[ 1 + O_K \left( b^{-1/2} \left( \frac{\beta_{K+2}}{\beta_{K+1}} \right)^b \right) \right]. \quad (43)$$

*In particular  $\mathcal{R}_{K,b} \sim A_{K+1,b}$ , with a positive exact remainder. The estimates hold for fixed  $K$ ; the identity (40) does not license summing sector asymptotic formulas.*

*Proof.* The integral representation  $f(x) = \int_0^1 (1-t+tx)^{-1} dt$  gives (38). For  $x = e^v$ ,  $R = R_b(v)$ ,  $E = x - R$ , and  $D = 1 - t + tx$ , the finite geometric identity gives

$$f(E) - f(x) - \sum_{r=1}^K a_r(x) R^r = R^{K+1} \int_0^1 \frac{t^{K+1} dt}{D^{K+1}(D - tR)}. \quad (44)$$

Since  $0 \leq tR/D \leq R/x < 1$ , this lies between  $a_{K+1}(x)R^{K+1}$  and  $(x/E)a_{K+1}(x)R^{K+1}$ . Expanding the infinite geometric series and using monotone convergence proves (40); in particular the sectors are integrable.

There is an absolute  $c > 0$  such that  $e^{-v}E_b(v) \geq c$  for  $0 \leq v \leq b$ . Indeed  $e^{-v}E_b(v)$  is decreasing, while Stirling's bounds show that the  $\lfloor \sqrt{b}/2 \rfloor + 1$  summands of  $e^{-b}E_b(b)$  immediately below degree  $b$  are each at least a constant times  $b^{-1/2}$ . For those terms use

$$e^{-b} \frac{b^{b-j}}{(b-j)!} = \frac{e^{-b} b^b}{b!} \prod_{h=0}^{j-1} (1 - h/b), \quad \log(1-u) \geq -2u \quad (0 \leq u \leq 1/2).$$

For sufficiently large  $b$  and  $v \geq b$ ,  $E_b(v) \geq v^b/b! \geq 2$  and  $\log E_b(v) \leq v$ . Thus

$$0 \leq \int_b^\infty [f(E_b(v)) - f(e^v)] dv \leq \frac{2b! b^{2-b}}{b-2} = O(b^{3/2} e^{-b}).$$

The integrated geometric bounds now give

$$A_{K+1,b} \leq \mathcal{R}_{K,b} \leq c^{-1} A_{K+1,b} + O(b^{3/2} e^{-b}). \quad (45)$$

Theorem 13.1 below, together with  $\beta_{K+1} > e^{-1}$ , proves (42). Applying the same bound with  $K+1$  and using  $\mathcal{R}_{K,b} = A_{K+1,b} + \mathcal{R}_{K+1,b}$  gives (43). Finally, the derivative of  $-r \log(1+1/r)$  is negative, by  $\log(1+x) > x/(1+x)$ , proving the asserted ordering of the bases.  $\square$

### 13 Every algebraic order within a fixed sector

**Theorem 13.1.** *For every fixed integer  $r \geq 1$  and every fixed  $N \geq 0$ , there are rational coefficients  $c_{r,j}$ , with  $c_{r,0} = 1$ , such that*

$$A_{r,b} = C_r b^{(3-r)/2} \beta_r^b \left( \sum_{j=0}^N \frac{c_{r,j}}{b^j} + O_{r,N}(b^{-N-1}) \right), \quad C_r = \frac{r^{r+3/2}}{(r+1)^2} (2\pi)^{(1-r)/2}. \quad (46)$$

*The first correction is*

$$c_{r,1} = \frac{3-r}{2} - \frac{r^2(r+1)}{2} + \frac{23r}{12} + \frac{13}{12r} - \frac{r+1}{r} H_r. \quad (47)$$

*In particular, the first three corrections in sectors two and three are*

$$A_{2,b} = \frac{8}{9\sqrt{\pi}} \sqrt{b} \left( \frac{4}{9} \right)^b \left( 1 - \frac{27}{8b} + \frac{5669}{128b^2} - \frac{962949}{1024b^3} + O(b^{-4}) \right), \quad (48)$$

$$A_{3,b} = \frac{81\sqrt{3}}{32\pi} \left( \frac{27}{64} \right)^b \left( 1 - \frac{43}{3b} + \frac{33952}{81b^2} - \frac{42671116}{2187b^3} + O(b^{-4}) \right). \quad (49)$$

*The first sector has the exact formula*

$$A_{1,b} = (b+1)(\zeta(b+2) - 1). \quad (50)$$

*Consequently its algebraic coefficients are  $c_{1,1} = 1$  and  $c_{1,j} = 0$  for  $j \geq 2$ ; its smaller exponential terms are retained by (50).*

### 13.1 Separate the derivative correction

Changing variables in (38) gives

$$a_r(e^v) = e^{-(r+1)v} \frac{v - \sum_{j=1}^r (1 - e^{-v})^j / j}{(1 - e^{-v})^{r+1}}, \quad (51)$$

with continuous value  $1/(r+1)$  at  $v = 0$ . For  $v \geq 1$  this equals  $e^{-(r+1)v}(v - H_r) + O_r((1+v)e^{-(r+2)v})$ . Define

$$J_{r,b} = \int_0^\infty (v - H_r) e^{-(r+1)v} R_b(v)^r dv. \quad (52)$$

The integrand of  $J_{r,b}$  need not be positive. Coefficientwise,  $R_b(v)^r \leq \sum_{k \geq rm} r^k v^k / k!$ . Hence, for fixed  $h > r$ ,

$$\int_0^\infty (1+v) e^{-hv} R_b(v)^r dv \leq \sum_{k \geq rm} \frac{r^k}{h^{k+1}} \left(1 + \frac{k+1}{h}\right) = O_{r,h}(m(r/h)^{rm}).$$

The bounded derivative factors on  $[0, 1]$  and  $R_b(v) \leq e/m!$  give

$$A_{r,b} = J_{r,b} + O_r \left( m \left( \frac{r}{r+2} \right)^{rm} + (m!)^{-r} \right). \quad (53)$$

The bases  $(r/(r+2))^r$  are at most  $1/3 < e^{-1}$ , so these corrections are below every fixed primary sector. All constants here may depend on the fixed  $r$ .

For completeness, (50) can be evaluated without asymptotics. Put  $F(k) = \int_0^\infty e^{-kv} R_b(v) dv = 1/[(k-1)k^m]$  for  $k \geq 2$ . Expanding  $a_1(e^v) = \sum_{k \geq 2} ((k-1)v - 1)e^{-kv}$  and integrating gives

$$A_{1,b} = \sum_{k \geq 2} (-(k-1)F'(k) - F(k)) = m \sum_{k \geq 2} k^{-m-1}.$$

Interchanging the sum and integral is valid by absolute integrability: replacing each linear factor by  $(k-1)v + 1$  gives an integrated summable bound. This also yields  $J_{1,b} = m/2^{m+1}$ .

[**write, 2 October 2026**] Since  $A_{1,b}$  is Part I's  $L_b$ , this is a second proof of the first identity in (30). Part I (Section 7, "Evaluation of the linear term") integrates the double series in  $(h, k)$  term by term; this Part integrates against  $F(k) = 1/((k-1)k^m)$  and differentiates in  $k$ .

### 13.2 Uniform tail expansion and saddle localization

Write  $s = v/m$  and

$$R_b(ms) = \frac{(ms)^m}{m!} S_m(s), \quad S_m(s) = \sum_{\ell \geq 0} s^\ell \prod_{i=1}^\ell (1 + i/m)^{-1}. \quad (54)$$

For every fixed  $d < 1$ , to arbitrary fixed order and with any fixed number of derivatives,

$$S_m(s) = \sum_{j=0}^N \frac{\sigma_j(s)}{m^j} + O_{N,d}(m^{-N-1}), \quad \sigma_0(s) = \frac{1}{1-s}, \quad \sigma_j(s) = -\frac{s}{1-s} \sigma'_{j-1}(s), \quad (55)$$

uniformly on  $0 \leq s \leq d$ . To justify the remainder, expand  $\prod_{i=1}^\ell (1 + iu)^{-1}$  at  $u = 0$ . Its absolute order- $N$  remainder at  $u = 1/m$  is bounded by  $m^{-N-1}(\ell(\ell+1)/2)^{N+1}$ . This follows by bounding its

alternating derivative at positive  $u$  by its value at zero. Summing the polynomial bound against  $s^\ell$  proves uniformity, also after fixed derivatives. The exact differential equation  $(1-s)S_m + (s/m)S'_m = 1$  then proves the recurrence in (55).

Substitution in (52) gives

$$J_{r,b} = \frac{m^{r+2}}{(m!)^r} \int_0^\infty (s - H_r/m) S_m(s)^r e^{m\phi_r(s)} ds, \quad \phi_r(s) = r \log s - (r+1)s. \quad (56)$$

Its unique nondegenerate saddle is  $s_r = r/(r+1)$ , with  $\lambda_r = -\phi_r''(s_r) = (r+1)^2/r$ . Choose fixed  $0 < a < s_r < d < 1$ . On  $[0, a]$ , use  $S_m \leq (1-a)^{-1}$  and the phase gap below the maximum. On  $[d, 1]$ , the term ratios in (54) give  $S_m \leq m+1$ , and another phase gap defeats that polynomial. Finally the original integral over  $v \geq m$  is  $O_r(me^{-m})$ , since  $R_b(v) \leq e^v$ . These contributions are exponentially smaller than the saddle term after Stirling's bounds are included.

On  $[a, d]$ , (55) and the interior Laplace expansion therefore apply to every fixed order. The leading term is

$$J_{r,b} \sim D_r m^{(3-r)/2} \beta_r^m, \quad D_r = r^{3/2} (r+1)^{r-2} (2\pi)^{(1-r)/2}.$$

Since  $D_r \beta_r = C_r$ , this proves the leading constant in (46). The same argument supplies a genuine remainder at each algebraic order, not merely a formal series. Together with (53), it establishes the all-orders assertion of Theorem 13.1.

### 13.3 A finite algorithm for all coefficients

The included author script uses the following exact operations to find  $c_{r,0}, \dots, c_{r,N}$ .

1. Set  $z = m^{-1/2}$  and  $s = s_r + uz$ . Expand  $(s - H_r z^2)(\sum_{j=0}^N \sigma_j(s) z^{2j})^r$  through degree  $2N$  in  $z$ .
2. Multiply by the truncated exponential of the higher phase and Stirling terms

$$\sum_{k \geq 3} \frac{r(-1)^{k-1}}{k} (u/s_r)^k z^{k-2} - r \sum_{k \geq 1} \frac{B_{2k}}{2k(2k-1)} z^{4k-2},$$

retaining only the terms needed through  $z^{2N}$ .

3. Apply normalized Gaussian moments:  $u^{2j} \mapsto (2j-1)!!/\lambda_r^j$ ,  $u^{2j+1} \mapsto 0$ . Divide by  $s_r(1-s_r)^{-r}$  to obtain relative coefficients  $d_{r,j}$  in powers of  $m^{-1}$ .
4. Put  $p = (3-r)/2$ . The coefficients in  $b$  are the coefficients of  $q^j$  in

$$\sum_{j=0}^N d_{r,j} q^j (1+q)^{p-j} \pmod{q^{N+1}}, \quad q = b^{-1}.$$

All these quantities are rational except for the already factored leading constant. In particular the first-order calculation gives

$$d_{r,1} = -\frac{r^2(r+1)}{2} + \frac{23r}{12} + \frac{13}{12r} - \frac{r+1}{r} H_r,$$

and  $c_{r,1} = d_{r,1} + p$ , proving (47). The explicit coefficients in (48)–(49) are independently reproduced as described in Section 15.

## 14 Reciprocation and the limits of exponential resolution

**Corollary 14.1.** *Use the combinatorial identification  $\mu_b = T_b^{-1}$  from the main report (Theorem 1.1), and write  $\mu_\infty = 1/\tau = 6/\pi^2$ . For every fixed  $K \geq 1$ ,*

$$\mu_\infty - \mu_b = \tau^{-2} \sum_{r=1}^K A_{r,b} + \tilde{\mathcal{R}}_{K,b}, \quad (57)$$

where

$$\tilde{\mathcal{R}}_{K,b} = \tau^{-2} A_{K+1,b} \left[ 1 + O_K \left( b^{-1/2} \left( \frac{\beta_{K+2}}{\beta_{K+1}} \right)^b \right) \right]. \quad (58)$$

Thus  $\tilde{\mathcal{R}}_{K,b} > 0$  for all sufficiently large  $b$  for each fixed  $K$ . No all- $b$  positivity of this reciprocal remainder is asserted.

*Proof.* Put  $\delta_b = T_b - \tau$ . The exact identity

$$\mu_\infty - \mu_b = \frac{\delta_b}{\tau^2} - \frac{\delta_b^2}{\tau^2(\tau + \delta_b)} \quad (59)$$

shows the scope of the linear sector transfer. Theorem 12.1 with  $K = 1$  and (50) give  $\delta_b = O(b2^{-b})$ , so the second term is  $O(b^2 4^{-b})$ . For fixed  $K$ , this is exponentially smaller even than  $A_{K+2,b}$ , since  $\beta_{K+2} > e^{-1} > 1/4$ . Substituting (43) in (59) therefore gives (58). This proof uses no expansion with a growing number of sectors.  $\square$

In particular,

$$\frac{6}{\pi^2} - \mu_b \sim \frac{9(b+1)}{\pi^4 2^b}.$$

Within each fixed primary sector, the coefficients of the deficit are  $\tau^{-2} C_r c_{r,j}$ . The exact reciprocal correction in (59) still exists on smaller exponential scales and must not be dropped in a purported complete transseries.

**The essential scale warning.** Theorems 12.1 and 14.1 require exact sectors, or approximations with errors controlled on the next exponential scale. If each  $A_{r,b}$ ,  $1 \leq r \leq K$ , is replaced by only  $N_r + 1$  terms of (46), one must add the error

$$O \left( \sum_{r=1}^K b^{(3-r)/2 - N_r - 1} \beta_r^b \right). \quad (60)$$

For any fixed truncation order, such a remainder in an earlier nonterminating sector generally exceeds all later exponential sectors. For example, subtracting only the displayed four terms of  $A_{2,b}$  does not isolate  $A_{3,b}$ . Subtracting the exact sector does. The exact first sector is a useful exceptional simplification, not a cure for this issue in the later sectors.

The word “transseries” here describes an exact positive-sector decomposition with fixed-sector all-orders expansions. It does not assert optimal truncation, summability, uniform estimates for  $r = r(b)$ , a justified sum of all sector asymptotic expansions, or a theorem at the accumulation scale  $e^{-b}$ . Reciprocation and any subsequent inversion must preserve these distinctions.

[write, 2 October 2026] The word is used here in this narrow sense only. This Part constructs no transseries in the sense of the repository’s transseries volume ([Analysis/Transseries/](#)), no Borel sum and no Stokes data, which is why it stays beside Part I in this collection.

## 15 Independent checks and reproducibility

The fixed-sector derivation [8] received an independent PASS audit [9] before this addendum was assembled. That audit verified the exact positive remainder, localization and uniform tail expansion, every leading constant, and the coefficients through order  $b^{-3}$ . It is an audit of the integral analysis; the growth-constant identification remains the separate theorem in the main report.

The independent coefficient calculation expands an exact tail-index sum against independent geometric moments, rather than importing the author’s local Gaussian calculation. It reproduces the coefficient arrays for  $r = 1, 2, 3, 4$  through degree three. A third included script checks  $r = 1, 2, 3$  using Gamma-centered moments. These are computational cross-checks of the finite coefficients; the all-orders remainder is justified by the proof above.

For numerical independence, the audit expands

$$R_b(v)^r = \sum_{k=0}^r (-1)^k \binom{r}{k} e^{(r-k)v} E_b(v)^k.$$

If  $E_b(v)^k = \sum_n d_{k,n} v^n / n!$ , exact binomial convolution gives integers  $d_{k,n}$  and the finite rational identity

$$J_{r,b} = \sum_{k=0}^r (-1)^k \binom{r}{k} \sum_n d_{k,n} \frac{(n+1) - (k+1)H_r}{(k+1)^{n+2}}. \quad (61)$$

Exact rational arithmetic avoids cancellation and uses no quadrature. Its normalized values for  $r = 2, 3$  and  $b = 50, 100, 200$  agree with the author’s saved values to at least 44 decimal places. The script also checks the derivative formula for  $r = 1, \dots, 6$ ,  $J_{1,b} = m/2^{m+1}$  for  $b = 0, \dots, 30$ , and three exact  $A_1$  values. The full author quadrature run additionally covers  $b = 500$  and exact  $A_2, A_3$  values; it reproduces the saved JSON byte for byte in the recorded environment.

| $b$ | $J_{2,b}/(C_2 b^{1/2} \beta_2^b)$ | $J_{3,b}/(C_3 \beta_3^b)$ |
|-----|-----------------------------------|---------------------------|
| 50  | 0.945325812081121543              | 0.807327309466966331      |
| 100 | 0.969943924143225415              | 0.886287443120262425      |
| 200 | 0.984129422497738385              | 0.936943942102952663      |
| 500 | 0.993420049029619244              | 0.972871415163468681      |

These decimals illustrate convergence. The quadratures are high-precision numerical checks, not interval-certified bounds or substitutes for the proof.

**Replay.** The ZIP contains this PDF and editable TeX, the unmodified derivation and audit sources, the check scripts, their reference results, and provenance hashes. With Python 3, SymPy 1.14.0, and mpmath 1.3.0 installed, run `./verify.sh`; this writes fresh results to `checks/replay/`, checks all assertions, and compares them with the recorded outputs. Run `./build.sh` to rebuild the PDF with PDFLaTeX. Run `sha256sum -c SHA256SUMS` before changing supplied files. PDF metadata uses a fixed date; byte identity across different TeX installations is not promised. The package is offline-replayable after dependencies are installed.

[**write, 2 October 2026**] This paragraph describes the delivered archive, which survives in commit 096ee7b87. In this repository the scripts are `code/52-sectors-*`, the saved results `data/52-sectors-*`, the derivation and audit `52-sectors-fixed-taylor-sectors.md` and `52-sectors-fixed-sector-independent-audit.md`; `verify.sh` uses the delivered layout and

must not be run here. The README gives a rerun recipe. At intake (2 October 2026) the independent audit, the Gamma-moment check, the reciprocal check and the exact coefficients of the author script reproduced the saved results; the author’s 45-digit quadratures were not rerun.

## 16 Questions left open

1. How do the fixed-sector bounds behave as  $r$  grows with  $b$ ? The saddle  $r/(r+1)$  approaches the endpoint 1, where the present compact-interval expansion loses uniformity. Identifying the correct joint regime is necessary before analyzing the accumulation near  $e^{-b}$ .
2. Can an optimally truncated or summable expansion for an earlier sector give a certified error below the next sector, and how do derivative exponentials and reciprocal products enter a complete smaller-scale description?
3. Can the refined exact-sector bounds yield a sharper, rounding-safe inverse for the cap at a prescribed deficit in  $\mu_b$ ? Such an inverse needs explicit, appropriately scaled errors; finite coefficient substitution alone does not establish it.
4. For the original counting problem, can one determine fixed-cap prefactors or normalized ratio limits, and then obtain estimates uniform in a growing cap? None of the constant-level expansions here resolves those length-asymptotic questions.

[**write, 5 October 2026**] The normalized ratio limits of Question 4 are proved for every fixed cap in Part III, Theorem 17.1. Fixed-cap prefactors and estimates uniform in a growing cap remain open. Questions 1–3 are restated in Part III, Section 26, whose envelopes  $R_K$  (Proposition 24.5) are a partial answer to Question 3: a sharper cap inverse, eventual rather than certified.

## Part III

# Ascent sequences with bounded multiplicities

*Source.* Bundle Report 99 (batch 102), archive

`Ascent_Sequences_with_Bounded_Multiplicities_Source.zip` (arrival commit 60f54ea06, placed by 6ab1f1979), delivered as `article.tex` (63 lines) with eleven section files, whose expansion `article_standalone.tex` (1,407 lines) is the text printed here; the delivered PDF has 28 pages. Dated 2 October 2026; author line “Research report prepared with OpenAI”. No commit pin, and no mention of this repository or of Parts I and II. Delivered Section  $k$  is Section  $k+16$  here, delivered equation  $(k)$  is  $(k+61)$ , and statement numbers follow their sections. Printed in full, with labels prefixed **amc:bm:** and dated [**write, 5 October 2026**] notes. No symbol was renamed: here the cap is  $r$ , the  $b$  of Parts I and II.

**Abstract (as delivered).** Let  $c_{N,r}$  count ascent sequences of length  $N$  in which every value occurs at most a fixed integer  $r \geq 2$  times. We prove both the factorial-normalized root limit and the consecutive-ratio limit, with common value

$$\mu_r = \left[ \int_0^\infty \frac{\log E_r(w)}{E_r(w) - 1} dw \right]^{-1}, \quad E_r(w) = \sum_{j=0}^r \frac{w^j}{j!}.$$

The proof uses an exact sorted residual-capacity recurrence, finite positive matrices, and a constructive lower bound from buffered distinct labels and Eulerian permutations. It gives polynomial-time enumeration for fixed  $r$ , exponential concentration of multiplicities and ascents, a full endpoint large-deviation principle, and a Lambert  $W$  inverse for large counting thresholds. A separate analysis of the constants as  $r \rightarrow \infty$  gives a positive Taylor-sector hierarchy and a sharp approach to  $6/\pi^2$ . Complete ordinary proofs and reproducible exact checks are supplied. The word-length limit holds for each fixed cap; the sector expansions concern the cap parameter alone. No coefficient prefactor, complete word-length asymptotic equivalent, or uniform growing-cap theorem is claimed.

[**write, 5 October 2026**] *Reading this Part.* Report 99 was written without knowledge of Parts I and II and proves what it needs for every fixed cap  $r \geq 2$ . Its letters differ from theirs (Guide, “Notation”): the cap is  $r$  (there  $b$ );  $R = r + 1$  is Part II’s  $m = b + 1$ ; the sector index is  $m$  (there  $r$ ); the sector  $T_m(R)$  is Part II’s  $A_{m,b}$  at  $R = b + 1$ , while Part I’s  $T_b$  is this Part’s  $J_r$ ;  $A = \zeta(2)$  is their  $\tau$ , while  $\tau$  here is a time variable or a threshold; and  $b_j$  are endpoint limits and  $b$  a block length, never a cap.

- *Second routes.* Lemma 18.1 and Proposition 18.2 re-prove Lemma 2.1 and its sorted reduction. The root half of Theorem 17.1 re-proves Theorem 1.1 with the weaker error  $o_r(N)$ , and covers  $r = 2$ . Proposition 17.3 re-proves part of Theorem 1.2 and  $T_2 = 3\pi^2/8$ . Corollary 23.2 is weaker than (28). Section 24 re-proves the large-cap results of Section 7 and Part II in the variable  $R = b + 1$ .
- *New.* The ratio limit for every fixed  $r \geq 3$ , which answers Question 2 of Section 10; the endpoint laws (Theorem 17.2); the weighted growth formula and the large-deviation principle (Theorems 22.1 and 22.2); the histogram refinement and the  $O_r(N^{r+2})$  algorithm (Corollary 18.3); the continuous cap, the exact-sector envelopes and their displacements, and the selection criterion of Corollary 24.6.
- *The cap two.* The case  $r = 2$  of this Part is bundle Report 97, delivered in the same bundle with the same date, author line and method, and printed as Part V of the sibling report **a202058-ascent-000-growth**. Report 97 adds what is special to that cap: a two-catalytic functional equation, closed forms of the endpoint constants, and exact coefficients through  $n = 395$ .

## 17 Definitions and principal results

An *ascent sequence* of length  $N \geq 1$  is a word  $a_1 \cdots a_N$  of nonnegative integers such that  $a_1 = 0$  and

$$a_i \leq 1 + \text{asc}(a_1 \cdots a_{i-1}) \quad (2 \leq i \leq N), \quad \text{asc}(a_1 \cdots a_k) = \#\{i < k : a_i < a_{i+1}\}.$$

Let  $c_{N,r}$  count these words with each value occurring at most  $r$  times, and set  $c_{0,r} = 1$ . Throughout the combinatorial proof,  $r \geq 2$  is fixed. The word length is denoted by  $N$ ; a separate cap index  $R = r + 1$  will be introduced explicitly in Section 24.

Put

$$E_j(w) = \sum_{k=0}^j \frac{w^k}{k!}, \quad g(v) = \frac{\log v}{v-1}, \quad g(1) = 1, \quad J_r = \int_0^\infty g(E_r(w)) dw, \quad \mu_r = J_r^{-1}. \quad (62)$$

The integral is positive and finite: its integrand is continuous at zero and is  $O_r(\log w/w^r)$  at infinity.



**Theorem 17.1.** *For every fixed integer  $r \geq 2$ ,*

$$\lim_{N \rightarrow \infty} \left( \frac{c_{N,r}}{N!} \right)^{1/N} = \lim_{N \rightarrow \infty} \frac{c_{N+1,r}}{(N+1)c_{N,r}} = \mu_r. \quad (63)$$

*In particular  $\log c_{N,r} = N \log N + N(\log \mu_r - 1) + o_r(N)$ .*

[write, 5 October 2026] The root half re-proves Part I's Theorem 1.1 for every  $r = b \geq 3$ , where Part I has the stronger error  $O_b(n/\log n)$ , and, at  $r = 2$ , the root limit of Part I of the sibling report `a202058-ascent-000-growth` (label `a58:fd:thm:main` there, error  $O(n^{9/13}(\log n)^3)$ ). The proof here covers  $r = 2$  with the other caps; Part I's barrier proof excludes it. The ratio half is new for every fixed  $r \geq 3$ . It answers Question 2 of Section 10 and supersedes the statements of Part I and of the Guide that no ratio limit is known for  $b \geq 3$ . At  $r = 2$  it is a second proof, without a rate, of the sibling's Part IV (label `a58:rt:thm:main`, rate  $O(n^{-1/4}(\log n)^{3/4})$ ). Report 97, printed as the sibling's Part V, proves the case  $r = 2$  by the same terminal-mark argument (Theorem 46.1 there).

**Theorem 17.2.** *For a uniform word counted by  $c_{N,r}$ , let  $N_k$  be the number of values occurring exactly  $k$  times, let  $D$  be its number of distinct values, and let  $A$  be its ascent count. Define*

$$b_j = \frac{\mu_r}{(r-j)!} \int_0^\infty [-g'(E_r(w))] w^{r-j} dw \quad (1 \leq j \leq r). \quad (64)$$

*Then  $b_j > 0$ ,  $\sum_j b_j = \mu_r$ , and*

$$\begin{aligned} \frac{N_k}{N} &\longrightarrow b_{r-k} & (1 \leq k < r), \\ \frac{N_r}{N} &\longrightarrow \frac{1 - \sum_{j < r} (r-j)b_j}{r}, \\ \frac{D}{N} &\longrightarrow \frac{1 + \sum_{j < r} j b_j}{r}, & \frac{A}{N} &\longrightarrow b_r + \frac{1 + \sum_{j < r} j b_j}{r}. \end{aligned} \quad (65)$$

*Every displayed convergence holds in mean and in probability, with exponentially small probabilities for any fixed positive deviation.*

Here “exponentially small” means that for each fixed deviation  $\eta > 0$  there are  $c > 0$  and  $N_0$ , depending on  $r$  and  $\eta$ , such that the relevant probability is at most  $e^{-cN}$  for  $N \geq N_0$ .

[write, 5 October 2026] *The case  $r = 2$  in closed form.* For  $r = 2$ ,

$$b_1 = 1 - \frac{8}{3\pi}, \quad b_2 = \frac{8(\pi+1)}{3\pi^2} - 1, \quad \frac{D}{N} \rightarrow 1 - \frac{4}{3\pi}, \quad \frac{A}{N} \rightarrow \frac{4}{3\pi} + \frac{8}{3\pi^2}.$$

These are the closed forms of Report 97, Corollary 46.4 and equation (276) of the sibling report `a202058-ascent-000-growth`. *Proof.* At  $r = 2$  the polynomial of (77) is  $P_{\mathbf{u}}(s) = s^2/2 + u_1 s + u_2$ , so  $\chi(u_1, u_2) = \int_0^\infty g(u_2 + u_1 s + s^2/2) ds$  is Report 97's  $\chi(s, t)$  with  $(s, t) = (u_1, u_2)$  (the second form of its equation (281)). Section 22 gives  $b_j = -\mu_2 \chi_{u_j}(1, 1)$ , and Report 97 evaluates this gradient as  $(\mu_2(1+K), -\mu_2 K)$  with  $K = 3\pi^2/8 - \pi - 1$  and  $\mu_2 = 8/(3\pi^2)$ . The last two limits follow from the formulas of Theorem 17.2. A 30-digit quadrature of (64) agrees (checked at the write). Here  $b_1$  is the limiting proportion of values used once ( $X_1 = N_1$ ) and  $b_2$  that of available unused values ( $X_2$ ). The endpoint laws are new in this report: Parts I and II have no statement about the shape of a typical word.

**Proposition 17.3.** *The constants  $\mu_r$  increase strictly to  $6/\pi^2$ . For  $r = 2$ ,*

$$\mu_2 = \frac{8}{3\pi^2}.$$

*For  $r = 1$  the only word of each length is  $01 \cdots (N-1)$ , so the factorial-normalized root limit is zero.*

*Proof.* The identity

$$g(v) = \int_0^1 \frac{dt}{1+t(v-1)} \quad (66)$$

shows that  $g$  is strictly decreasing. Hence  $E_{r+1}(w) > E_r(w)$  for  $w > 0$  implies  $J_{r+1} < J_r$ . The integrable function  $g(E_2(w))$  dominates all subsequent integrands. Dominated convergence and then Tonelli give

$$J_r \longrightarrow \int_0^\infty \frac{w}{e^w - 1} dw = \sum_{k \geq 1} \int_0^\infty w e^{-kw} dw = \sum_{k \geq 1} \frac{1}{k^2} = \frac{\pi^2}{6}.$$

For the evaluation at  $r = 2$ , substitute  $s = w + 1$  and define

$$I(a) = \int_1^\infty \frac{2 \log((s^2 + a)/(1 + a))}{s^2 - 1} ds.$$

The substitution  $z = 1/s$  and a geometric-series integral yield  $I(0) = \pi^2/2$ . Differentiation under the integral gives

$$\frac{d}{dv} I(v^2) = -\frac{4 \arctan v}{1 + v^2}.$$

Thus  $I(a) = \pi^2/2 - 2(\arctan \sqrt{a})^2$ , and  $J_2 = I(1) = 3\pi^2/8$ . Finally, distinctness forces each next entry in an ascent sequence to be its next new maximum; induction proves the  $r = 1$  statement.  $\square$

[**write, 5 October 2026**] Second routes. The strict monotone convergence  $\mu_r \uparrow 6/\pi^2$  is part of Theorem 1.2, proved in Section 7 by the same dominated convergence. Part I takes  $T_2 = 3\pi^2/8$  from the sibling report; the evaluation of  $J_2$  given here is Report 97's Lemma 44.2 there, by the same substitution. In the table,  $\mu_2, \dots, \mu_6$  and  $\mu_8$  agree with Section 9 to all printed digits, and  $\mu_7$  is new; at intake  $\mu_2, \dots, \mu_8$  were recomputed to 18 digits.

| Cap $r$ | $\mu_r$ | numerical quadrature |
|---------|---------|----------------------|
| 2       |         | 0.270189823046234057 |
| 3       |         | 0.452065717728321870 |
| 4       |         | 0.534247641664251966 |
| 5       |         | 0.571991298656956259 |
| 6       |         | 0.589970111415399178 |
| 7       |         | 0.598798801567128511 |
| 8       |         | 0.603232139970701640 |

These decimal values illustrate the exact integral formula and are not used as proof.

## 17.1 Relation to existing work

Ascent sequences were introduced in the study of Fishburn objects; see Bousquet-Mélou, Claesson, Dukes and Kitaev [2]. Bounded multiplicity is equivalent to avoiding a constant subsequence of length  $r + 1$ . The whole array is OEIS A294220 [4];  $r = 2$  and  $r = 3$  are A202058 and A317784, respectively. Duncan and Steingrímsson [3] studied pattern avoidance in ascent sequences. Conway, Conway, Elvey Price and Guttmann [1] gave a polynomial algorithm for  $r = 2$  and conjectured the constant  $8/(3\pi^2)$  numerically.

The adjacent-label idea in their Section 2.6 is the starting point here. We prove that it works for arbitrary residual capacities, and supply finite positive estimates that identify the growth constants. The arguments below are self-contained; no separate report for  $r = 2$  is needed. A literature search through 2 October 2026 located the array and those antecedents, but no general theorem of the form (63). This limited search does not certify a priority claim.

[write, 5 October 2026] *Stale at its date.* The root half of (63) was already proved for every fixed  $b \geq 3$  by Part I of this report (source 47, dated 1 October 2026, in this repository since aa7345800 of 2 October 2026), and for  $b = 2$  by Part I of a202058-ascent-000-growth (also dated 1 October 2026). Report 99 cites neither. Its ratio half was new for every  $r \geq 3$ ; for  $r = 2$  the sibling's Part IV, dated 2 October 2026, proves the ratio limit with a rate. The sentence is kept as the record of Report 99's search. "No separate report for  $r = 2$  is needed" remains true: the proof of this Part covers  $r = 2$ .

## 17.2 Logical structure and scope

Section 18 proves exact compaction. Sections 19–21 establish the root limit using a finite upper matrix and an explicit lower construction. Section 22 obtains the ratio and endpoint laws from two-sided terminal marks. Section 23 inverts the word-length growth. Section 24 analyzes the different limit  $r \rightarrow \infty$  of the constants themselves. The verification package checks finite identities, independently of the asymptotic arguments. No interchange of  $N \rightarrow \infty$  and  $r \rightarrow \infty$  is made.

## 18 Exact sorting of residual capacities

After a nonempty prefix, the allowed original labels are  $0, \dots, A + 1$ , where  $A$  is its ascent count. Delete labels already used  $r$  times, and renumber the remaining labels in increasing order. Give each remaining label its residual capacity in  $\{1, \dots, r\}$ . Capacity  $r$  means unused. Let  $L$  be the number of surviving labels less than or equal to the previous last label. A next compact label  $i$  creates an ascent exactly when  $i \geq L$ .

Selecting capacity  $j$  decreases that capacity by one. For  $j = 1$  the label is deleted and the new threshold is  $i$ ; for  $j \geq 2$  it survives and the new threshold is  $i + 1$ . Every ascent adds a new capacity- $r$  label at the right. These rules describe the original process exactly before any sorting.

**Lemma 18.1** (Adjacent exchange). *Suppose two adjacent available labels have compact indices  $i, i + 1$  with  $i + 1 < L$ . Exchanging their residual capacities preserves the number of legal suffixes of every length, refined by the final residual-capacity histogram and the total number of new ascents.*

*Proof.* Temporarily keep exhausted labels as permanently forbidden positions rather than deleting them. The two selected labels are adjacent among usable labels; any intervening forbidden original labels cannot occur in a future suffix. Both selected labels are already available and are at most the previous last label.

In any legal suffix, find each maximal contiguous run containing only these two labels. Reverse the run and interchange the two labels. Reversal followed by complementation preserves its number of internal ascents. Comparisons at its ends with a label outside the pair are unchanged, because every other usable label lies below both or above both. At the initial boundary neither selected label can make an ascent. Therefore the total new ascent count is preserved.

Inside a transformed run, the intermediate ascent counts can differ. This causes no illegality: both run labels were already available initially and availability only increases. On exiting the run, accumulated ascent counts agree, so every subsequent external label has the same bound as before. The usages of the two selected labels are exchanged. Exchanging their initial residual capacities therefore preserves the final capacity constraints; partial usages cannot exceed their total usages. The final residual capacities are exchanged as well, preserving their histogram. The transformation is an involution, so it is a bijection. Deleting forbidden labels afterward merely restores the compact indexing.  $\square$

**Proposition 18.2** (Sorted recurrence). *Let  $X_j$  count labels of residual capacity  $j$ , and put  $C_j = \sum_{k \leq j} X_k$ ,  $C_0 = 0$ . Starting from*

$$\mathbf{X} = \mathbf{e}_{r-1} + \mathbf{e}_r, \quad L = 1$$

*after the initial zero, the exact transfer consists, for every  $j$  and every integer  $C_{j-1} \leq i < C_j$ , of*

$$\mathbf{X}' = \mathbf{X} - \mathbf{e}_j + \mathbf{1}_{\{j \geq 2\}} \mathbf{e}_{j-1} + \mathbf{1}_{\{i \geq L\}} \mathbf{e}_r, \quad L' = i + \mathbf{1}_{\{j \geq 2\}}. \quad (67)$$

*It counts  $c_{N,r}$  and preserves its full terminal capacity histogram.*

*Proof.* Assume capacities are sorted increasingly. If capacity  $j \geq 2$  is selected at index  $i$ , its new capacity is  $j - 1$ . Bubble it left through earlier members of its old category  $j$  until it reaches the category boundary. Each exchanged pair has upper index at most  $i$ , while the new threshold is  $i + 1$ . Lemma 18.1 applies to every exchange, leaving the threshold fixed. Thus the suffix count is unchanged and the capacities are sorted again. For  $j = 1$  deletion already preserves sorting; any fresh capacity- $r$  label belongs at the right. The initial state is sorted. Backward induction on suffix length proves the transfer, and the same induction with histogram refinements proves its marginal interpretation.  $\square$

The threshold remains a comparison marker; it does not follow the capacity being moved during the suffix-count equivalence. This distinction prevents an incorrect literal-history interpretation of the sorting step.

At level  $N$ , write  $N_k$  for the original multiplicity counts and  $D = \sum_k N_k$ . The histogram refinement gives

$$X_j = N_{r-j} \ (j < r), \quad X_r = A + 2 - D, \quad N = rN_r + \sum_{j < r} (r - j)X_j. \quad (68)$$

Also  $M = \sum_j X_j \leq N + 1$ , every coordinate changes by at most one in a step, and each cumulative boundary  $C_j$  changes by at most one. For  $j < r$  the latter change is  $\mathbf{1}_{\{\text{category } j+1\}} - \mathbf{1}_{\{\text{category } 1\}}$ ; for  $j = r$  it is  $\mathbf{1}_{\{\text{ascent}\}} - \mathbf{1}_{\{\text{category } 1\}}$ .

**Corollary 18.3** (Polynomial enumeration). *For fixed  $r$ , the coefficients through length  $N$  can be computed using  $O_r(N^{r+2})$  arithmetic operations and  $O_r(N^{r+1})$  memory.*

*Proof.* At level  $k$ , there are  $O_r(k^r)$  possible vectors  $\mathbf{X}$  with coordinate sum at most  $k+1$ , each with  $O(k)$  last thresholds. For each vector form prefix sums of its counts over  $L$ . A choice  $i$  sends the prefix sum over  $L \leq i$  to its ascent endpoint in (67), and the complementary sum to its nonascent endpoint. There are  $O(k)$  choices over all categories together. Thus one vector costs  $O_r(k)$ , one level costs  $O_r(k^{r+1})$ , and all levels cost  $O_r(N^{r+2})$ . Only two consecutive levels are stored. This counts arithmetic operations rather than bit operations on integers of increasing size.  $\square$

[write, 5 October 2026] New: Parts I and II state no complexity bound, and Part I's sorted reduction counts words without the histogram refinement (68). At  $r = 2$  the state  $(X_1, X_2, L)$  and the transfer (67) are exactly Report 97's state  $(m, h, L)$  and its repeat and new transitions (sibling report, Section 43): a capacity-one label is a once-seen value, a capacity-two label an unused one. The corollary therefore proves the count of  $O(N^4)$  operations and  $O(N^3)$  stored entries that Report 97 states in its Section 49 without proof.

## 19 Finite upper and lower block estimates

The two bounds in this section count finite collections of paths. Their asymptotic use later requires no limiting eigenvalue theorem or path large-deviation principle.

### 19.1 An explicit positive matrix

**Lemma 19.1.** *Let  $v > 0$ ,  $w_0, \dots, w_D \geq 0$ ,  $W = \sum_z w_z > 0$ ,  $w_* = \max_z w_z$ , and  $F(v) = (v-1)/\log v$ , with  $F(1) = 1$ . Define*

$$K_{\geq}(L, z) = w_z v^{\mathbf{1}_{\{z \geq L\}}}, \quad K_{>}(L, z) = w_z v^{\mathbf{1}_{\{z > L\}}}.$$

*Use  $K = K_{\geq}$  if  $v \geq 1$  and  $K = K_{>}$  if  $v < 1$ . Then, for every integer  $b \geq 0$ ,*

$$K^b \mathbf{1} \leq \max(v, v^{-1}) \Lambda^b \mathbf{1}, \quad \Lambda \leq F(v)W + |v-1|w_*. \quad (69)$$

*Proof.* Suppose first that  $v > 1$ , and put  $c = v - 1$ . The equation

$$\prod_{z=0}^D (1 - cw_z/\lambda) = 1/v, \quad \lambda > cw_*$$

has one solution, since the product increases continuously from zero to one. Set  $f_0 = 1$  and  $f_{z+1} = (1 - cw_z/\lambda)f_z$ . Then  $f_{D+1} = 1/v$  and  $1/v \leq f_z \leq 1$ . Telescoping gives  $\sum_z w_z f_z = \lambda/v$ ; the zeroth row and successive row differences now give  $K_{\geq} f = \lambda f$ . Positivity yields  $K_{\geq}^b \mathbf{1} \leq v \lambda^b \mathbf{1}$ .

At  $B = cW/\log v + cw_*$ , the inequality  $-\log(1-t) \leq t/(1-t)$  gives

$$\sum_z -\log(1 - cw_z/B) \leq \frac{cW}{B - cw_*} = \log v.$$

Monotonicity puts  $B$  at or above the root, proving the stated bound for  $\lambda$ .

If  $0 < v < 1$ , put  $d = 1 - v$ . The product  $\prod_z (1 + dw_z/\lambda)$  decreases continuously from infinity to one, so its equation with  $1/v$  again has one root. The same recurrence, now  $f_{z+1} = (1 + dw_z/\lambda)f_z$ , gives  $1 \leq f_z \leq 1/v$  and  $K_{\geq} f = \lambda f$ . For  $B = dW/\log(1/v) = F(v)W$ ,

$$\sum_z \log(1 + dw_z/B) \leq dW/B = \log(1/v),$$

so  $\lambda \leq B$ . Since  $K_{>} = K_{\geq} + d \operatorname{diag}(w_z)$ ,

$$K_{>} f \leq (\lambda + dw_*) f.$$

The bounds on  $f$  give (69). At  $v = 1$  the matrix has identical rows and the result follows with  $\Lambda = W$ .  $\square$

[**write, 5 October 2026**] This lemma, for  $v \geq 1$  and for  $v < 1$ , is the finite Perron estimate of Report 97 (Lemmas 44.1 and 46.2 of the sibling report), there written for the cap two. Part I reaches the root limit by another method: positive-operator barriers and a stopped Feynman–Kac comparison (Sections 3–5).

## 19.2 Uniform domination of every tilted block

Use arbitrary positive marks  $u_1, \dots, u_r$ , and set  $u_0 = 1$ ,  $v = u_r$ ,  $c_j = u_{j-1}/u_j$ . A category- $j$  transition has weight  $c_j v^{\mathbf{1}_{\{i \geq L\}}}$ ; the product over a path is  $\mathbf{u}^{\mathbf{X}' - \mathbf{X}}$ .

Consider  $b$  steps starting at  $\mathbf{X}$ , and use its initial cumulative boundaries  $C_j$ . Every output threshold  $z$  of category  $j$  lies in the fixed interval

$$I_j = [C_{j-1} - b + \mathbf{1}_{\{j \geq 2\}}, C_j + b - 1 + \mathbf{1}_{\{j \geq 2\}}] \cap \mathbb{N}_0.$$

Its size is at most  $X_j + 2b + 1$ . This follows from the one-step cumulative-boundary bound after (68). Use a finite matrix index range containing every initial and output threshold, and set  $w_z = \sum_j c_j \mathbf{1}_{\{z \in I_j\}}$ .

For category 1 the exact ascent indicator is  $[z \geq L]$ ; for every other category it is  $[z > L]$ . Thus the weak-threshold matrix dominates when  $v \geq 1$ , and the strict-threshold matrix dominates when  $v < 1$ . If two category intervals overlap, their weights add; this simply allows extra positive paths. Backward induction on the number of remaining steps proves domination uniformly over every hidden category vector with a given threshold. Lemma 19.1 therefore gives

$$\sum_{\text{block paths}} \mathbf{u}^{\mathbf{X}' - \mathbf{X}} \leq \max(v, v^{-1}) \Lambda^b, \quad \Lambda \leq N \rho(\mathbf{X}/N; \mathbf{u}) + C(b+1), \quad (70)$$

where

$$\rho(\mathbf{x}; \mathbf{u}) = \frac{1}{g(u_r)} \sum_{j=1}^r \frac{u_{j-1}}{u_j} x_j. \quad (71)$$

Here  $C$  is uniform on a fixed compact set of positive marks, since  $\sum_z w_z \leq \sum_j c_j (X_j + 2b + 1)$  and  $w_* \leq \sum_j c_j$ . Boundary states with zero coordinates cause no difficulty.

## 19.3 Buffered distinct labels

Write  $A(m, k)$  for the number of permutations of  $m$  objects with  $k$  ascents.

**Lemma 19.2.** *Start at a sorted state with  $L \leq X_1$ . Fix  $b \geq 2$  and integers  $q_j \geq 0$  with  $\sum_j q_j = b$ ,  $q_1, q_r \geq 1$ . Provided the binomial arguments are large enough, for every  $0 \leq k \leq b - 3$  there are at least*

$$A(b-2, k) \prod_{j=1}^r \binom{X_j - 2b}{q_j} \quad (72)$$

paths with endpoint increments

$$\Delta X_j = q_{j+1} - q_j \ (j < r), \quad \Delta X_r = k + 1 - q_r, \quad (73)$$

and with final threshold  $L' \leq X'_1$ . The same statement uses  $A(0, 0) = 1$  when  $b = 2$ .

*Proof.* Choose  $q_j$  distinct integer labels from  $C_{j-1} + b, \dots, C_j - b - 1$ . Each remains in category  $j$  whenever it is selected, because every boundary has moved by at most  $b$ . This argument concerns integer positions in the sorted transfer, so it remains true after capacities are rearranged. All selected labels are distinct.

Place the largest selected label first and the smallest last. The first belongs to category  $r$  and makes an ascent since  $L \leq X_1$ . Every later comparison of distinct labels is an ascent precisely when the next integer is larger than the preceding integer: the previous threshold equals that integer or that integer plus one. The first-to-middle and middle-to-last comparisons are descents. Hence a middle permutation with  $k$  ascents gives exactly  $k + 1$  block ascents. All such paths have the increments (73). The last label belongs to category 1 and is less than the initial  $X_1 - b$ , whereas  $X'_1 \geq X_1 - b$ , so  $L' \leq X'_1$ . Multiplying the independent label choices by the Eulerian count proves (72).  $\square$

There is no sign restriction on the increments in this lemma. Positive drifts will only be needed for repeating a fixed initial block.

#### 19.4 Eulerian entropy without a local limit theorem

Let  $U$  be uniform on  $[0, 1]$ , and define

$$M(t) = \mathbb{E}e^{tU} = \frac{e^t - 1}{t}, \quad M(0) = 1, \quad I(a) = \sup_{t \in \mathbb{R}} \{ta - \log M(t)\}, \quad 0 < a < 1.$$

The tilted mean  $M'(t)/M(t)$  increases strictly from zero to one, since its derivative is the positive tilted variance. Thus there is a unique tilt with mean  $a$ .

**Lemma 19.3.** *For every fixed  $a \in (0, 1)$  there are integers  $k_m = ma + O(m^{2/3})$  such that*

$$\log A(m, k_m) \geq \log(m!) - mI(a) + o(m). \quad (74)$$

*Proof.* For independent uniforms  $U_1, \dots, U_m$  with sum  $S_m$ , the cube-slice identity is

$$\mathbb{P}(k \leq S_m \leq k + 1) = A(m, k)/m!.$$

Here is an algebraic verification. Inclusion–exclusion for the unit cube inside a simplex gives

$$\text{vol}\{S_m \leq t\} = \frac{1}{m!} \sum_{j=0}^m (-1)^j \binom{m}{j} (t - j)_+^m.$$

Subtracting the values at  $k + 1$  and  $k$  gives  $m!^{-1} \sum_{j=0}^{k+1} (-1)^j \binom{m+1}{j} (k + 1 - j)^m$ . This is the Eulerian formula: it has the initial value  $A(1, 0) = 1$  and, by substitution, satisfies  $A(m, k) = (k + 1)A(m - 1, k) + (m - k)A(m - 1, k - 1)$ , obtained by inserting the largest element in a permutation. This proves the identity.

Tilt each uniform by density  $e^{tu}/M(t)$ , with mean  $a$ . Its variance is finite, so Chebyshev's inequality gives probability tending to one to  $|S_m - ma| \leq m^{2/3}$ . This window meets  $O(m^{2/3})$  unit

bins. At least one has tilted probability at least a positive constant times  $m^{-2/3}$ . Returning to the original measure on that bin costs at most

$$\exp\{-tma + m \log M(t) - |t|O(m^{2/3})\}.$$

The logarithm is  $-mI(a) + o(m)$ . Multiplying by  $m!$  proves the claim. The bin is selected, not prescribed in advance.  $\square$

For positive  $\mathbf{x}$ , choice proportions  $p_j > 0$  summing to one, and  $a \in (0, 1)$ , put

$$\mathcal{L}(\mathbf{x}; \mathbf{p}, a) = \sum_j p_j \log x_j - \sum_j p_j \log p_j - I(a). \quad (75)$$

Let  $b = \delta N + O(1)$ ,  $X_j = Nx_j + o(N)$ , and  $q_j = bp_j + O(1)$  with their sum adjusted to  $b$ . Lemma 19.3 supplies a middle-permutation bin with  $k + 1 = ba + O(b^{2/3})$ . Stirling's formula and (72) give

$$\log Q \geq b \log N + b\mathcal{L}(\mathbf{x}; \mathbf{p}, a) - O_r \left( N\delta^2 \sum_j x_j^{-1} \right) + o(N). \quad (76)$$

Indeed, expanding the falling factorial in  $\binom{Nx_j - 2b + o(N)}{q_j}$  costs  $O(b^2/(Nx_j)) + o(N)$  when  $\delta$  is small compared with  $x_j$ . The difference between  $(b - 2)!$  and  $b!$  costs only  $O(\log N)$ . The estimate is uniform on positive compact subsets for a fixed finite partition of time.

## 20 The characteristic and its initial boundary

For positive  $\mathbf{u} = (u_1, \dots, u_r)$ , set  $u_0 = 1$  and define

$$P_{\mathbf{u}}(s) = \sum_{j=0}^r \frac{u_j s^{r-j}}{(r-j)!}, \quad \chi(\mathbf{u}) = \int_0^\infty g(P_{\mathbf{u}}(s)) \, ds, \quad D = \sum_{j=1}^r u_{j-1} \partial_{u_j}. \quad (77)$$

This integral and the derivatives used below are locally uniformly convergent on the positive orthant. Indeed  $P_{\mathbf{u}}(s)$  has positive leading coefficient  $1/r!$ , while its derivatives are polynomials of degree at most  $r - 1$ . Formula (66) bounds derivatives of  $g$ , or direct differentiation gives the required integrable tails. Since  $DP_{\mathbf{u}} = \partial_s P_{\mathbf{u}}$ ,

$$D\chi = -g(u_r). \quad (78)$$

Along  $u_j(w) = E_j(w)$ , the binomial identity gives  $P_{E(w)}(s) = E_r(w + s)$ . Define

$$\tau(w) = \mu_r \int_w^\infty g(E_r(z)) \, dz, \quad x_j(w) = -\mu_r E_j(w) \chi_{u_j}(E(w)). \quad (79)$$

Then  $\tau$  decreases from one to zero as  $w$  increases from zero to infinity. In particular

$$x_j(w) = \frac{\mu_r E_j(w)}{(r-j)!} \int_0^\infty [-g'(E_r(w+s))] s^{r-j} \, ds > 0. \quad (80)$$

The derivative is negative by (66), with  $g'(1) = -1/2$ .



**Lemma 20.1.** *Writing derivatives with respect to  $\tau$  by dots, define*

$$p_j = \frac{(u_{j-1}/u_j)x_j}{\mu_r g(u_r)}, \quad a = -\frac{u_r g'(u_r)}{g(u_r)} = \frac{u_r}{u_r - 1} - \frac{1}{\log u_r}, \quad (81)$$

*with  $a = 1/2$  at  $u_r = 1$ . Then every  $p_j > 0$ ,  $\sum_j p_j = 1$ ,  $0 < a < 1$ , and*

$$\dot{x}_j = p_{j+1} - p_j \quad (j < r), \quad \dot{x}_r = a - p_r. \quad (82)$$

*If  $\theta_j = \log u_j$ , then  $\theta(1) = 0$  and, for every  $\mathbf{y} \geq 0$ ,*

$$\rho(\mathbf{y}; \mathbf{u}(\tau))/\mu_r = -\dot{\boldsymbol{\theta}}(\tau) \cdot \mathbf{y}. \quad (83)$$

*On the path,  $\rho(\mathbf{x}; \mathbf{u}) = \mu_r$  and  $\dot{\boldsymbol{\theta}} \cdot \mathbf{x} = -1$ .*

*Proof.* Equation (78) gives the sum of the  $p_j$ . The value of  $a$  is the mean of a uniform variable tilted by  $t = \log u_r$ , so it lies strictly between zero and one. Differentiating (78) gives

$$D\chi_{u_j} + \chi_{u_{j+1}} = -\mathbf{1}_{\{j=r\}}g'(u_r), \quad \chi_{u_{r+1}} = 0.$$

Use  $u'_j(w) = u_{j-1}$  and  $\tau'(w) = -\mu_r g(u_r)$  to differentiate (79); the result is (82). Also

$$\dot{\theta}_j = -\frac{u_{j-1}/u_j}{\mu_r g(u_r)},$$

which proves (83). Substituting the sum of the  $p_j$  proves the last claims.  $\square$

## 20.1 The singular origin

**Lemma 20.2.** *As  $w \rightarrow \infty$ ,*

$$\tau(w) \sim \frac{\mu_r r! r}{r-1} w^{1-r} \log w, \quad (84)$$

$$\frac{x_j(w)}{\tau(w)} \longrightarrow \alpha_j = \frac{(r-1)r!(r+j-2)!}{j!(2r-1)!} > 0, \quad (85)$$

$$p_j(w) \longrightarrow \frac{j\alpha_j}{r-1}, \quad \sum_j j\alpha_j = r-1.$$

*Furthermore  $\mathbf{x}(0) = 0$ ,  $\boldsymbol{\theta}(\tau) = O_r(\log(1/\tau))$ , and  $\boldsymbol{\theta} \cdot \mathbf{x} \rightarrow 0$  as  $\tau \downarrow 0$ .*

*Proof.* The elementary estimates

$$g(E_r(z)) \sim \frac{r! r \log z}{z^r}, \quad -g'(E_r(z)) \sim \frac{(r!)^2 r \log z}{z^{2r}}, \quad E_j(w) \sim w^j/j!$$

prove (84) by integration. In (80), substitute  $s = wt$ . After division by its leading power and logarithm, dominated convergence applies with a constant multiple of  $t^{r-j}(1+t)^{-2r}(1+\log(1+t))$ , which is integrable. The beta integral gives

$$\frac{1}{(r-j)!} \int_0^\infty t^{r-j}(1+t)^{-2r} dt = \frac{(r+j-2)!}{(2r-1)!}.$$

This proves (85). Since  $u_{j-1}/u_j \sim j/w$ , the formula for  $p_j$  follows; taking limits in  $\sum p_j = 1$  gives the displayed sum. The remaining conclusions follow from  $x_j = O(w^{1-r} \log w)$ ,  $\theta_j = O(\log w)$ , and (84). A crude power bound on the latter also directly gives  $\log w = O(\log(1/\tau))$ .  $\square$

For a small fixed  $\varepsilon > 0$ , set  $\beta = \mathbf{x}(\varepsilon)/\varepsilon$ . Define the straight-line controls  $\mathbf{q}, a_0$  by

$$\beta_j = q_{j+1} - q_j \ (j < r), \quad \beta_r = a_0 - q_r, \quad \sum_j q_j = 1. \quad (86)$$

They are unique:  $q_j = q_1 + \sum_{k < j} \beta_k$ , and the sum fixes  $q_1$ .

**Lemma 20.3.** *For all sufficiently small  $\varepsilon$ , every  $\beta_j > 0$ , every  $q_j > 0$ , and  $0 < a_0 < 1$ . Moreover*

$$1 - a_0 \sim \frac{1}{r \log w(\varepsilon)}, \quad \varepsilon \mathcal{L}(\beta; \mathbf{q}, a_0) \rightarrow 0.$$

*Proof.* Integrate (82) from zero to  $\varepsilon$  and use  $\mathbf{x}(0) = 0$ . The unique controls in (86) are the time averages of  $\mathbf{p}$  and  $a$ . Their positivity and strict feasibility follow from Lemma 20.1; positivity of  $\beta_j$  follows from (85). This avoids relying only on the limiting relation  $a_0 \rightarrow 1$ .

The exact identity

$$(1 - a(v))g(v) = \frac{1}{v - 1} - \frac{\log v}{(v - 1)^2}$$

shows that its composition with  $v = E_r(z)$  has tail integral asymptotic to  $r!w^{1-r}/(r - 1)$ . Divide by  $\tau(w)/\mu_r$  in (84) to obtain the asserted average. For  $a \geq 1/2$ , the maximizing tilt in  $I(a)$  is nonnegative. Since  $M(t) \geq \int_a^1 e^{tu} du \geq (1 - a)e^{ta}$ ,

$$0 \leq I(a) \leq \log \frac{1}{1 - a}.$$

Thus  $I(a_0) = O(\log \log w)$ . The other terms in (75) stay bounded because  $\beta$  and  $\mathbf{q}$  tend to strictly positive vectors. Finally  $\varepsilon \log \log w(\varepsilon) \rightarrow 0$  by (84).  $\square$

## 21 Matching growth bounds

### 21.1 Finite seeds with every category large

For every fixed  $B$  we need a reachable state with all coordinates much larger than  $B$  and with  $L \leq X_1$ . This requirement can be met by explicit words.

If  $r \geq 3$ , take  $M$  large and begin with  $0, 1, \dots, (r - 1)M - 1$ . Partition these labels into  $r - 1$  groups of  $M$  labels. For each target residual capacity  $j = 1, \dots, r - 1$ , append that group in increasing order exactly  $r - 1 - j$  times. All appended labels were already available, and each appears at most  $r - 1$  times overall. The final counts are  $X_j = M$  for every  $j < r$ . The group destined for capacity 1 produces at least  $M - 1$  additional ascents, so  $X_r \geq M$ . Histogram-preserving compaction gives a canonical path to such counts. Selecting compact label zero once then gives  $L = 0$  and leaves every category count at least  $M - 1$ .

For  $r = 2$ , use the three increasing runs

$$0, \dots, M; \quad 0, \dots, M; \quad M + 1, \dots, 2M + 1,$$

then append  $M + 1$ . The result is an ascent sequence with each multiplicity at most two. Its terminal state is  $X_1 = M$ ,  $X_2 = M + 1$ ,  $L = 0$ . These seeds have length independent of the later word length  $N$ .

## 21.2 A repeated straight-line prefix

Fix a sufficiently small  $\varepsilon$  and the feasible controls of Lemma 20.3. Choose a large but fixed block length  $B$ . Pick integers  $q_j^{(B)} = Bq_j + O(1)$  summing to  $B$ , and choose the favorable Eulerian bin from Lemma 19.3 at mean  $a_0$ . The resulting increments

$$d_j = q_{j+1}^{(B)} - q_j^{(B)} \quad (j < r), \quad d_r = k_B + 1 - q_r^{(B)}$$

satisfy  $d_j/B \rightarrow \beta_j$ . They are all strictly positive for sufficiently large  $B$ . Start with a sufficiently large fixed seed and repeat this block. Every block ends with  $L \leq X_1$ , so Lemma 19.2 applies uniformly to all paths produced.

At block number  $\ell$ ,  $X_j = \ell d_j + O_B(1)$ . Its exact lower count is therefore

$$\ell^B C_B (1 + O_B(1/\ell)), \quad C_B = A(B-2, k_B) \prod_j \frac{d_j^{q_j^{(B)}}}{q_j^{(B)}!}.$$

Taking products over  $m$  blocks gives logarithm  $B \log(m!) + m \log C_B + O_B(\log m)$ . Stirling's formula yields the factorial-normalized rate per letter

$$\frac{\log C_B}{B} - \log B \rightarrow \mathcal{L}(\beta; \mathbf{q}, a_0) \quad (B \rightarrow \infty).$$

A prefix of length  $\varepsilon N + O_B(1)$  consequently contributes, after subtracting its  $\varepsilon N \log N$  term, the action

$$\varepsilon \log \varepsilon - \varepsilon + \varepsilon \mathcal{L}(\beta; \mathbf{q}, a_0), \quad (87)$$

in the successive limits  $N \rightarrow \infty$  and  $B \rightarrow \infty$ . This tends to zero as  $\varepsilon \downarrow 0$ .

At fixed  $B$  the actual prefix slopes are  $\beta_B = \mathbf{d}/B$ , not exactly  $\beta$ . Translate the nominal later path by  $\varepsilon(\beta_B - \beta)$ . For large enough fixed  $B$  it remains in a positive compact region. Its velocities do not change, and its action converges uniformly to that of the desired path. This explicitly handles rounded or irrational endpoints.

## 21.3 The lower bound

On the characteristic, substitute (81) into (75). The equality  $I(a) = a \log v + \log g(v)$  gives

$$\mathcal{L}(\mathbf{x}; \mathbf{p}, a) = \log \mu_r - \boldsymbol{\theta} \cdot \dot{\mathbf{x}}. \quad (88)$$

For example, the categorical part equals  $\log \mu_r + \log g(v) - \sum_j p_j \log(u_{j-1}/u_j)$ ; its remaining mark contribution is exactly  $\boldsymbol{\theta} \cdot \dot{\mathbf{x}}$  by (82). Integration by parts and Lemma 20.1 give

$$\int_{\varepsilon}^1 \mathcal{L}(\mathbf{x}; \mathbf{p}, a) d\tau = (1 - \varepsilon)(\log \mu_r - 1) + \boldsymbol{\theta}(\varepsilon) \cdot \mathbf{x}(\varepsilon). \quad (89)$$

The right side tends to  $\log \mu_r - 1$ .

To realize this action with finite paths, partition  $[\varepsilon, 1]$  into a fixed finite mesh. On each interval take the averages of  $p_j$  and  $a$ . The drift equations are linear, so these averages give exactly the nominal endpoint increment. Round the category counts to integers and choose the favorable Eulerian bin. Every block then has a fixed category-vector endpoint, apart from the accumulated  $o(N)$  errors from bin selection and rounding; its final threshold always satisfies  $L \leq X_1$ . The errors are  $o(N)$  because the partition is fixed before  $N$  grows. The path remains in a positive compact set,

where entropy is uniformly continuous. Applying (76) and multiplying gives the sum of the interval actions, with error  $O_{r,\varepsilon}(N\delta) + o(N)$  as the mesh  $\delta$  decreases.

A precise order of limits is: fix  $\varepsilon$ , fix an accurate prefix block  $B$ , fix a sufficiently fine finite later mesh, let  $N \rightarrow \infty$ , refine the mesh, improve  $B$ , and finally let  $\varepsilon \downarrow 0$ . The last two finite refinements can be interchanged on the positive compact region. Any bounded number of leftover positions is filled only at the very end by new maxima; the extension is injective and changes the normalized logarithm by  $o(N)$ . We do not need a uniform estimate as  $B$  grows with  $N$ .

Combining (87) and (89) proves

$$\liminf_{N \rightarrow \infty} \frac{1}{N} \log \frac{c_{N,r}}{N!} \geq \log \mu_r,$$

since  $\log N! = N \log N - N + o(N)$ .

## 21.4 The upper block telescope

Fix  $\varepsilon > 0$ . On  $[\varepsilon, 1]$  the mark curve and its first two derivatives are bounded. Partition the steps from  $k = \lceil \varepsilon N \rceil$  to  $N$  into finitely many blocks of lengths  $b \leq \delta N$ . At the start of one block write  $\boldsymbol{\theta} = \boldsymbol{\theta}(\tau)$  and  $\Delta \boldsymbol{\theta} = \boldsymbol{\theta}(\tau + b/N) - \boldsymbol{\theta}(\tau)$ . Since  $|X'_j - X_j| \leq b$ , (70) implies

$$\sum_{\text{paths}} e^{\boldsymbol{\theta}(\tau + b/N) \cdot \mathbf{X}'} \leq e^{\boldsymbol{\theta} \cdot \mathbf{X} + \Delta \boldsymbol{\theta} \cdot \mathbf{X} + b \|\Delta \boldsymbol{\theta}\|_1} \max(v, v^{-1}) \Lambda^b.$$

Use  $\log z \leq z - 1$  on  $z = \Lambda/(N\mu_r)$ , together with (83), to get

$$\Lambda^b \leq (N\mu_r/e)^b \exp \left\{ -\frac{b}{N} \dot{\boldsymbol{\theta}}(\tau) \cdot \mathbf{X} + C_{r,\varepsilon} \frac{b(b+1)}{N} \right\}. \quad (90)$$

The factor  $e^{-b}$  is essential: it is the constant  $-1$  in  $\log z \leq z - 1$ .

Taylor's theorem gives  $\Delta \boldsymbol{\theta} = (b/N) \dot{\boldsymbol{\theta}} + O_{r,\varepsilon}(b^2/N^2)$  and  $\|\Delta \boldsymbol{\theta}\|_1 = O_{r,\varepsilon}(b/N)$ . Since  $\|\mathbf{X}\|_1 \leq N+1$ , the linear state terms cancel. Thus

$$\sum_{\text{paths}} e^{\boldsymbol{\theta}(\tau + b/N) \cdot \mathbf{X}'} \leq e^{\boldsymbol{\theta}(\tau) \cdot \mathbf{X}} (N\mu_r/e)^b \exp\{C_{r,\varepsilon}(b^2/N + 1)\}. \quad (91)$$

This is uniform even at zero coordinates.

Iterate, using  $\boldsymbol{\theta}(1) = 0$ . There are at most  $k!$  initial prefixes: at position  $i$  the ascent bound allows at most  $i$  choices, so every ascent sequence is bounded by this elementary factorial count. The initial mark is at most  $\exp((k+1)\|\boldsymbol{\theta}(k/N)\|_1)$ . For a fixed partition with  $J$  blocks,

$$c_{N,r} \leq k! \exp((k+1)\|\boldsymbol{\theta}(k/N)\|_1) (N\mu_r/e)^{N-k} \exp\{C_{r,\varepsilon}(N\delta + J)\}.$$

After division by  $N!$  and Stirling's formula,

$$\limsup_{N \rightarrow \infty} \frac{1}{N} \log \frac{c_{N,r}}{N!} \leq \varepsilon \log \varepsilon + (1 - \varepsilon) \log \mu_r + \varepsilon \|\boldsymbol{\theta}(\varepsilon)\|_1 + C_{r,\varepsilon} \delta.$$

First let  $\delta \downarrow 0$  at fixed  $\varepsilon$ , then  $\varepsilon \downarrow 0$ . Lemma 20.2 makes the prefix term vanish. This proves the matching upper bound and the root assertion of Theorem 17.1.

## 22 Weighted growth and endpoint laws

Define the histogram polynomial

$$H_N(\mathbf{u}) = \sum_{\text{length-}N \text{ compact paths}} \prod_{j=1}^r u_j^{X_j}.$$

By Proposition 18.2, this is also the corresponding original-word histogram polynomial. All marks in this section are fixed positive real numbers, independent of  $N$ .

**Theorem 22.1** (Full weighted growth). *For every  $\mathbf{u} \in (0, \infty)^r$ ,*

$$\lim_{N \rightarrow \infty} \frac{1}{N} \log \frac{H_N(\mathbf{u})}{N!} = -\log \chi(\mathbf{u}). \quad (92)$$

*The function  $F(\mathbf{q}) = -\log \chi(e^{q_1}, \dots, e^{q_r})$  is finite and smooth on all of  $\mathbb{R}^r$ .*

*Proof.* Fix terminal marks  $\mathbf{u}^0 > 0$ , and put  $\nu = 1/\chi(\mathbf{u}^0)$ . Replace the earlier mark flow by

$$u_j(w) = \sum_{\ell=0}^j \frac{u_{j-\ell}^0 w^\ell}{\ell!}, \quad u_0^0 = 1, \quad Q(w) = u_r(w) = P_{\mathbf{u}^0}(w), \quad \tau(w) = \nu \int_w^\infty g(Q(z)) dz. \quad (93)$$

All marks stay positive,  $u'_j = u_{j-1}$ , and  $P_{\mathbf{u}(w)}(s) = Q(w+s)$ . Set

$$x_j(w) = -\nu u_j(w) \chi_{u_j}(\mathbf{u}(w)) = \frac{\nu u_j(w)}{(r-j)!} \int_w^\infty (z-w)^{r-j} [-g'(Q(z))] dz > 0. \quad (94)$$

Equations (81)–(83) hold with  $\nu$  in place of  $\mu_r$ , while  $\boldsymbol{\theta}(1) = \log \mathbf{u}^0$ . In particular the legal drift controls satisfy  $p_j > 0$ ,  $\sum p_j = 1$ , and  $0 < a < 1$  even when  $v = u_r < 1$ . On every fixed cutoff interval they are bounded away from their respective boundaries.

The leading terms  $u_j(w) \sim w^j/j!$  and  $Q(w) \sim w^r/r!$  are unchanged. The dominated beta-integral calculation in Lemma 20.2 therefore gives the same initial slopes  $\alpha_j$ , with  $\nu$  replacing  $\mu_r$  in the time scale. Averaging the positive controls still gives a strictly feasible positive-drift prefix. The identity used in Lemma 20.3 still proves its vanishing action. Although the terminal ascent fraction can be below  $1/2$ , the initial one tends to one; the Eulerian bin argument permits all interior fractions anyway.

For the upper bound, (70) already holds on both sides of  $v = 1$ . The calibration gives  $\rho(\mathbf{y}; \mathbf{u})/\nu = -\boldsymbol{\theta} \cdot \mathbf{y}$ . The exact calculation (90)–(91) therefore yields

$$\sum_{\text{paths}} e^{\boldsymbol{\theta}_{\text{next}} \cdot \mathbf{X}'} \leq e^{\boldsymbol{\theta} \cdot \mathbf{X}} (N\nu/e)^b \exp\{C_{\varepsilon, \mathbf{u}^0}(b^2/N + 1)\}.$$

No sign condition on  $\boldsymbol{\theta}$  is used: its increments and the initial mark are bounded by absolute values. The terminal weight is now  $(\mathbf{u}^0)^{\mathbf{X}}$ . Summation over the factorially bounded prefix and the same nested limits give the upper bound  $-\log \chi(\mathbf{u}^0)$ .

For the lower bound use the explicit seeds, repeated prefix blocks, and buffered later blocks of Section 21 on the path (94). At fixed cutoff and fixed prefix size  $B$ , translate the later path by  $\Delta_B = \varepsilon(\boldsymbol{\beta}_B - \boldsymbol{\beta})$ . The terminal normalized vector is uniformly  $\mathbf{x}(1) + \Delta_B + o(1)$  over every path selected by this construction. Hence its terminal logarithmic weight divided by  $N$  is

$$\log \mathbf{u}^0 \cdot (\mathbf{x}(1) + \Delta_B) + o(1).$$

This is a two-sided estimate and remains valid for marks below one: the absolute error is bounded by  $\|\log \mathbf{u}^0\|_1$  times the maximum coordinate error. Final deterministic padding changes it by only  $O(1/N)$ .

The action identity is now  $\mathcal{L} = \log \nu - \boldsymbol{\theta} \cdot \dot{\mathbf{x}}$ , so

$$\int_{\varepsilon}^1 \mathcal{L} \, d\tau = (1 - \varepsilon)(\log \nu - 1) - \log \mathbf{u}^0 \cdot \mathbf{x}(1) + \boldsymbol{\theta}(\varepsilon) \cdot \mathbf{x}(\varepsilon).$$

The negative terminal term cancels the positive terminal weight. First let  $N$  grow with all finite choices fixed, then refine the later partition and improve  $B$ , then let  $\varepsilon \downarrow 0$ . The origin and prefix terms vanish exactly as before. After factorial normalization, this gives the lower bound  $\log \nu$ , proving (92).

Finally, a mixed derivative of order  $k$  of  $\chi$  has integrand

$$g^{(k)}(P_{\mathbf{u}}(s)) \prod_{\ell=1}^k \frac{s^{r-j_{\ell}}}{(r-j_{\ell})!}.$$

Uniformly on a compact positive mark neighborhood, its tail is  $O(\log s / s^{r+\sum_{\ell} j_{\ell}})$ , and its bounded-interval part is smooth and dominated. Differentiation of every finite order under the integral is valid. Thus  $\chi$  is positive and smooth, proving the last assertion.  $\square$

## 22.1 Concentration and the consecutive ratio

Let  $\mathbb{P}_N$  be the uniform endpoint law and put  $Y_N = \mathbf{X}/N$ . Theorem 22.1 implies

$$\Lambda(\mathbf{q}) := \lim_{N \rightarrow \infty} \frac{1}{N} \log \mathbb{E}_N e^{N\mathbf{q} \cdot Y_N} = F(\mathbf{q}) - F(0). \quad (95)$$

In particular it is a finite differentiable convex function: convexity also follows as the pointwise limit of the finite logarithmic moment-generating functions. At zero,

$$\partial_j F(0) = -\mu_r \chi_{u_j}(1) = b_j.$$

Equation (78) at the all-one point gives  $\sum_j b_j = \mu_r$ .

For any  $\eta > 0$ , differentiability makes  $F(te_j) - F(0) - t(b_j + \eta) < 0$  for sufficiently small  $t > 0$ . Markov's inequality gives an exponentially small upper tail for  $X_j/N \geq b_j + \eta$ . The same argument with  $t < 0$  gives the lower tail. A finite union bound proves exponential concentration of  $Y_N$  at  $\mathbf{b}$ . Uniform boundedness of  $Y_N$  also gives convergence in mean.

From a state there are exactly  $M = \sum_j X_j$  possible next labels. Therefore

$$\frac{c_{N+1,r}}{c_{N,r}} = \mathbb{E}_N M. \quad (96)$$

Since  $M/N \rightarrow \sum_j b_j = \mu_r$  in mean, this proves the ratio limit in Theorem 17.1. The original-word identities (68) yield (65); their fixed affine transformations preserve the exponential concentration. This proves Theorem 17.2.

## 22.2 A full large-deviation principle

The weighted result gives more than concentration. We include an ordinary finite-dimensional proof, so no unstated boundary version of a large-deviation theorem is needed.

**Theorem 22.2.** *The laws of  $Y_N$  satisfy a full large-deviation principle of speed  $N$  with good convex rate*

$$\mathcal{I}_r(\mathbf{x}) = \sup_{\mathbf{q} \in \mathbb{R}^r} \{\mathbf{q} \cdot \mathbf{x} - F(\mathbf{q}) + F(0)\}. \quad (97)$$

For every open  $G$  and closed  $C$  in  $\mathbb{R}^r$ ,

$$\begin{aligned} \liminf_N N^{-1} \log \mathbb{P}_N(Y_N \in G) &\geq -\inf_G \mathcal{I}_r, \\ \limsup_N N^{-1} \log \mathbb{P}_N(Y_N \in C) &\leq -\inf_C \mathcal{I}_r. \end{aligned}$$

Its unique zero is  $\mathbf{b} = \nabla F(0)$ . Its effective domain lies in  $S = \{\mathbf{x} \geq 0 : \sum_j x_j \leq 1\}$ ; no claim that all of  $S$  has finite rate is made.

*Proof.* Write  $\Lambda = F - F(0)$ . The conjugate (97) is lower semicontinuous and convex. The finite supports satisfy  $Y_N \geq 0$  and  $\sum_j Y_{N,j} \leq 1 + 1/N$ . Thus  $\Lambda(\mathbf{q}) \leq \sup_{\mathbf{x} \in S} \mathbf{q} \cdot \mathbf{x}$ . Separation followed by positive scaling shows  $\mathcal{I}_r = +\infty$  outside  $S$ . Its closed sublevel sets are therefore compact.

For the upper bound, fix  $K < \infty$  and  $\eta > 0$ . At each  $\mathbf{x}$  in the compact set  $C \cap [0, 2]^r$ , choose  $\mathbf{q}$  with

$$\mathbf{q} \cdot \mathbf{x} - \Lambda(\mathbf{q}) \geq \min(\mathcal{I}_r(\mathbf{x}), K) - \eta.$$

A neighborhood  $U_{\mathbf{x}}$  satisfies the same inequality with  $2\eta$  for every point in it. Markov's inequality and (95) bound the probability of  $Y_N \in U_{\mathbf{x}}$  by the corresponding exponential rate. A finite subcover exists; taking the largest exponential term yields

$$\limsup_N N^{-1} \log \mathbb{P}_N(Y_N \in C) \leq -\min(\inf_C \mathcal{I}_r, K) + 2\eta.$$

All supports lie in  $[0, 2]^r$ , so no probability was omitted. Send  $\eta \downarrow 0$  and  $K \rightarrow \infty$ .

For the lower bound, first take  $\mathbf{x} = \nabla \Lambda(\mathbf{q})$ . Tilt the law by density  $e^{N\mathbf{q} \cdot Y_N} / \mathbb{E}_N e^{N\mathbf{q} \cdot Y_N}$ . Its limiting logarithmic moment-generating function is  $\Lambda(\mathbf{q} + \mathbf{h}) - \Lambda(\mathbf{q})$ , differentiable at zero. The coordinate Chernoff argument just given proves concentration under the tilt at  $\mathbf{x}$ . Changing back to the original law on a radius- $\delta$  ball around  $\mathbf{x}$  gives

$$\liminf_N N^{-1} \log \mathbb{P}_N(Y_N \in B_\delta(\mathbf{x})) \geq \Lambda(\mathbf{q}) - \mathbf{q} \cdot \mathbf{x} - \|\mathbf{q}\|\delta.$$

Convexity makes  $\mathcal{I}_r(\mathbf{x}) = \mathbf{q} \cdot \mathbf{x} - \Lambda(\mathbf{q})$ . If an open set contains  $\mathbf{x}$ , send  $\delta \downarrow 0$  inside that set.

It remains to cover finite-rate points that might not themselves be gradients. Fix  $\mathbf{x}$  with  $\mathcal{I}_r(\mathbf{x}) < \infty$ . For  $\epsilon > 0$  maximize

$$\mathbf{q} \cdot \mathbf{x} - \Lambda(\mathbf{q}) - \frac{\epsilon}{2} \|\mathbf{q}\|^2.$$

A maximizer  $\mathbf{q}_\epsilon$  exists: the affine tangent lower bound on  $\Lambda$  makes the negative quadratic dominate at infinity. Let its maximum be  $J_\epsilon$ . Then  $J_\epsilon \uparrow \mathcal{I}_r(\mathbf{x})$  as  $\epsilon \downarrow 0$ , and

$$\frac{\epsilon}{2} \|\mathbf{q}_\epsilon\|^2 \leq \mathcal{I}_r(\mathbf{x}) - J_\epsilon \longrightarrow 0.$$

The first-order equation gives  $\mathbf{x}_\epsilon = \nabla \Lambda(\mathbf{q}_\epsilon) = \mathbf{x} - \epsilon \mathbf{q}_\epsilon \rightarrow \mathbf{x}$ . At these gradient points,

$$\mathcal{I}_r(\mathbf{x}_\epsilon) = J_\epsilon - \frac{\epsilon}{2} \|\mathbf{q}_\epsilon\|^2 \longrightarrow \mathcal{I}_r(\mathbf{x}).$$

Every open neighborhood of  $\mathbf{x}$  eventually contains  $\mathbf{x}_\epsilon$ , so its lower bound follows from the gradient case. Taking the infimum over an arbitrary open set proves the full lower bound, including finite-rate boundary points.

The tangent inequality for  $\Lambda$  gives  $\mathcal{I}_r(\mathbf{b}) = 0$ . If  $\mathbf{x} \neq \mathbf{b}$ , a sufficiently small dual vector in the direction  $\mathbf{x} - \mathbf{b}$  makes  $\mathbf{q} \cdot \mathbf{x} - \Lambda(\mathbf{q}) > 0$ , by differentiability at zero. This proves uniqueness.  $\square$

For a fixed terminal tilt  $\mathbf{u}^0 = \mathbf{e}^{q^0}$ , the same proof gives the rate

$$\mathcal{I}_{r,q^0}(\mathbf{x}) = \mathcal{I}_r(\mathbf{x}) - \mathbf{q}^0 \cdot \mathbf{x} + F(\mathbf{q}^0) - F(0),$$

with unique typical endpoint

$$b_j(\mathbf{u}^0) = \partial_j F(\mathbf{q}^0) = -\frac{u_j^0 \chi_{u_j}(\mathbf{u}^0)}{\chi(\mathbf{u}^0)} = x_j(1).$$

Continuous affine images give rate functions obtained by minimizing  $\mathcal{I}_r$  over their fibers, for the original multiplicities, distinct values, and ascents. This follows directly by applying the open- and closed-set bounds to inverse images; compact support handles closed images. The  $2/N$  correction for ascents vanishes deterministically and does not change the bounds.

Smooth limiting pressure alone does not establish convergence of its finite-size second derivatives, a central limit theorem, a local limit theorem, or a coefficient prefactor. Those are separate problems.

**[write, 5 October 2026]** New in the repository: Theorems 22.1 and 22.2, and the endpoint laws derived from them. At  $r = 2$  the upper half of Theorem 22.1 is Report 97's Proposition 46.3 (sibling report), which proves only  $\limsup \leq -\log \chi$ ; the theorem here gives the limit for every positive mark vector and the smoothness of  $F$ .

## 23 Monotonicity and inversion

**Lemma 23.1.** *One has  $c_{n+1,r} > c_{n,r}$  for every  $n \geq 1$ .*

*Proof.* The maximum of an ascent sequence is at most its total number of ascents. This follows inductively: a new maximum is necessarily reached by an ascent, and its value is at most one plus the previous ascent count. If a word has maximum  $M$ , append the fresh value  $M + 1$ . The ascent bound permits it, and it cannot create an occurrence beyond the cap of any value. Removing the last entry recovers the original word, so this is an injection  $\mathcal{A}_{n,r} \rightarrow \mathcal{A}_{n+1,r}$ . Its image ends in a positive new maximum. The valid word

$$0, 1, \dots, n-1, 0$$

of length  $n+1$  ends in zero, and is outside the image. For  $n = 1$  this word is 00. Thus the inequality is strict. The equality  $c_{0,r} = c_{1,r} = 1$  is the only initial tie relevant here.  $\square$

Fix  $r \geq 2$ . For real  $y > 1$  define the threshold

$$N_r(y) = \min\{n \geq 0 : c_{n,r} \geq y\}, \quad Y = \log y.$$

Let  $W_0$  denote the positive real branch of Lambert's function, characterized by  $W_0(z)e^{W_0(z)} = z$  for  $z > 0$ .

**Corollary 23.2.** *As  $y \rightarrow \infty$ ,*

$$N_r(y) = \frac{Y}{W_0(\mu_r Y/e)} + o\left(\frac{Y}{(\log Y)^2}\right). \quad (98)$$

*Equivalently,*

$$N_r(y) = \frac{Y}{\log Y} + \frac{Y(\log \log Y + 1 - \log \mu_r)}{(\log Y)^2} + o\left(\frac{Y}{(\log Y)^2}\right). \quad (99)$$



*Proof.* Set

$$F(x) = x \log(\mu_r x / e).$$

Theorem 17.1 and Stirling's formula give  $\log c_{n,r} = F(n) + o(n)$ . If  $N = N_r(y)$ , the threshold inequalities

$$\log c_{N-1,r} < Y \leq \log c_{N,r}$$

and  $F(N) - F(N-1) = O(\log N)$  imply

$$Y = F(N) + o(N). \tag{100}$$

This does not require a coefficient-ratio estimate. It follows first that  $N \sim Y / \log Y$ .

The positive solution  $\nu$  of  $F(\nu) = Y$  is exactly

$$\nu = Y / W_0(\mu_r Y / e).$$

Indeed, writing  $w = W_0(\mu_r Y / e)$  gives  $\mu_r \nu / e = e^w$  and  $\nu w = Y$ . Also  $\nu \sim Y / \log Y$ , so  $N / \nu \rightarrow 1$ . On the interval between them,

$$F'(x) = \log(\mu_r x) \sim \log Y.$$

The mean value theorem applied to (100) now gives

$$N - \nu = o(N / \log Y) = o(Y / (\log Y)^2),$$

proving (98). Finally, its defining relation  $w + \log w = \log Y + \log \mu_r - 1$  first gives  $w \sim \log Y$ . Substituting this estimate once more gives

$$W_0(\mu_r Y / e) = \log Y - \log \log Y + \log \mu_r - 1 + O\left(\frac{\log \log Y}{\log Y}\right).$$

Expanding the reciprocal yields (99); its next explicit error is  $O(Y(\log \log Y)^2 / (\log Y)^3)$ , which is smaller than the error already allowed in (98).  $\square$

*Remark 23.3.* The factor in the Lambert argument is  $\mu_r Y / e$ . It follows directly from  $F(x) = x \log(\mu_r x / e)$ ; replacing it by  $Y / (e \mu_r)$  would change the second term incorrectly. The error in (98) is generally much larger than one, so neither a floor nor a ceiling of the displayed approximation is justified as an exact threshold rule.

[**write, 5 October 2026**] Second route, weaker. With  $Y = \log y$ , which is Part I's  $L = \log X$ , the leading term  $Y / W_0(\mu_r Y / e)$  is Part I's  $n_0 = L / W_0(L / (e T_b))$ , because  $\mu_r = 1 / T_b$ . Part I's error  $O_b(L / (\log L)^3)$  in (28) is smaller than the  $o(Y / (\log Y)^2)$  of (98), for every  $b \geq 3$ ; for  $r = 2$  the sibling report's Parts II and IV give  $x - O(\log x) \leq N \leq x + O(\sqrt{x \log x})$ . Lemma 23.1 sharpens the monotonicity used in Section 6 ("the counts are nondecreasing") to strict increase; the injection is the same.

## 24 Asymptotics as the cap grows

This section studies the constants  $J_r = \mu_r^{-1}$  as the cap  $r$  tends to infinity. Throughout this section

$$R = r + 1, \quad A = \zeta(2) = \frac{\pi^2}{6}.$$

The parameter  $R$  is unrelated to the word length  $N$ . Every sector index and every inverse- $R$  truncation order below is fixed. In particular, these results do not assert a uniform word-length theorem with a growing cap.

**[write, 5 October 2026]** *Second routes in this section.* Section 24 re-proves, in the variable  $R = b + 1$ , the large-cap results of Section 7 and Part II. Its  $J_r$  is  $T_b$ , its  $A$  is  $\tau$ , and its sector  $T_m(R)$  at  $R = b + 1$  is Part II's  $A_{m,b}$  (sector index  $m$  here,  $r$  there). Thus (114) is the first identity of (30) and (50) ( $L(b + 1) = L_b = A_{1,b}$ ); (113) at integer  $R$  is (40); Proposition 24.2 is (42); (127) and (128) are (43) with the leading term of Theorem 13.1; Theorem 24.4 is Theorem 13.1, whose  $D_r$  and  $d_{r,j}$  are this section's  $C_m$  and  $c_{m,j}$ ; (136) is Part II's  $d_{r,1}$ ; (104) is (4); and (103) is the sharp remainder of Part II (equations (43), (48)). The coefficient algorithm (133)–(135) uses centred gamma moments and is a different finite algorithm from the Gaussian-moment algorithm after Theorem 13.1. New here: the continuous cap (107)–(111) with the strict monotonicity of  $\mathcal{J}$  (Section 8 assumes “no interpolation of the exact sequence  $T_b$  to real  $b$ ”; here the interpolation is an analytic device), the derivative control (138)–(140), the exact-sector envelopes with Proposition 24.5, the displacement (147) and the selection criterion of Corollary 24.6.

**Theorem 24.1** (Sharp large-cap expansion). *As  $R = r + 1 \rightarrow \infty$ ,*

$$J_r - A = R(\zeta(R + 1) - 1) + \frac{2}{\sqrt{\pi}} \sqrt{R} \left( \frac{4}{9} \right)^R \left( 1 - \frac{31}{8R} + O(R^{-2}) \right). \quad (101)$$

*Consequently,*

$$J_r - A = (r + 1)(\zeta(r + 2) - 1) + O(\sqrt{r}(4/9)^r), \quad (102)$$

$$J_r - A - (r + 1)(\zeta(r + 2) - 1) \sim \frac{8}{9\sqrt{\pi}} \sqrt{r}(4/9)^r, \quad (103)$$

$$\frac{6}{\pi^2} - \mu_r \sim \frac{9(r + 1)}{\pi^4 2^r}. \quad (104)$$

**[write, 5 October 2026]** The coefficient  $-31/8$  in (101) belongs to the variable  $R = b + 1$ ; in Part II's variable  $b$  the same expansion has  $-27/8$ , as in (48). There is no contradiction: see the conversion after (137). The power  $\sqrt{r}$  in (102) is sharper than the  $b^{5/2}$  of (3), whose constants are explicit; the sharp power is Part II's.

The proof gives an exact positive sector decomposition, all inverse- $R$  orders within each fixed sector, and a rigorous inversion for the minimum integer cap needed to achieve a specified gap from  $6/\pi^2$ .

## 24.1 Derivative bounds and a continuous cap parameter

Recall  $g(v) = \log v / (v - 1)$ , with its removable value  $g(1) = 1$ . Its integral representation gives, for  $v > 0$  and integers  $m \geq 0$ ,

$$\frac{(-1)^m}{m!} g^{(m)}(v) = \int_0^1 \frac{t^m}{[1 + t(v - 1)]^{m+1}} dt > 0. \quad (105)$$

For  $v \geq 1$ ,  $tv \leq 1 + t(v - 1)$ , and therefore

$$\frac{|g^{(m)}(v)|}{m!} \leq \frac{g(v)}{v^m} \leq \frac{1 + \log v}{v^{m+1}}. \quad (106)$$

The second inequality follows from  $vg(v) = \log v + \log v / (v - 1) \leq 1 + \log v$ . These estimates include  $v = 1$  and require no exception near  $x = 0$ .

It is useful to extend the parameter to real  $R > 2$ . Define

$$\mathcal{E}(R, x) = e^x \frac{\Gamma(R, x)}{\Gamma(R)}, \quad \delta_R(x) = e^x \frac{\gamma(R, x)}{\Gamma(R)}, \quad \mathcal{J}(R) = \int_0^\infty g(\mathcal{E}(R, x)) dx. \quad (107)$$

Here  $\Gamma(R, x)$  and  $\gamma(R, x)$  are the upper and lower incomplete gamma functions. At integer  $R = r + 1$ ,  $\mathcal{E}(R, x) = E_r(x)$  and  $\mathcal{J}(R) = J_r$ . For  $R \geq 1$ ,

$$\mathcal{E}(R, x) = \frac{1}{\Gamma(R)} \int_0^\infty (x + u)^{R-1} e^{-u} du \geq \max \left( 1, \frac{x^{R-1}}{\Gamma(R)} \right). \quad (108)$$

For each fixed  $R$ ,  $\mathcal{E}(R, x) \sim x^{R-1} / \Gamma(R)$  as  $x \rightarrow \infty$ , by dominated convergence in this integral. Thus  $g(\mathcal{E}(R, x)) = O(\log x / x^{R-1})$ , and  $\mathcal{J}(R)$  is finite for  $R > 2$ .

The gamma convolution identity couples a gamma random variable of shape  $R + h$  as the sum of independent gamma variables of shapes  $R$  and  $h$ . Consequently  $\Gamma(R, x) / \Gamma(R)$ , a gamma upper-tail probability, is strictly increasing in  $R$  for every  $x > 0$ . Hence  $\mathcal{J}(R)$  is strictly decreasing. Continuity follows by dominated convergence on compact  $R$ -intervals, dominated by the integrable value at the smallest parameter. Also  $\mathcal{E}(R, x) < e^x$ , so  $\mathcal{J}(R) > A$ . As  $R \rightarrow \infty$ , the gamma lower tail at fixed  $x$  tends to zero, whence

$$\mathcal{J}(R) \downarrow \int_0^\infty \frac{x}{e^x - 1} dx = \sum_{k \geq 1} k^{-2} = A.$$

The interchange in the last equality is Tonelli's theorem. The continuous extension is an analytic device; no combinatorial interpretation of a noninteger cap is asserted.

Expanding  $e^{x(1-t)}$  in a beta integral gives the identity

$$\delta_R(x) = \frac{x^R}{\Gamma(R+1)} F_R(x), \quad F_R(x) = \sum_{k \geq 0} \frac{x^k}{(R+1)_k}. \quad (109)$$

Indeed,

$$\delta_R(x) = \frac{x^R}{\Gamma(R)} \int_0^1 t^{R-1} e^{x(1-t)} dt,$$

and the termwise integrals are beta integrals. For any fixed  $0 < \alpha < 1$ ,

$$F_R(x) \leq (1 - \alpha)^{-1} \quad (0 \leq x \leq \alpha R). \quad (110)$$

Moreover, if  $G \sim \text{Gamma}(R, \text{rate } 1)$ , then

$$e^{-x} \delta_R(x) = \Pr(G \leq x) \leq \exp[-R(\alpha - 1 - \log \alpha)] \quad (x \leq \alpha R). \quad (111)$$

To prove this, apply Markov's inequality to  $e^{-sG}$  and use  $\mathbb{E}e^{-sG} = (1 + s)^{-R}$  with  $s = 1/\alpha - 1$ . Thus  $\mathcal{E}(R, x) \geq e^x / 2$  uniformly below  $\alpha R$  for all sufficiently large  $R$ .

## 24.2 Exact positive sectors and their finite remainders

Define

$$T_m(R) = \frac{(-1)^m}{m!} \int_0^\infty g^{(m)}(e^x) \delta_R(x)^m dx \quad (m \geq 1). \quad (112)$$

Each sector is positive and finite: its integrand is at most  $(1+x)e^{-x}$  by (106) and  $\delta_R \leq e^x$ . For  $v = e^x$ ,  $D = 1 + t(v-1)$ , and  $\delta = \delta_R(x)$ ,

$$\frac{1}{D - t\delta} = \frac{1}{D} \sum_{m \geq 0} \left( \frac{t\delta}{D} \right)^m, \quad 0 \leq \frac{t\delta}{D} \leq \frac{\delta}{v} < 1.$$

All terms are nonnegative. Integrating first in  $t$  and then in  $x$ , Tonelli's theorem proves the exact identity

$$\mathcal{J}(R) - A = \sum_{m \geq 1} T_m(R). \quad (113)$$

In particular every finite remainder is nonnegative and is bounded pointwise by  $g(\mathcal{E}(R, x))$  before integration.

The first sector is explicit for every real  $R > 0$ . Put  $P(R, x) = \gamma(R, x)/\Gamma(R)$ . Integration by parts gives

$$\begin{aligned} T_1(R) &= - \int_0^\infty \frac{d}{dx} g(e^x) P(R, x) dx \\ &= \frac{1}{\Gamma(R)} \int_0^\infty \frac{x^R e^{-x}}{e^x - 1} dx = R(\zeta(R+1) - 1) =: L(R). \end{aligned} \quad (114)$$

The boundary terms vanish because  $P(R, 0) = 0$  and  $g(e^x) \rightarrow 0$  at infinity; expanding  $(e^x - 1)^{-1}$  then gives the zeta identity by Tonelli. In particular,

$$L(R) = L_0(R) + O(R3^{-R}), \quad L_0(R) = R2^{-R-1}. \quad (115)$$

Set

$$a_m = \left( \frac{m}{m+1} \right)^m, \quad C_m = (2\pi)^{(1-m)/2} m^{3/2} (m+1)^{m-2}. \quad (116)$$

The bases  $a_m$  decrease to  $e^{-1}$  and are all strictly larger than  $e^{-1}$ .

**Proposition 24.2** (Finite-sector remainder). *For every fixed  $M \geq 1$ ,*

$$\mathcal{J}(R) - A = \sum_{m=1}^M T_m(R) + O_M\left(R^{(2-M)/2} a_{M+1}^R\right). \quad (117)$$

*Proof.* Choose  $\alpha < 1$  so close to 1 that  $1/(\alpha e) < a_{M+1}$ . Below  $\alpha R$ , Taylor's theorem and (106), with  $\mathcal{E}(R, x) \geq e^x/2$ , bound the remainder after  $M$  terms by

$$2^{M+2}(1+x)e^{-(M+2)x}\delta_R(x)^{M+1}.$$

Use (109)–(110) and extend the resulting integral to infinity. Up to a fixed multiplicative constant its integral is

$$\frac{1}{\Gamma(R+1)^{M+1}} \int_0^\infty (1+x)x^{(M+1)R} e^{-(M+2)x} dx = O_M\left(R^{(2-M)/2} a_{M+1}^R\right),$$

by the gamma integral and Stirling's formula. For the upper tail, (106) and (108) imply, when  $R > 3$  and  $\alpha R \geq 1$ ,

$$\int_{\alpha R}^{\infty} g(\mathcal{E}(R, x)) dx \leq \frac{2\Gamma(R)(\alpha R)^{3-R}}{R-3} = O_{\alpha}(R^{3/2}(\alpha e)^{-R}). \quad (118)$$

Here  $\log \mathcal{E}(R, x) \leq x$  was used. The strict exponential inequality in the choice of  $\alpha$  absorbs this tail into the required bound. Positivity of the exact sector series bounds the tail of the remainder by this same integral.  $\square$

### 24.3 Uniform tail-factor expansions to every order

Let  $h_{\ell}(1, \dots, k)$  denote the complete homogeneous symmetric polynomial of degree  $\ell$  in  $1, \dots, k$ , with  $h_0 = 1$ . Define

$$f_{\ell}(y) = (-1)^{\ell} \sum_{k \geq 0} h_{\ell}(1, \dots, k) y^k. \quad (119)$$

For the empty alphabet,  $h_{\ell} = 0$  when  $\ell > 0$ .

**Lemma 24.3** (Uniform expansion of the tail factor). *For fixed  $0 < \alpha < 1$  and nonnegative integers  $q, d$ , as real  $R \rightarrow \infty$ ,*

$$F_R(Ry) = \sum_{\ell=0}^q f_{\ell}(y) R^{-\ell} + O_{\alpha, q, d}(R^{-q-1}) \quad (120)$$

*uniformly in the  $C^d$  norm on  $0 \leq y \leq \alpha$ .*

*Proof.* For  $p_k(z) = \prod_{j=1}^k (1 + jz)^{-1}$  and  $z \geq 0$ , the product rule gives

$$\frac{(-1)^u p_k^{(u)}(z)}{u!} = \sum_{b_1 + \dots + b_k = u} \prod_{j=1}^k \frac{j^{b_j}}{(1 + jz)^{b_j+1}} \leq h_u(1, \dots, k).$$

Taylor's theorem at  $z = 0$  therefore bounds the error through degree  $q$ , at  $z = 1/R$ , by

$$R^{-q-1} h_{q+1}(1, \dots, k) \leq R^{-q-1} \left( \frac{k(k+1)}{2} \right)^{q+1}. \quad (121)$$

Multiply by  $y^k$  and sum. Differentiating  $d$  times in  $y$  adds only a fixed polynomial in  $k$  to the geometrically summable majorant. This proves the stated uniform remainder.  $\square$

The coefficient  $h_{\ell}(1, \dots, k)$  is a polynomial in  $k$  of degree at most  $2\ell$ , computable from

$$\ell h_{\ell} = \sum_{u=1}^{\ell} S_u(k) h_{\ell-u}, \quad S_u(k) = \sum_{j=1}^k j^u. \quad (122)$$

Thus  $f_{\ell}$  is rational, with denominator dividing  $(1-y)^{2\ell+1}$ . An effective construction replaces each power of  $k$  by the same power of the Euler operator  $y d/dy$  acting on  $(1-y)^{-1}$ , and multiplies by  $(-1)^{\ell}$ . For example,

$$f_0 = \frac{1}{1-y}, \quad f_1 = -\frac{y}{(1-y)^3}, \quad f_2 = \frac{y(1+2y)}{(1-y)^5}. \quad (123)$$

## 24.4 Gamma concentration and the sector saddle

For  $v > 1$ , differentiate the normally convergent series  $g(v) = \sum_{k \geq 1} (\log v) v^{-k}$ . The differentiated summand is

$$\frac{(-1)^m}{m!} \frac{d^m}{dv^m} [(\log v) v^{-k}] = \binom{k+m-1}{m} v^{-k-m} \left( \log v - \sum_{j=0}^{m-1} \frac{1}{k+j} \right).$$

The  $k = 1$  term contains  $H_m = \sum_{j=1}^m 1/j$ . The remaining coefficients grow only polynomially in  $k$ , so for  $x \geq \log 2$ ,

$$\frac{(-1)^m}{m!} g^{(m)}(e^x) = e^{-(m+1)x} (x - H_m) + O_m((1+x)e^{-(m+2)x}). \quad (124)$$

Put

$$\begin{aligned} I_{R,m} &= \frac{\Gamma(mR+2)}{(m+1)^{mR+2} \Gamma(R+1)^m}, & y_m &= \frac{m}{m+1}, \\ X &\sim \text{Gamma}(mR+2, \text{rate } m+1), & Y &= X/R. \end{aligned} \quad (125)$$

The gamma moment-generating function yields exponential concentration of  $Y$  at  $y_m$ . More explicitly, for  $c \neq y_m$ , the appropriate one-sided probability is at most

$$K_{m,c} e^{-R\mathcal{I}_m(c)}, \quad \mathcal{I}_m(c) = (m+1)c - m - m \log \frac{(m+1)c}{m} > 0. \quad (126)$$

Use the exponential parameter  $t = (m+1) - m/c$ , positive for the upper tail and negative for the lower tail, in  $(1 - t/(m+1))^{-mR-2}$ . The extra shape parameter 2 contributes a fixed factor. Fixed polynomial moments on a discarded tail are likewise exponentially small, by this exponential-moment bound or by Cauchy–Schwarz with a slightly weakened exponent.

Choose  $0 < \beta < y_m < \alpha < 1$ , with  $\alpha > -\log a_m$ . Such a choice exists since  $a_m > e^{-1}$ . On  $\beta R \leq x \leq \alpha R$ , the expansions (120) and (124) are uniform. Below  $\beta R$ , the derivative bound and the bound on  $F_R$  reduce the integral to lower gamma tails of shapes  $mR+1$  and  $mR+2$ . Above  $\alpha R$ , the sector integrand is at most  $(1+x)e^{-x}$ , whose integral is  $O(Re^{-\alpha R})$ . These errors, as well as the last term in (124), are smaller than every inverse power of  $R$  relative to  $I_{R,m}$ . It follows first that

$$T_m(R) \sim (m+1)^m I_{R,m},$$

and hence, by Stirling's formula,

$$T_m(R) \sim C_m R^{(3-m)/2} a_m^R. \quad (127)$$

Combining this with Proposition 24.2 at  $M$  and at  $M+1$  proves

$$\mathcal{J}(R) - A - \sum_{m=1}^M T_m(R) \sim T_{M+1}(R) \sim C_{M+1} R^{(2-M)/2} a_{M+1}^R. \quad (128)$$

This holds for every fixed  $M$ ; it does not resolve the accumulation of bases at  $e^{-1}$  when the number of sectors grows.

## 24.5 All inverse-cap orders and an effective coefficient algorithm

**Theorem 24.4** (Every fixed sector to all algebraic orders). *For each fixed  $m \geq 1$  and  $q \geq 0$ , there are rational coefficients  $c_{m,j}$  such that*

$$T_m(R) = C_m R^{(3-m)/2} a_m^R \left( \sum_{j=0}^q c_{m,j} R^{-j} + O_{m,q}(R^{-q-1}) \right). \quad (129)$$

*They are effectively computable by finite polynomial and rational operations.*

All expectations of rational functions in the following proof are restricted to a fixed compact saddle interval inside  $(0, 1)$ . Their unrestricted gamma expectations need not exist because of the pole at  $Y = 1$ . Full gamma moments enter only after expansion into polynomials.

*Proof.* Let  $z$  be a formal variable and define

$$\begin{aligned} A_{m,\ell}(y) &= [z^\ell] \left( \sum_{u \geq 0} f_u(y) z^u \right)^m, \\ B_{m,\ell}(y) &= A_{m,\ell}(y) - \frac{H_m}{y} A_{m,\ell-1}(y), \quad A_{m,-1} = 0. \end{aligned} \quad (130)$$

Only finitely many coefficients are used at a specified order. The exact centered gamma moment polynomial is

$$M_{m,d}(z) = \sum_{j=0}^d \binom{d}{j} (-y_m)^{d-j} \frac{\prod_{i=0}^{j-1} [m + (i+2)z]}{(m+1)^j}. \quad (131)$$

At  $z = 1/R$  this equals  $\mathbb{E}[(Y - y_m)^d]$ . For  $d \geq 1$  its first possible power is  $z^{\lceil d/2 \rceil}$ , as seen from

$$\mathbb{E}e^{t(Y-y_m)} = \exp \left[ \sum_{u \geq 1} z^u \left( \frac{2t^u}{u(m+1)^u} + \frac{mt^{u+1}}{(u+1)(m+1)^{u+1}} \right) \right]. \quad (132)$$

For the requested order  $q$ , form the coefficients through degree  $q$  of

$$\mathcal{P}_{m,q}(z) = \sum_{\ell=0}^q z^\ell \sum_{d=0}^{2(q-\ell)} \frac{B_{m,\ell}^{(d)}(y_m)}{d!} M_{m,d}(z). \quad (133)$$

Let  $B_{2k}$  denote Bernoulli numbers and form the formal Stirling correction

$$\mathcal{S}_m(z) = \left( 1 + \frac{z}{m} \right) \exp \left[ \sum_{k \geq 1} \frac{B_{2k}}{2k(2k-1)} (m^{1-2k} - m) z^{2k-1} \right]. \quad (134)$$

The coefficient algorithm is

$$c_{m,j} = [z^j] \frac{\mathcal{S}_m(z) \mathcal{P}_{m,q}(z)}{(m+1)^m}, \quad 0 \leq j \leq q. \quad (135)$$

Its inputs are rational, so its outputs are rational.

To prove the remainder, raise the uniform expansion in Lemma 24.3 to the fixed power  $m$  on the compact saddle interval. Equation (124) then reduces  $T_m/I_{R,m}$ , up to exponentially small errors, to the localized expectation of  $\sum_{\ell=0}^q R^{-\ell} B_{m,\ell}(Y)$  with uniform error  $O(R^{-q-1})$ . For each  $\ell$ ,

Taylor-expand  $B_{m,\ell}$  about  $y_m$  through degree  $2(q-\ell)+1$ . The remainder has expectation bounded by a constant times  $\mathbb{E}|Y - y_m|^{2(q-\ell)+2} = O(R^{-q+\ell-1})$ . The signed moment of degree  $2(q-\ell)+1$  is itself of that order, by (132), and may be omitted. The difference between localized and full polynomial moments is exponentially small by (126). This proves (133) with the required uniform error.

Finally, Stirling's formula with its finite-order remainder gives

$$I_{R,m} = \frac{C_m}{(m+1)^m} R^{(3-m)/2} a_m^R \left( \mathcal{S}_m(1/R) \text{ through degree } q + O_{m,q}(R^{-q-1}) \right).$$

The factor  $1+z/m$  comes from  $\Gamma(mR+2) = (mR+1)\Gamma(mR+1)$ ; subtracting  $m$  copies of  $\log \Gamma(R+1)$  gives the Bernoulli coefficients in (134). This proves the theorem. Uniform majorants are supplied by (121), (126), Taylor's theorem on the fixed compact interval, and Stirling's remainder. No convergence of the infinite inverse- $R$  series is asserted.  $\square$

The first correction has the closed form

$$c_{m,1} = \frac{23m}{12} + \frac{13}{12m} - \frac{m^2(m+1)}{2} - \frac{(m+1)H_m}{m}. \quad (136)$$

For example, the coefficient lists through degree three are

$$\begin{aligned} (c_{1,0}, c_{1,1}, c_{1,2}, c_{1,3}) &= (1, 0, 0, 0), \\ (c_{2,0}, c_{2,1}, c_{2,2}, c_{2,3}) &= \left(1, -\frac{31}{8}, \frac{5437}{128}, -\frac{896281}{1024}\right), \\ (c_{3,0}, c_{3,1}, c_{3,2}, c_{3,3}) &= \left(1, -\frac{43}{3}, \frac{32791}{81}, -\frac{40869055}{2187}\right). \end{aligned} \quad (137)$$

All  $c_{1,j}$  for  $j \geq 1$  vanish, consistently with (114).

**[write, 5 October 2026]** *Conversion to Part II's variable.* Write (129) as  $T_m(R) \sim C_m R^{(3-m)/2} a_m^R \sum_j c_{m,j} R^{-j}$ . At  $R = b+1$ , with  $q = 1/b$ ,

$$T_m(b+1) = C_m a_m b^{(3-m)/2} a_m^b (1+q)^{(3-m)/2} \sum_j c_{m,j} \left(\frac{q}{1+q}\right)^j,$$

so Part II's constant is  $C_m a_m$  and its coefficients in  $b^{-j}$  are the coefficients of  $q^j$  on the right. For  $m = 2$  the list  $(1, -\frac{31}{8}, \frac{5437}{128}, -\frac{896281}{1024})$  in  $R$  becomes  $(1, -\frac{27}{8}, \frac{5669}{128}, -\frac{962949}{1024})$  in  $b$ , and  $C_2 a_2 = 8/(9\sqrt{\pi})$ : this is (48). For  $m = 3$  the list  $(1, -\frac{43}{3}, \frac{32791}{81}, -\frac{40869055}{2187})$  becomes  $(1, -\frac{43}{3}, \frac{33952}{81}, -\frac{42671116}{2187})$ , and  $C_3 a_3 = 81\sqrt{3}/(32\pi)$ : this is (49). The conversion was checked exactly with SymPy at the write. In general  $c_{m,j}$  here equals Part II's  $d_{m,j}$ , the coefficients “in powers of  $m^{-1}$ ” there, whose  $m$  is this section's  $R$ ; Part II's  $c_{r,j}$  are the converted ones.

For an explicit check of the quadratic coefficient,  $Y$  in (125) has  $\mathbb{E}(Y - 2/3) = 2/(3R)$  and  $\mathbb{E}(Y - 2/3)^2 = 2/(9R) + O(R^{-2})$ . Using  $f_0, f_1$  in (123) gives

$$\frac{T_2(R)}{I_{R,2}} = 9 - \frac{153}{4R} + O(R^{-2}) = 9 \left(1 - \frac{17}{4R} + O(R^{-2})\right),$$

while

$$I_{R,2} = \frac{2}{9\sqrt{\pi}} \sqrt{R}(4/9)^R \left(1 + \frac{3}{8R} + O(R^{-2})\right).$$



Thus  $c_{2,1} = -31/8$ . Proposition 24.2 at  $M = 2$  has error  $O((27/64)^R)$ , exponentially smaller than the displayed algebraic error in the quadratic sector. This proves (101). If  $\Delta(R) = \mathcal{J}(R) - A$ , then

$$A^{-1} - \mathcal{J}(R)^{-1} = A^{-2}\Delta(R) + O(\Delta(R)^2).$$

Since  $\Delta(R) \sim R2^{-R-1}$ , this proves (104). Moreover  $\Delta(R)^2 = O(R^2 4^{-R})$  is exponentially smaller than every fixed sector, since  $a_m > e^{-1} > 1/4$ . Therefore every finite-sector expansion for the integral gap transfers to the reciprocal gap by multiplying its sectors by  $36/\pi^4$ .

## 24.6 Derivative control for threshold inversion

For every fixed  $m$ ,

$$T'_m(R) \sim (\log a_m)T_m(R), \quad |T'_m(R)| = O_m(T_m(R)). \quad (138)$$

Here is a justification which does not differentiate an unqualified asymptotic error. At fixed  $x \leq \alpha R$ , differentiate the absolutely convergent series (109):

$$\partial_R F_R(x) = - \sum_{k \geq 0} \frac{x^k}{(R+1)_k} \sum_{j=1}^k \frac{1}{R+j}, \quad |\partial_R F_R(x)| \leq R^{-1} \sum_{k \geq 0} k \alpha^k = O_\alpha(R^{-1}).$$

Since  $F_R \geq 1$  and  $\psi(R+1) = \log R + O(R^{-1})$ ,

$$\partial_R \log \delta_R(x) = \log(x/R) + O_\alpha(R^{-1}). \quad (139)$$

On the compact saddle interval, differentiation of the sector integrand introduces the factor  $m \log(x/R) + O(R^{-1})$ , tending at the saddle to  $m \log y_m = \log a_m$ .

For the upper tail, differentiation of the gamma density gives

$$\partial_R P(R, x) = \mathbb{E}[(\log G - \psi(R)) \mathbf{1}_{\{G \leq x\}}], \quad G \sim \text{Gamma}(R, \text{rate } 1).$$

The variance of  $\log G$  is  $\psi_1(R) = O(R^{-1})$ , so Cauchy–Schwarz bounds this derivative uniformly in  $x$  by  $O(R^{-1/2})$ . By (106), the absolute derivative of the sector integrand is at most  $O_m(R^{-1/2})(1+x)e^{-x}$ . Its tail beyond  $\alpha R$  is negligible when  $\alpha > -\log a_m$ . On the lower tail, (139) adds only  $1 + |\log(x/R)|$  to the previous gamma bound; use  $|\log y| \leq C(y^{-1/2} + y^{1/2})$  and gamma tails with shapes shifted by  $1/2$ . These integrable estimates justify differentiation and prove (138). The derivative  $\partial_R P(R, x)$  is strictly negative: it is the covariance of the strictly increasing function  $\log G$  and the decreasing nonconstant indicator  $\mathbf{1}_{\{G \leq x\}}$  under a nondegenerate gamma density.

For fixed  $K \geq 1$ , write

$$\Delta_K(R) = \sum_{m=1}^K T_m(R), \quad \lambda = \log 2.$$

Direct differentiation of the exact linear term, together with (138), gives

$$\Delta'_K(R) = -(\lambda - 1/R)L_0(R) \left[ 1 + O_K(R^{-1/2}(8/9)^R) \right]. \quad (140)$$

Indeed, differentiating the nonleading zeta modes gives an error  $O(R3^{-R})$ , and all remaining fixed sectors and their derivatives are dominated by the quadratic sector. For  $K = 1$  the stated common bound can be improved.

[**write, 5 October 2026**] *The remark proved.* The sentence above is stated without proof. For  $K = 1$  the bracket in (140) is in fact  $1 + O((2/3)^R)$ , smaller than  $O(R^{-1/2}(8/9)^R)$ :

**Proposition (write).** *For every real  $R \geq 2$ ,*

$$\Delta'_1(R) = -(\lambda - 1/R)L_0(R)[1 + \varepsilon_1(R)], \quad |\varepsilon_1(R)| \leq \frac{18S}{\lambda - 1/2} \left(\frac{2}{3}\right)^R < 10.4 \left(\frac{2}{3}\right)^R,$$

where  $S = \sum_{k \geq 3} k^{-3} \log k < 0.1115$ .

*Proof.* By (114),  $\Delta_1(R) = T_1(R) = L(R) = \sum_{k \geq 2} Rk^{-R-1}$  for real  $R > 0$ . On a compact interval  $[R_0, R_1] \subset (0, \infty)$  the derivatives  $\frac{d}{dR}(Rk^{-R-1}) = k^{-R-1}(1 - R \log k)$  are bounded by  $(1 + R_1 \log k)k^{-R_0-1}$ , which is summable over  $k \geq 2$ , so termwise differentiation is valid:

$$L'(R) = \sum_{k \geq 2} k^{-R-1}(1 - R \log k).$$

The term  $k = 2$  is  $2^{-R-1}(1 - R \log 2) = -(\lambda - 1/R)L_0(R)$ , with  $L_0(R) = R2^{-R-1}$  as in (115). For  $R \geq 2$  and  $k \geq 3$  one has  $R \log k \geq 2 \log 3 > 1$ , hence  $|1 - R \log k| \leq R \log k$ , and  $(3/k)^{R+1} \leq (3/k)^3$ . So the remaining terms satisfy

$$\left| \sum_{k \geq 3} k^{-R-1}(1 - R \log k) \right| \leq R 3^{-R-1} \sum_{k \geq 3} \log k \left(\frac{3}{k}\right)^{R+1} \leq 27S R 3^{-R-1}.$$

Since  $\lambda - 1/R \geq \lambda - 1/2 > 0$ , dividing by  $(\lambda - 1/R)L_0(R)$  gives  $|\varepsilon_1(R)| \leq \frac{27S}{\lambda - 1/2} (2/3)^{R+1} = \frac{18S}{\lambda - 1/2} (2/3)^R$ . Numerically  $S = 0.11148\dots$  and  $\lambda - 1/2 = 0.19314\dots$ , so the constant is below 10.4.  $\square$

Thus the common bound of (140) improves at  $K = 1$  from relative order  $R^{-1/2}(8/9)^R$  to  $(2/3)^R$ , the order of the first nonleading zeta mode. For  $K \geq 2$  the error term is set by the quadratic sector and is not improved by this argument.

## 24.7 Canonical envelopes for the minimum cap

Given  $0 < \epsilon < 1/A$ , set

$$\tau(\epsilon) = \frac{A^2 \epsilon}{1 - A\epsilon}. \quad (141)$$

The desired inequality is equivalent to

$$A^{-1} - \mathcal{J}(R)^{-1} \leq \epsilon \iff \Delta(R) \leq \tau(\epsilon).$$

For all sufficiently small  $\epsilon$  there is a unique real  $R_*(\epsilon) > 2$  with  $\Delta(R_*) = \tau(\epsilon)$ , and strict monotonicity yields the exact integer answer

$$r_{\min}(\epsilon) = \lceil R_*(\epsilon) \rceil - 1. \quad (142)$$

For each fixed  $K \geq 1$ , let  $R_K(\epsilon)$  be the unique sufficiently large root

$$\Delta_K(R_K) = \tau(\epsilon). \quad (143)$$

These envelopes use exact sectors. In particular  $R_1$  solves  $R(\zeta(R+1) - 1) = \tau$ , incorporating the full linear zeta term and the exact reciprocal transformation;  $R_2$  adds the exact quadratic integral. Positivity gives  $R_1 < R_2 < \dots < R_K < R_*$ .

**Proposition 24.5** (Displacement from an exact-sector envelope). *For every fixed  $K \geq 1$  and  $q \geq 0$ , with  $m = K + 1$ ,*

$$R_* - R_K = \frac{2C_m}{\lambda - 1/R_K} R_K^{(1-m)/2} (2a_m)^{R_K} \left( \sum_{j=0}^q c_{m,j} R_K^{-j} + O_{K,q}(R_K^{-q-1}) \right). \quad (144)$$

*In particular,*

$$R_* - R_2 \sim \frac{12\sqrt{3}}{\pi\lambda R_2} \left( \frac{27}{32} \right)^{R_2}. \quad (145)$$

*Proof.* The omitted tail at  $R_K$  is asymptotic to  $T_{K+1}(R_K)$  by (128). On a bounded neighborhood of  $R_K$ , (140) is uniformly asymptotic to  $-\lambda L_0$ , while the omitted tail is uniformly  $o(L_0)$ . Evaluating at  $R_K$  and at  $R_K$  plus a sufficiently large constant times  $T_{K+1}(R_K)/L_0(R_K)$  first brackets  $R_*$ , proving that its displacement tends to zero. The mean value theorem then applies on this shrinking interval. Equation (138) shows that evaluating the leading omitted sector at  $R_*$  rather than  $R_K$  makes only a relative exponentially small change. Subsequent sectors, the error in (140), and the displacement effect are all exponentially smaller than every fixed inverse power of  $R_K$ . Insert Theorem 24.4 and divide by  $(\lambda - 1/R_K)L_0(R_K)$  to obtain (144).  $\square$

The first inverse- $R_2$  correction in (145) is

$$\frac{12\sqrt{3}}{\pi(\lambda - 1/R_2)R_2} \left( \frac{27}{32} \right)^{R_2} \left( 1 - \frac{43}{3R_2} + O(R_2^{-2}) \right).$$

Because  $R_2$  uses the exact quadratic sector, this genuinely isolates the cubic exponential scale. A finite inverse- $R$  approximation to  $T_2$  alone cannot do so.

## 24.8 Lambert W inversion and the integer boundary

The leading reciprocal gap is  $C_0 R 2^{-R}$ , where  $C_0 = 18/\pi^4$ . Its inverse on the decreasing branch is

$$x_\epsilon = -\frac{1}{\lambda} W_{-1} \left( -\frac{\lambda\epsilon}{C_0} \right). \quad (146)$$

If  $U = \log(C_0/(\lambda\epsilon))$ , then

$$\lambda x_\epsilon = U + \log U + \frac{\log U}{U} + O((\log U)^2/U^2).$$

The Lambert expression itself is exact for the leading envelope.

Uniformly for large real  $R$ ,

$$\frac{A^{-1} - \mathcal{J}(R)^{-1}}{C_0 R 2^{-R}} = 1 + \frac{4}{\sqrt{\pi}} R^{-1/2} (8/9)^R \left( 1 - \frac{31}{8R} + O(R^{-2}) \right).$$

The exact-zeta correction has relative order  $O((2/3)^R)$  and the reciprocal nonlinearity has relative order  $O(R 2^{-R})$ , both exponentially smaller than the positive quadratic correction. Thus the true gap eventually exceeds the leading envelope. The shrinking-interval argument used above gives

$$R_* - x_\epsilon = \frac{4}{\sqrt{\pi}(\lambda - 1/x_\epsilon)} x_\epsilon^{-1/2} (8/9)^{x_\epsilon} \left( 1 - \frac{31}{8x_\epsilon} + O(x_\epsilon^{-2}) \right). \quad (147)$$

In particular the displacement is positive and tends to zero. With denominator  $\lambda$  factored out, its first relative inverse- $x_\epsilon$  coefficient is  $1/\lambda - 31/8$ . At any fixed order the last factor may be replaced by the corresponding finite series of coefficients  $c_{2,j}$  with its proved remainder.

**Corollary 24.6** (At most two eventual integer candidates). *For all sufficiently small positive  $\epsilon$ ,*

$$r_{\min}(\epsilon) \in \{\lceil x_\epsilon \rceil - 1, \lceil x_\epsilon \rceil\}. \quad (148)$$

*If  $x_\epsilon$  is an integer, the larger candidate is the answer. Otherwise put  $d_\epsilon = \lceil x_\epsilon \rceil - x_\epsilon$ ; the smaller candidate is correct exactly when  $R_* - x_\epsilon \leq d_\epsilon$ .*

*Proof.* For sufficiently small  $\epsilon$ , (147) gives  $0 < R_* - x_\epsilon < 1$ . Apply (142). Equality at an integer  $R_*$  is included because the original gap inequality is non-strict.  $\square$

[write, 5 October 2026] *Relation to Section 8.* With  $\lambda = \log 2$  and  $C_0 = 18/\pi^4 = 2D$ , where  $D = 9/\pi^4$  is the constant of (33), the Lambert root (146) is  $x_\epsilon = b_0 + 1$  for the  $b_0$  of (34), and  $r_{\min}(\epsilon)$  is Part I's  $b_\epsilon$ . So the corollary says  $b_\epsilon \in \{\lceil b_0 \rceil, \lceil b_0 \rceil + 1\}$  for all sufficiently small  $\epsilon$ . Part I's bracket (36) is effective for  $b_0 \geq 13$ ; this corollary is eventual only. When  $b_0$  is not an integer, Part I's lower bracket  $\lfloor b_0 \rfloor + 1$  is the smaller candidate here. Part I's counterexamples at  $\epsilon = h(n)$  are the case of an integer  $x_\epsilon$ , where the corollary selects the larger candidate. That the true gap exceeds the leading envelope, eventually here, is proved in Section 8 for every  $b \geq 3$  ( $d_b > h(b)$ ). New here: the exact selection criterion  $R_* - x_\epsilon \leq d_\epsilon$  and the displacement asymptotics (147).

An analogous, narrower two-candidate bracket uses  $R_2$  in place of  $x_\epsilon$ . For any fixed  $0 < \eta < 1$ , (145) gives the eventual interval

$$R_2 + (1 - \eta)H_2(R_2) < R_* < R_2 + (1 + \eta)H_2(R_2), \quad H_2(u) = \frac{12\sqrt{3}}{\pi\lambda u}(27/32)^u.$$

Once a rigorous validity bound for this interval has been established at a particular  $\epsilon$ , equal ceilings of its endpoints certify the integer cap. The eventual asymptotic statement alone does not provide a numerical certificate at a specified finite  $\epsilon$ . If validity is not certified, or an integer lies between certified endpoints, a certified direct bound on the gap at the candidate is needed. At an exact threshold the corresponding integer cap qualifies; arbitrarily close approaches to such thresholds cannot be decided from a fixed asymptotic truncation alone.

*Remark 24.7* (Limits of exponential inversion). Formula (144) is centered at the exact envelope root  $R_K$ . It does not permit linear superposition of arbitrarily many exponential shifts about  $x_\epsilon$ . Nonlinear inversion products interleave the sector scales: for example  $(8/9)^2$  lies strictly between  $2(6/7)^6$  and  $2(7/8)^7$ . Neither these fixed-sector formulas nor the all-orders algebraic expansions resolve a growing number of sectors or the accumulation layer at exponential base  $e^{-1}$ .

## 25 Reproducible finite verification

The accompanying package contains an offline, standard-library-only exact verification suite. Python 3.9 or later and a POSIX shell suffice. From any working directory, run

```
sh /path/to/bounded_ascent_report/run_checks.sh.
```

The script exits unsuccessfully on any failed check. It runs the full suite both normally and under `python -O`, then runs the negative-regression harness itself under optimization. No correctness condition relies on a removable Python assertion statement. Setup, data provenance, and detailed coverage are documented in `checks/README.md`.

**Three finite models and complete marginals.** For each  $r = 2, 3, 4, 5, 6$ , the suite independently computes  $c_{n,r}$  by literal original-word enumeration, an ordered residual-capacity dynamic program that never sorts, and the compact category-count recurrence. All three agree for  $0 \leq n \leq 10$ , and every nonempty endpoint histogram of  $X = (X_1, \dots, X_r)$  agrees for  $1 \leq n \leq 10$ . The 55 count values and all 50 endpoint histograms are also compared with frozen integer fixtures; the latter contain 985 nonzero bins in total. This is the complete category-count marginal, not an assertion that sorted and unsorted last-threshold marginals coincide. For  $r = 2$  the three-model and full-histogram comparison additionally extends through  $n = 11$ , although the frozen endpoint fixtures stop at  $n = 10$ . The literal enumerations visit 919,973 nonempty prefixes altogether.

The literal enumerator also records the full joint distribution of  $(N_1, \dots, N_r, D, A)$ , where  $D$  counts distinct values and  $A$  counts ascents. The endpoint identities in the report recover this distribution exactly from the compact histograms. In addition, the finite identity

$$c_{n+1,r} = \sum_X \left( \sum_{j=1}^r X_j \right) h_{n,r}(X)$$

is checked for  $n = 1, \dots, 9$  when  $r = 3, \dots, 6$ , and for  $n = 1, \dots, 10$  when  $r = 2$ .

**Published overlap.** The package transcribes  $c_{n,2}$  for  $0 \leq n \leq 11$  from OEIS A202058, including  $c_{11,2} = 77451$ , and the displayed square-array entries for  $r = 2, \dots, 6$ ,  $0 \leq n \leq 8$  from OEIS A294220. This gives 57 source entries, with overlap retained deliberately. The pages were checked on 2 October 2026. A294220 is the whole array; its  $r = 3$  column is A317784. The package does not label its locally computed  $r = 3$ ,  $n = 9, 10$  values as independently verified published terms. The nonpublished frozen histograms are regression data generated by a separate literal-word audit implementation, not independent proof certificates.

**Local structural checks.** The adjacent-swap tests exhaust 12,220 cases:  $r = 2, 3, 4$ ; initial capacity-list length  $d = 2, 3, 4$ ; all lists in  $\{1, \dots, r\}^d$ ; all thresholds  $L = 2, \dots, d$ ; all adjacent pairs with upper index below  $L$ ; and suffix lengths  $0, \dots, 4$ . Each test compares the full joint distribution of final  $X$  and new ascent count, not merely a total. Equal-capacity swaps and empty suffixes are included in this count.

The positive-block tests check 78,570 deterministic buffered paths, covering  $r = 2, \dots, 6$ , block lengths  $3, \dots, 8$ , six specified category-usage patterns, every middle permutation, and three initial thresholds. They verify category retention, exact coordinate drift, the permutation-ascent formula, and final-threshold closure; the complete middle-permutation ascent bins are also compared with independently computed Eulerian numbers. Five further blocks, one for each tested  $r$ , have strictly positive drift in every coordinate. These tests use no random sampling.

**Fail-closed data handling.** The validators require exact fields, fixed coverage, integer types excluding booleans, valid dimensions and indices, positive bin counts, feasible endpoint statistics, and the correct histogram totals. Duplicate JSON keys, duplicate rows or bins, missing data, extra or shortened value lists, altered declared coverage, and nonfinite constants are rejected. Required coverage is fixed in the validator rather than inferred from whatever data happen to be present. The 64 negative-regression cases run in 128 separate processes, half under `-O`. They include syntactically valid but false count values with matching corrupted histogram totals, false histogram bins preserving the correct total, and a false published value. Thus merely retaining correct schemas or aggregate totals cannot make those corruptions pass.

**Scope of the evidence.** All checks above use exact integer arithmetic, but their coverage is finite. They corroborate the recurrence and its implementations; they do not prove the exchange lemma in general, the asymptotic root or ratio limit, concentration, a coefficient equivalent, or a convergence rate. The proofs in the report supply the mathematical arguments. An optional, separately invoked `mpmath` program evaluates the characteristic integrals and finite-difference residuals. It is not part of `run_checks.sh`, has no interval-certified error bounds, and is explicitly labeled numerical diagnostics rather than proof.

**Optional exact large-cap algebra.** The separate `checks/analytics/` directory requires SymPy and is excluded from the standard-library runner. Its own runner checks, both normally and under `-O`, all printed rational  $c_{m,j}$  for  $m = 1, 2, 3$  and  $j = 0, 1, 2, 3$ , consistency under changing the finite truncation order, and  $c_{m,1} = \kappa_m$  for  $m = 1, 2, 3, 4, 5, 8, 12$ . It also checks the first three rational tail-factor functions and rejects 26 invalid index/order inputs. These are exact finite algebraic comparisons; the large-cap remainder estimates and inversion results rely on the analytic arguments, not on the replay. An additional optional `mpmath` sector-quadrature diagnostic is likewise excluded from both exact runners and supplies no certified error bounds.

[write, 5 October 2026] In this repository the suite is shipped as `code/53-bounded-*.py` and `code/53-bounded-*.sh`, its data as `data/53-bounded-data-*.json`, its two READMEs as `53-bounded-README.md` and `53-bounded-analytics-README.md`, and the provenance record as `data/53-bounded-PROVENANCE.json`. The delivered build script is byte-identical to the one of Report 97 and is shipped once, in the sibling report, as `code/38-growth-build_local.sh`. The delivered PDF, README, manuscript (printed as this Part) and `MANIFEST.sha256` (31 entries, all verified at placement) are not shipped. The scripts use the delivered layout (`checks/`, `checks/data/`) and must not be run here; the README gives a rerun route from the arrival commit. At intake (5 October 2026, on copies) `run_checks.sh` passed in about 5 minutes and reproduced every count stated above, the analytic runner passed in about 2 minutes, and the three `mpmath` diagnostics ran; none of them writes an output file to compare. The frozen counts agree with those printed in Section 9 for  $b = 3, 4$ .

## 26 Further questions and limits of the results

### 26.1 Subexponential word length asymptotics

Theorem 17.1 identifies  $N!\mu_r^N$  only at exponential accuracy. Determining a factor  $B_r(N)$  in an equivalent  $c_{N,r} \sim N!\mu_r^N B_r(N)$  is a separate problem. Even for  $r = 2$ , a precise amplitude and a complete correction hierarchy require estimates beyond the block action. A useful next step is a controlled discrete transport analysis around the characteristic, with particular care near its singular origin. Numerical evidence or a formal transport equation alone would not prove such a factor.

### 26.2 Fluctuations and local enumeration

The smooth limiting pressure gives exponential concentration and the full large-deviation principle, but does not justify differentiating finite-size limits twice. A central limit theorem would require additional uniform control of the weighted coefficients near a fixed mark. One could seek a nondegenerate covariance matrix from the Hessian of  $F$ , then prove that it is the actual covariance limit and establish a local limit theorem. The latter would refine the enumeration of words with prescribed multiplicity histograms.

### 26.3 The feasible endpoint region

The rate function is specified by a finite-dimensional convex dual, but its finite-rate boundary is not classified here. Besides nonnegativity and  $\sum_j x_j \leq 1$ , the exact multiplicity identity imposes  $\sum_{j < r} (r - j)x_j \leq 1$ . Identifying all attainable limiting endpoints, and determining which have finite rate, would give a more geometric interpretation of the pressure and of atypical ascent counts. Such a classification is unnecessary for the full large-deviation proof above.

### 26.4 A cap growing with the word length

The fixed- $r$  theorem and the large- $r$  expansion are two successive results, not a uniform joint limit. They do not allow substitution of  $r = r(N)$  into a coefficient asymptotic. The exponentially small gap  $6/\pi^2 - \mu_r$  suggests investigating logarithmic cap scales, but even the correct transition window requires estimates uniform in  $r$  and control of the unrestricted maximum multiplicity. The constants in the present seeds, block dimensions, and error estimates depend on  $r$ .

### 26.5 Sector accumulation and effective inversion

For every fixed sector and every fixed inverse-cap order, the analytic expansion is rigorous. The sector bases accumulate at  $e^{-1}$ , so letting the number of sectors grow with the cap requires new estimates. A finite inverse-power truncation of one sector also cannot isolate an exponentially smaller sector. The exact-sector threshold envelopes avoid that loss, while concrete decisions arbitrarily close to an integer threshold still need certified evaluation rather than asymptotic rounding.

Certified integral bounds and faster compact enumeration are practical next steps that do not require a full word-length expansion.

### 26.6 Further questions and research (added at the write)

[**write, 5 October 2026**] Under the repository's standing rule every claim of Report 99 that is not proved in it is listed here, with its source, its sketch and what is missing. No claim of Report 99 was found to be wrong.

1. *A word-length prefactor*  $B_r(N)$  in  $c_{N,r} \sim N! \mu_r^N B_r(N)$  (Section 26, first subsection; also Part I's Question 1). Sketch offered: a discrete transport analysis around the characteristic, careful at its singular origin. Missing: any estimate below the exponential scale for  $r \geq 3$ . For  $r = 2$  the sibling report **a202058-ascent-000-growth** proves  $-O(\sqrt{n}(\log n)^{3/2}) \leq \log(c_{n,2}/(n!\mu_2^n)) \leq O((\log n)^2)$  and the lim sup and cumulative laws with constant  $2/3$ ; the pointwise law is open there too.
2. *Fluctuations*: a central limit theorem and a local limit theorem for  $Y_N$ , with covariance the Hessian of  $F$  at the origin (second subsection). Missing: uniform control of the weighted coefficients near a fixed mark; the pressure limit alone does not allow differentiating twice.
3. *The finite-rate endpoint region* of Theorem 22.2 (third subsection). Known constraints:  $\mathbf{x} \geq 0$ ,  $\sum_j x_j \leq 1$  and  $\sum_{j < r} (r - j)x_j \leq 1$ . Missing: a classification of the attainable limits and of the points of finite rate.
4. *A cap growing with the length*,  $r = r(N)$  (fourth subsection; Part I's Question 4 and Part II's Question 4). Missing: estimates uniform in  $r$ , including the seeds, block dimensions and error terms, and control of the unrestricted maximum multiplicity.
5. *A growing number of sectors* and the accumulation of the bases  $a_m$  at  $e^{-1}$  (fifth subsection; Part II's Questions 1 and 2). Missing: estimates uniform in the sector index.

6. *Certified integer cap decisions* near integer transitions (fifth subsection and the discussion after Corollary 24.6; Part II’s Question 3). Part I’s bracket (36) is effective but does not choose between the two candidates. Missing: certified evaluation of the gap at the candidate.
7. The remark after (140), “for  $K = 1$  the stated common bound can be improved”, is stated without proof. *Resolved at the write*: the proposition in the note after it gives the relative error  $O((2/3)^R)$ .
8. *Practical next steps* named in Report 99: certified integral bounds (all quadratures in this Part, including the table of  $\mu_r$ , are uncertified) and faster compact enumeration.

## References

- [1] A. R. Conway, M. Conway, A. Elvey Price and A. J. Guttmann, *Pattern-Avoiding Ascent Sequences of Length 3*, Electronic Journal of Combinatorics **29**(4) (2022), P4.25. doi:10.37236/11266. Published paper; arXiv:2111.01279. Part III consulted the final publication, especially Sections 2.2, 2.6 and 5.
- [2] M. Bousquet-Mélou, A. Claesson, M. Dukes and S. Kitaev,  $(2 + 2)$ -free posets, ascent sequences and pattern avoiding permutations, Journal of Combinatorial Theory, Series A **117**(7) (2010), 884–909. doi:10.1016/j.jcta.2009.12.007. Author preprint: <https://arxiv.org/abs/0806.0666>. [Cited by Part III.]
- [3] P. Duncan and E. Steingrímsson, *Pattern avoidance in ascent sequences*, preprint (2011), arXiv:1109.3641. Part III cites the published version: Electronic Journal of Combinatorics **18**(1) (2011), Paper P226, doi:10.37236/713.
- [4] OEIS Foundation Inc., A294220, maximum-letter-multiplicity array. Retrieved 1 October 2026. Part III: A294220: Number of ascent sequences of length  $n$  where no letter multiplicity is larger than  $k$ ; the entry identifies columns  $k = 2, 3$  with A202058 and A317784 and supplies the small comparison table used in Part III’s verification package. Consulted 2 October 2026.
- [5] OEIS Foundation Inc., A202058: Number of ascent sequences avoiding the pattern 000. <https://oeis.org/A202058>. Linked exact data by A. Conway and M. Conway: <https://oeis.org/A202058/b202058.txt>. Part III’s package uses the explicitly stated finite overlap, not an assertion that its general-cap computation reproduces the entire long  $r = 2$  table.
- [6] OEIS Foundation Inc., A317784, cap-three column. Identification verified from A294220 on 1 October 2026.
- [7] *Factorial growth of ascent sequences avoiding 000*, companion proof and reproducible report, 1 October 2026. The separate deliverable is `a202058-report.pdf`; its SHA-256 and source hashes are recorded in this archive’s `provenance.json`. [Write note, 2 October 2026: batch-77 manuscript 37, printed in this repository as Part I of `SetTheory/Cardinals/docs/reports/generating-functions-and-asymptotics/oeis-sequence-asymptotics/a202058-ascent-000-growth`; the record is shipped as `data/47-caps-provenance.json`.]
- [8] *Fixed Taylor sectors and their all-orders large-cap expansions*. Derivation dated 1 October 2026. Included as `sources/fixed-taylor-sectors.md`. [Shipped here as `52-sectors-fixed-taylor-sectors.md`.]
- [9] *Independent audit of the fixed Taylor sectors*. PASS audit dated 1 October 2026. Included as `sources/fixed-sector-independent-audit.md`; exact version recorded in `provenance.json`. [Shipped here as `52-sectors-fixed-sector-independent-audit.md` and `data/52-sectors-provenance.json`.]